

ESTRELA ACAPUZHNO REDD+ PROJECT



Project title	ESTRELA ACAPUZHNO REDD+
Project ID	4761
Crediting period	08/11/2021 – 07/11/2051; 30 years
Project lifetime	08/11/2021 – 07/11/2051; 30 years
(CCB) GHG accounting period	08/11/2021 – 07/11/2051; 30 years
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Version	1.0
VCS Standard version	Version number of the VCS Standard used by the project
CCB Standards version	Version number of the CCB Standards used by the project
Project location	Municipalities of Jutai, Fonte Boa and Juruá; Department of Amazonas; Brasil
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Validation/verification body	<i>Not available</i>
History of CCB status	<i>Not applicable.</i>
Gold Level criteria	Exceptional Community Benefits The project will generate important benefits for the community, related to the increase in welfare and quality of life conditions. The Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho have restricted access to basic services such as energy, health care, education, transport, water and sanitation. The project will invest in social infrastructure, and in income generating productive activities, so that incentives for deforestation are reduced. It is expected that the project will help raise the economic conditions of the community while forest governance capacity is enhanced.

Expected verification schedule	November 2024
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1 SUMMARY OF PROJECT BENEFITS

1.1 Unique Project Benefits

Outcome or impact estimated by the end of project lifetime	Section reference
1) Strengthening of local governance and management of forests	2.4
2) Protection of High Conservation Values	5.2
3) Improvement of economic conditions of local communities	4
4) Enhancement of social infrastructure and access to basic services	4
1) Strengthening of local governance and management of forests	2.4

1.2 Standardized Benefit Metrics

Category	Metric	Estimated by the end of project lifetime	Section reference
GHG emission reductions or carbon dioxide removals	<i>Net estimated removals in the project area, measured against the without-project scenario</i>	Not applicable	
	<i>Net estimated reductions in the project area, measured against the without-project scenario</i>	11,217,391	2.1.13
Forest ¹ cover	<i>For REDD² projects: Estimated number of hectares of reduced forest loss in the project area measured against the without-project scenario</i>	Not applicable	
	<i>For ARR³ projects: Estimated number of hectares of forest cover increased in the project area measured against the without-project scenario</i>	8,481	2.1.13
Improved land management	<i>Number of hectares of existing production forest land in which IFM⁴ practices are expected to occur as a result of project activities, measured against the without-project scenario</i>	Not applicable	Not applicable
	<i>Number of hectares of non-forest land in which improved land management practices are expected to occur as a result of project activities, measured against the without-project scenario</i>	Not applicable	Not applicable

¹ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO, or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (VCS Program Definitions)

² Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (VCS Program Definitions)

³ Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (VCS Program Definitions)

⁴ Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood, and fuelwood (VCS Program Definitions)

Training	Total number of community members who are expected to have improved skills and/or knowledge resulting from training provided as part of project activities	390	2.1.17
	Number of female community members who are expected to have improved skills and/or knowledge resulting from training as part of project activities	320	2.1.17
Employment	Total number of people expected to be employed in project activities ⁵ , expressed as number of full-time employees ⁶	160	2.1.17
	Number of women expected to be employed as a result of project activities, expressed as number of full-time employees	160	2.1.17
Livelihoods	Total number of people expected to have improved livelihoods ⁷ or income generated as a result of project activities	80	2.1.17
	Number of women expected to have improved livelihoods or income generated as a result of project activities	320	2.1.17
Health	Total number of people for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	160	2.1.17
	Number of women for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	320	2.1.17

⁵ Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

⁶ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from the UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])

⁷ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. The Sustainable Livelihood Approach to Poverty Reduction. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

Education	Total number of people for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	160	2.1.17
	Number of women and girls for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	480	2.1.17
Water	Total number of people who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	240	2.1.17
	Number of women who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	160	2.1.17
Well-being	Total number of community members whose well-being ⁸ is expected to improve as a result of project activities	80	2.1.17
	Number of women whose well-being is expected to improve as a result of project activities	1,280	2.1.17
Biodiversity conservation	Expected change in the number of hectares managed significantly better by the project for biodiversity conservation ⁹ , measured against the without-project scenario	640	2.1.17
	Expected number of globally Critically Endangered or Endangered species ¹⁰ benefiting from reduced threats as a result of	390	2.1.17

⁸ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Livelihoods, Health, Education and Water), and may also include other benefits such as strengthened legal rights to resources, increased food security, conservation of access to areas of cultural significance, etc.

⁹ Managed for biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation, e.g. enhancing the status of endangered species

¹⁰ Per IUCN's Red List of Threatened Species

	<i>project activities¹¹, measured against the without-project scenario</i>		
		24	5.1.2

¹¹ In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

2 PROJECT DETAILS

2.1 Project Goals, Design and Long-Term Viability

2.1.1 Summary Description of the Project (VCS, 3.2, 3.6, 3.10, 3.11, 3.13, 3.14; CCB, G1.2)

The Estrela Acapuzihno REDD+ Project seeks the conservation of the existing forests in the indigenous territories (*Terra Indígena*) of Acapuri de Cima, Estrela da Paz and Riozinho, located within the municipalities of Fonte Boa, Jutai and Juruá, in the Amazon state of Brazil. The project area has a total extension of 391,225 ha. The territories of these communities are part of the northwestern geographic area of the country characterized for its high biodiversity and rain tropical forests and for the presence of deforestation activities and social conflicts. Terra Indígena Acapuri de Cima has 1 indigenous community (*aldeia*), Terra Indígena Estrela da Paz has 7 aldeias and Terra Indígena Riozinho has 6 aldeias, with a total of 1,600 people approximately.

In the last years, it has been a national concern the high deforestation rates and associated social and biodiversity impacts in this region. Local understanding of the REDD+ mechanism consists in that the more standing forest that their territory maintains, more benefits will be delivered by the project. Deforestation drivers include mining activities, subsistence agriculture, logging and agricultural frontier expansion. Social issues include absence of the state and lack of sanitation, energy, education and health infrastructure and lack of technical guidance regarding sustainable production models. No natural areas were drained or deforested before projects began to generate carbon credits.

The project aims to mitigate the effects caused by climate change by reducing CO₂ emissions derived from deforestation and forest degradation, restore degraded forest lands, promote sustainable management of forests and strength and develop productive activities to improve community welfare. The project also seeks to protect biodiversity and high conservation values, maintain ecological connectivity and secure natural habitats of vulnerable species (including those within some category of threat and those that represent ecological interest).

The climate, community and biodiversity objectives are detailed next:

- The project's climate objective is to mitigate climate change through measures that address deforestation and forest degradation drivers and reduce forest loss. The expected results include decrease in illegal logging, recovery of degraded forests, reduction of forest conversion to other land uses and enhance forest carbon stocks over time.
- The project's community objectives are constructed with the community and aim to protect their culture, identity and improve living conditions, including housing, transport, education, health, electricity, sanitation and water supply.

- Regarding biodiversity conservation, the project objective is protect forest and its diversity, contribute to maintain its connectivity and improve management of production systems to favour biodiversity, as compared to baseline scenario conditions.

To achieve the above mentioned objectives, project activities to be developed and implemented include four main components: i) strength productive systems and develop economic alternatives compatible with community well-being and nature conservation, ii) increase social investment to benefit living conditions and access to education, health, transport, sanitation and communications, iii) strength territorial governance and cultural protection, iv) monitor and protect forest and biodiversity within the boundaries of the Estrela Acapuzihno REDD+ project.

Moreover, expected GHG reductions correspond to an annual average of 373,913 tCO₂e, so for the lifetime of 30 years it could demonstrate reductions up to 11,217,391 tCO₂e. The project is not within a REDD+ jurisdictional program.

2.1.2 Audit History (VCS, 4.1)

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	15-November-2024	VERRA - CCB	Not defined at the moment.	One year

2.1.3 Sectoral Scope and Project Type (VCS, 3.2)

Sectoral scope	14: Agriculture, forestry, and other land use
AFOLU project category ¹²	Reduced Emissions from Deforestation and Degradation (REDD)
Project activity type	Avoiding Unplanned Deforestation.

¹² See Appendix 1 of the VCS Standard

2.1.4 Project Eligibility (VCS, 3.1, 3.6, 3.8, 3.18, 4.1; CCB Program Rules, 4.2.4, 4.6.4)

- This project uses an eligible methodology under the VCS Program. The Methodology VM0007 has been fully applied, including any referenced tools or modules.
- The project activities are referenced by international consensus of relevant GHG programs and agreements.
- The ODS gases, such as CFCs and HCFCs are not applicable for this specific project once the emissions' characteristics are solely related to REDD+ AFOLU.
- Although this project is not part of a Jurisdictional REDD+ program, this is aligned to the most recent criteria and trends for an integrated network of carbon initiatives, including the latest methodologies' updates from VERRA, and the recent evolution of an official Carbon Market in Brasil.
- The proponent has contacted and hired a qualified validation/verification body (VVB), which will perform rigorous assessments of VCS methodologies and methodology elements to confirm that they conform with VCS Program requirements and AFOLU-specific best practices.
 - The proponent has agreed alongside the VVB about the validation/verification deadlines.
 - The project is eligible under the VCS Program
 - The project applies the latest version of the applicable methodology in all cases, ensuring compliance unless a grace period applies. The project updates to the latest version of the methodology when reassessing the baseline or renewing a crediting period. This project does not violate any applicable local or regional laws, regardless of whether or not the law is enforced.
- The project and its implementation activities do not violate any applicable law, regardless of whether or not the law is enforced.

2.1.5 Transfer Project Eligibility (VCS, 3.23, Appendix 2)

Not applicable.

2.1.6 Project Design (VCS, 3.6)

Indicate if the project has been designed as:

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)

☒ Grouped project

2.1.6.1 Eligibility Criteria for Grouped Projects (VCS, 3.6; CCB, G1.14)

Project corresponds to a grouped project and is expecting to integrate other areas as new instances in the future.

The project corresponds to a grouped project. Neighboring indigenous territories have worked and began their protection activities to ensure forest and culture protection. These includes the following indigenous territories: Terra Indígena Rio Bía, Terra Indígena Macarrão, Terra Indígena São Domingos do Jacapari e Estação, e Terra Indígena Espírito Santo. These territories may be included as future instances of this project. Details of this geographic areas are presented in Annex 11.

In order to include further instances, the new areas shall ensure that they follow the same conditions and characteristics of the first instance, expressed in meeting the following set of conditions:

- a) Meet the applicability conditions of the methodology applied to the first instance.
- b) Use and apply the same technologies or measures that are describe for the first instance.
- c) The baseline scenario of the first instance can be applied to the new instance according to the specified project activity and geographic area.
- d) Demonstrate that additionality defined for the first instance applies or is consistent with the characteristics of the new instance.

2.1.7 Project Proponent (VCS, 3.7; CCB, G1.1)

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2.1.8 Other Entities Involved in the Project

Organization name	CONSELHO DOS POVOS INDIGENAS DE JUTAI - COPIJU	
Role in the project	Field Coordination with the Terras Indigenas	
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Email

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The email address domain must match that of the organization

2.1.9 Project Ownership (VCS, 3.2, 3.7, 3.10; CCB, G5.8)

The project proponent are the Indigenous Territories: Terra Indigena Acapuri de Cima, Terra Indigena Estrela da Paz and Terra Indigena Riozinho. The entire project area is located within the areas awarded to these communities through Ordinance no. 287 of April 13, 2000; Decree 000 of July 3, 1995; and Ordinance No. 651, of May 17, 2012, respectively (Section 2.5.1).

These land rights are protected by Constitutional Law of the Federative Republic of Brazil, in its articles 231 and 232; Decree No. 1775, of January 8, 1996 and Ordinance MJ No. 14, of January 9, 1996. These lands cannot be expropriated nor have liens set on them. The Right of Use is exclusively placed on the local communities belonging to the Indigenous Territories

2.1.10 Project Start Date (VCS, 3.8)

Project start date	08-november-2021
Justification	With the aim of conserving themselves culturally as a Indigenous People of the Amazon, the leaders, caciques and community members created and promulgated on the first week of November 2021, the establishment of the Indigenous Environmental Guard of Amazonas (GAIAM), in the Indigenous Village Bugaio, at Estrela da Paz Territory, in accordance with the Internal Norms of Self-Government of the Indigenous Communities of Amazonas. The purpose of the GAIAM is to protect, conserve, restore and defend the flora and fauna of their territories, including the social component, as well as to promote the professional training of GAIAM students, to qualify them in all scientific areas in which there is a need for qualification and professionalization for the training of students of the Indigenous Environmental Guard, in information-research of the

original peoples of Amazonas and literary constructions, as well as promoting the development of environmental and sustainable programs, projects, vocational courses and services, events, aimed at articulating with indigenous communities and their leaders, to combat climate warming and the greenhouse effect and all kinds of environmental damage and stop environmental catastrophes in the lungs of the planet, among other very important points and aspects.

The project start date is November 08, 2021. This is the date on which the GAIAM students (caciques, leaders and members) began their first training course to move towards their objectives as a community initiative. The students were part of the first activity developed under one the main GAIAM educational strategy denominated the Differentiated Indigenous University of the Amazonian Peoples (UIDPA). This course also included an Intercultural meeting of indigenous peoples of other indigenous territories within Jutai and congregated members from Ticuna, Kokama, Kambaba, Katukina and Kanamari ethnicities.

The course addressed environment related topics, guiding how to take care of the environment where the indigenous communities live, and focussing on preservation of the natural forest and the land. The course also promoted cultural practices, including dances, preparation of clothing for cultural presentations, traditional body painting, highlighting the need of rescuing traditional language, the role of the warrior in pedagogy.

2.1.11 Benefits Assessment and Project Crediting Period (VCS, 3.9; CCB, G1.9)

Crediting period	The crediting period goes from November 08 of 2021, to November 07 of 2051, resulting in an expected lifetime of 30 years.
Start date of first or fixed	08-November-2021 to 07-November-2051

crediting period	
CCB benefits assessment period	30 years

2.1.12 Differences in Assessment/Project Crediting Periods (CCB, G1.9)

There are no differences between the GHG emissions accounting neither other assessment nor the crediting period.

2.1.13 Project Scale and Estimated Reductions or Removals (VCS, 3.10)

Indicate the estimated annual GHG emission reductions/carbon dioxide removals (ERRs) of the project:

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Complete the table below for the first (if renewable) or fixed crediting period:

Calendar year of crediting period	Estimated reductions or removals (tCO ₂ e)
08-November-2021 to 31-December-2021	22.927
01-January-2022 to 31-December-2022	142.295
01-January-2023 to 31-December-2023	170.690
01-January-2024 to 31-December-2024	199.084

01-January-2025 to 31-December-2025	227.479
01-January-2026 to 31-December-2026	255.874
01-January-2027 to 7-November-2027	236.890
Total estimated ERRs during the first or fixed crediting period	1.232.312
Total number of years	6
Average annual ERRs	209.206.

2.1.14 Physical Parameters (CCB, G1.3)

The three Indigenous Territories (*Terras Indígenas*) are located within the north-western zone of Brazil. Acapuri de Cima is located to the north of the Amazon River (*Rio Solimões*) belonging 94% to the municipality of Fonte Boa, and 6% to the municipality of Jutai; Estrela da Paz and Riozinho are located to the south of the Amazon River. The territory of Estrela da Paz is located in the municipality of Jutai. 62% of the territory of Riozinho is in the municipality of Jutai and 38% in the municipality of Juruá. However, in the case of Riozinho the administrative management and benefits are received in Jutai.

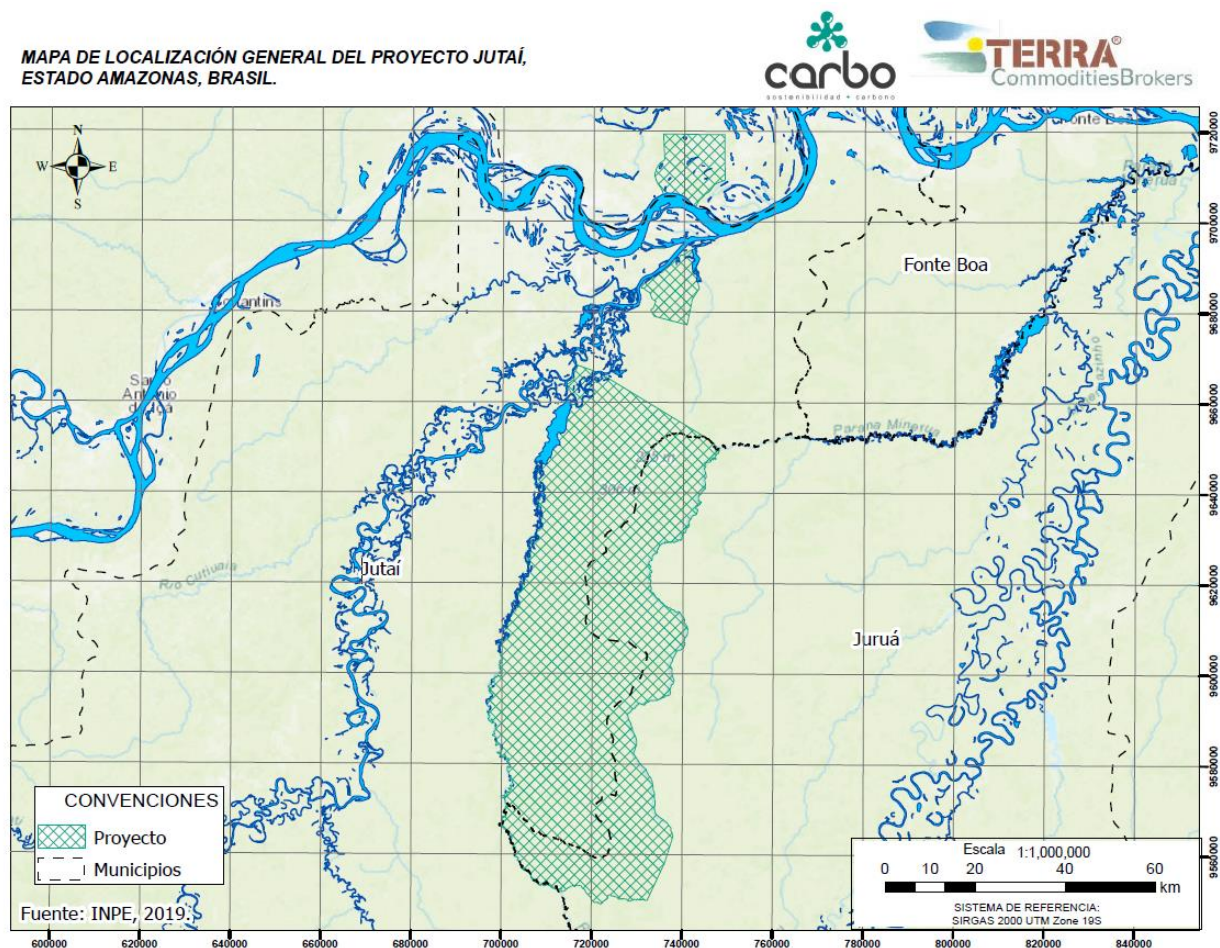
The project area is located within the Amazon state, known as the largest state of Brazil bordering with Colombia, Perú and Venezuela, and known for being tropical rain forest, with most of its land occupied by forest reserves and water. Access to this area is mostly by waterways or by plane. The municipality of Jutai is the fifth largest municipality of the Amazon state and is in the Juruá-Purus moist forests ecoregion.

The geographic coordinates of the indigenous territories are:

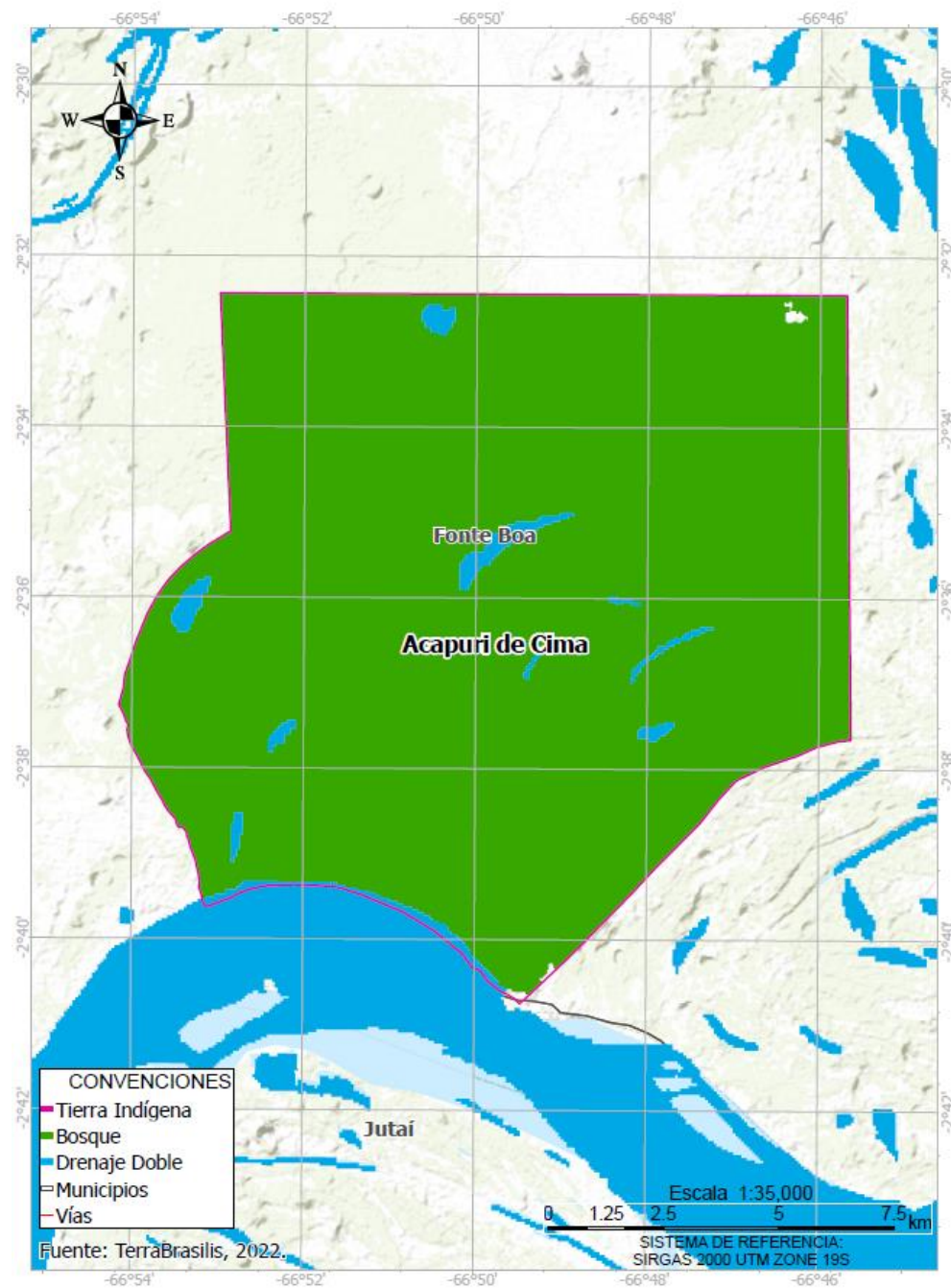
Indigenous territory	North	South	West	East
Acapuri da Cima	02° 32' 15.075984"S 66° 53' 0.928651"WGr	02° 37' 42.076374"S 66° 45' 38.018211"WGr	02° 32' 16.684942"S 66° 48' 9.636686"WGr	02° 39' 12.644736"S 66° 53' 10.924560"WGr

Estrela da Paz	02°50'40.744"S 66°55'37.946"WGr	02°54'51.970"S 66°50'58.972"WGr	02°52'17.859"S 66°53'47.912"WGr	02°49'28.311"S 66°48'32.132"WGr
Riozinho	03°01'05,9"S 66°59'30,0"WGr	03°59'31,3"S 66°53'07,9"WGr	03°04'45,0"S 67°03'18,7"WGr	03°10'10,3"S 66°46'28,9"WGr

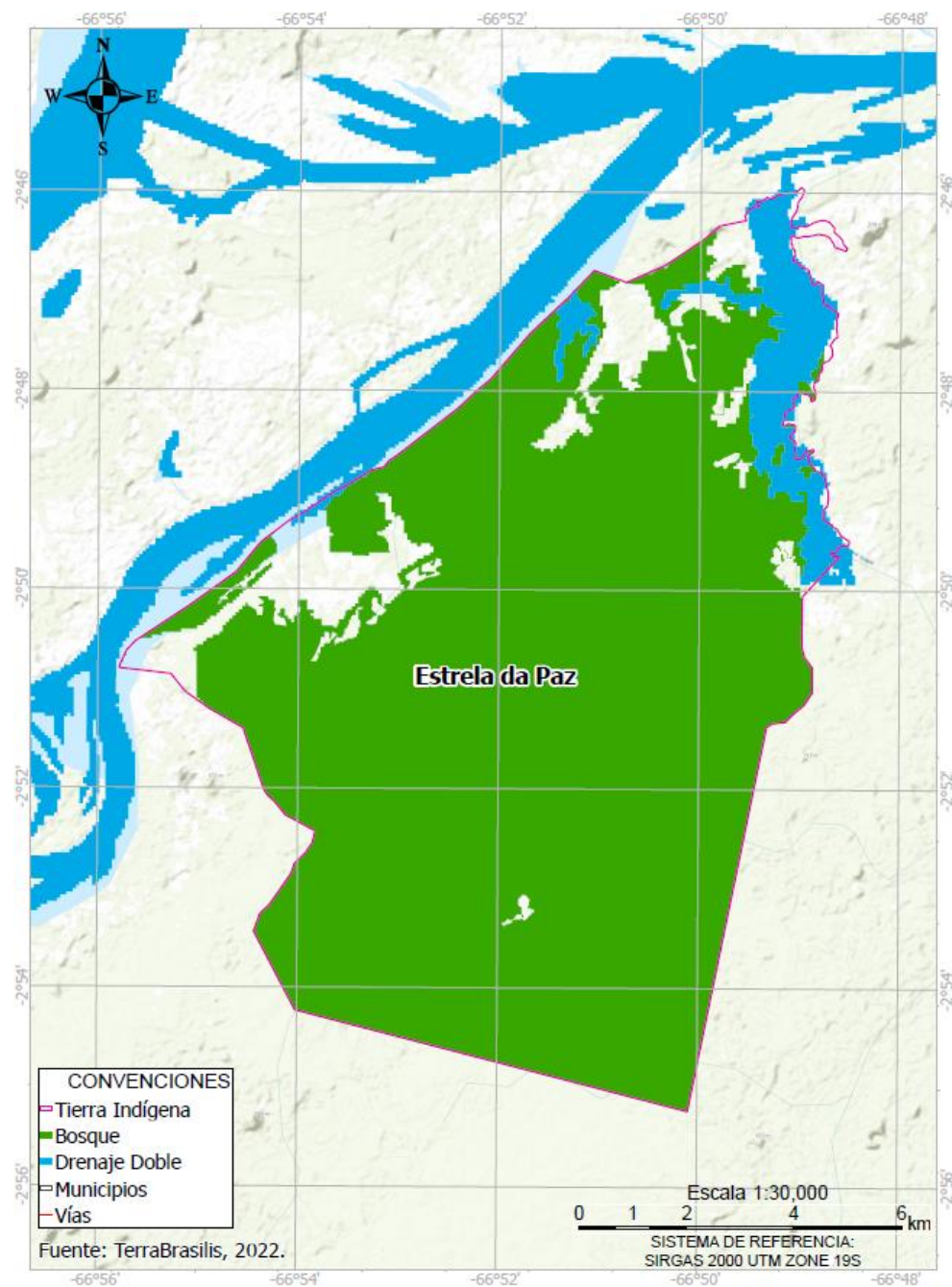
MAPA DE LOCALIZACIÓN GENERAL DEL PROYECTO JUTAI,
ESTADO AMAZONAS, BRASIL.



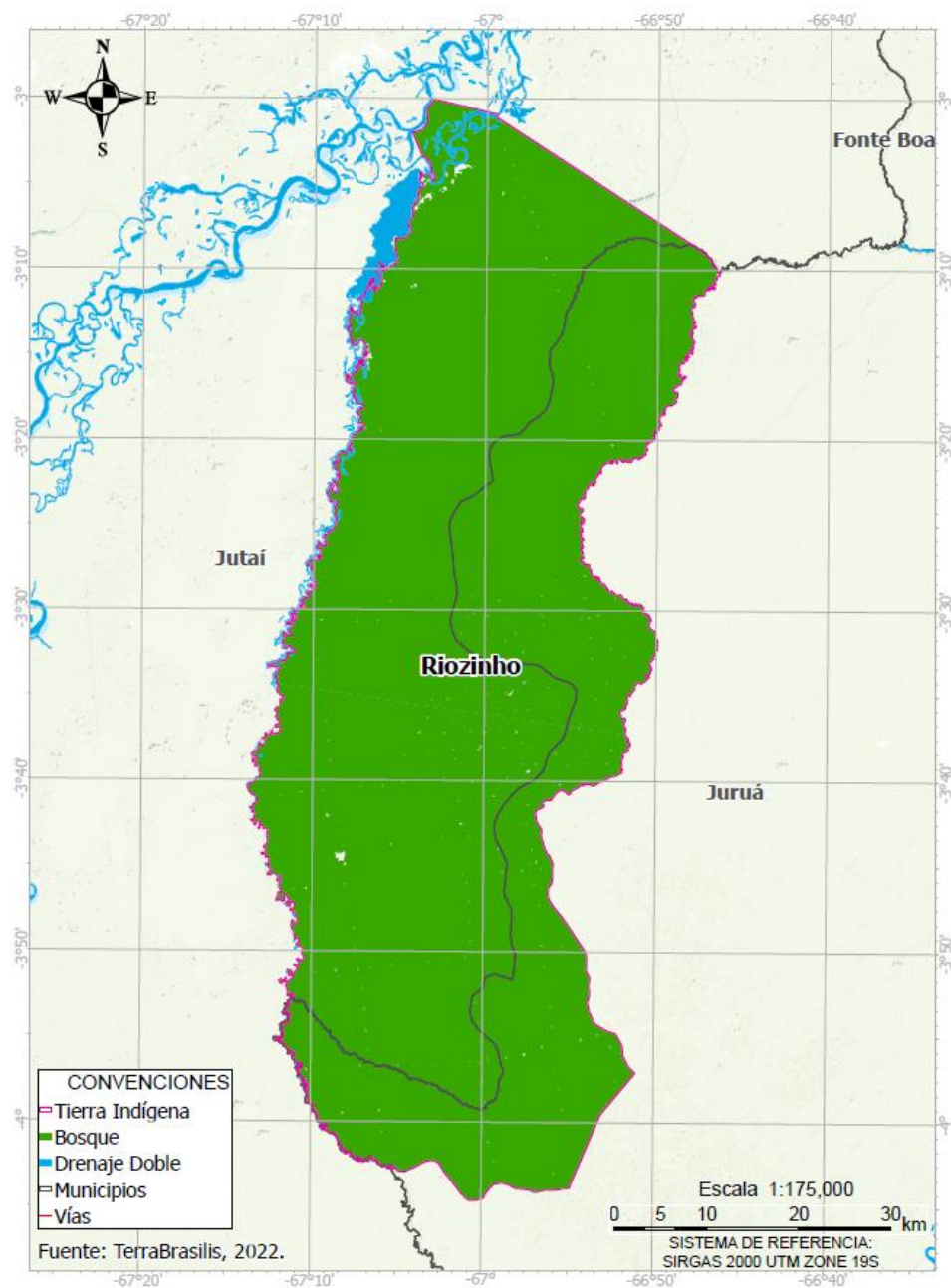
Map 1 Location of the Indigenous Territories (Terras Indígenas)



Map 2. Location of the Indigenous Territory (Terra Indígena) of Acapuri de Cima



Map 3. Location of the Indigenous Territory (Terra Indígena) of Estrela da Paz

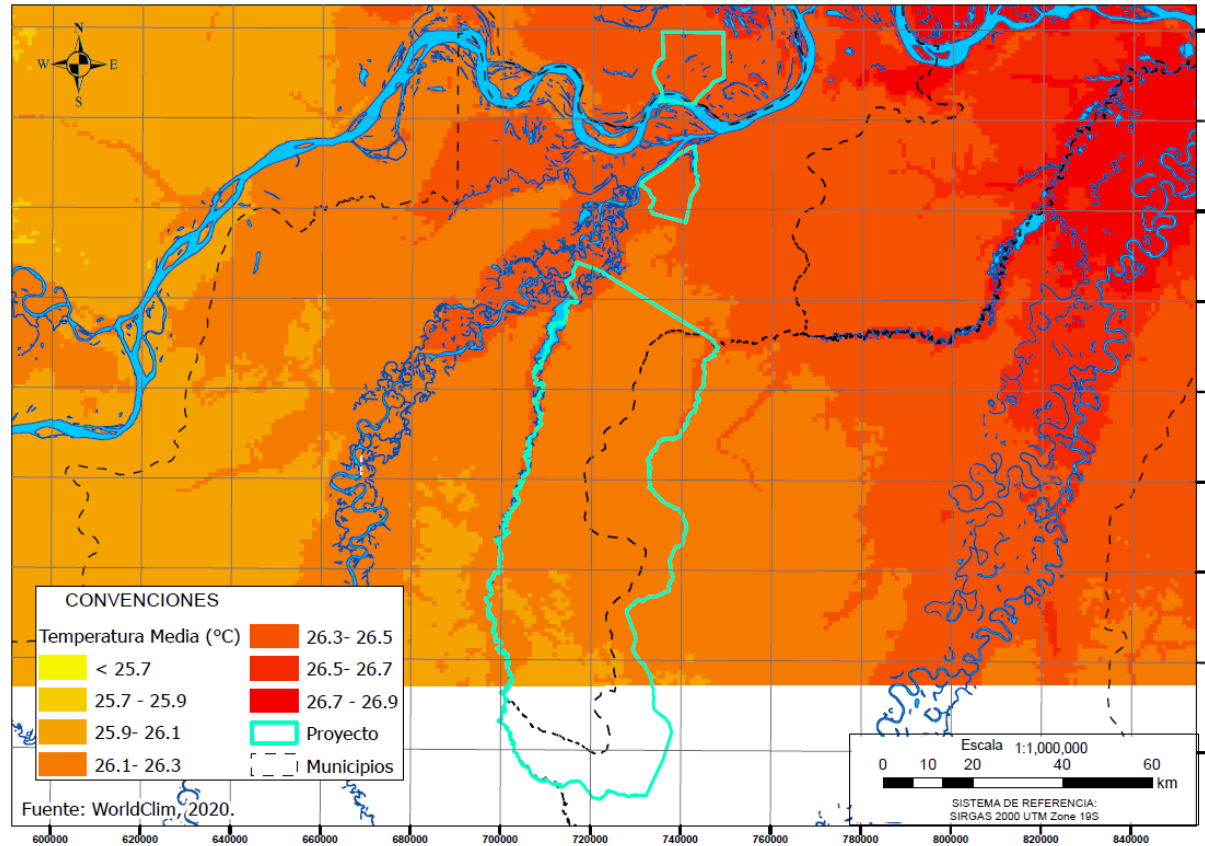


Map 4. Location of the Indigenous Territory (Terra Indígena) of Riozinho

Climate

Due to its location the temperature varies between 22°C and 32°C, being the average temperature 26°, and highest temperatures registered of 38°C.

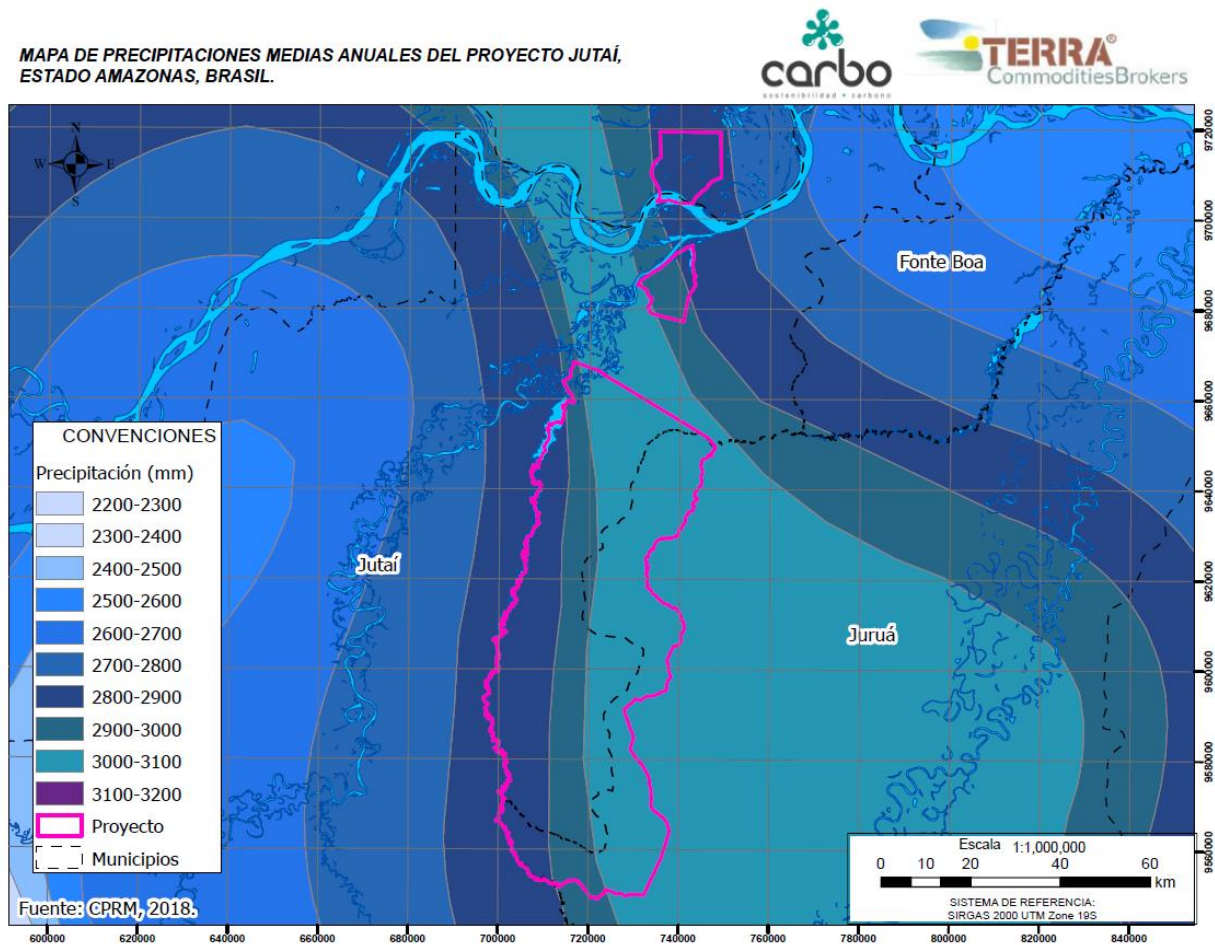
MAPA TEMPERATURA MEDIA DEL PROYECTO JUTAI,
ESTADO AMAZONAS, BRASIL.



Map 5. Temperature in project area.

Average annual rainfall corresponds to 2,509 mm and average annual relative humidity 89%. The differences between more and less rainy seasons are smaller and is represented mainly in the annual change of the water level of the rivers, which are influenced by the rainfall regime and also by the annual flood pulse of the Solimões and Jutai rivers.

Considering these conditions, the climate of the project area corresponds to tropical rain forest.



Map 6. Precipitation in project area.

Hydrology

Within the municipalities of Jutai and Fonte Boa there are important water sources, including the Amazon River (*Solimões river*) and the Jutai River.

The Amazon River is known for being the largest river in the world by discharge volume of water, flowing from Perú, passing by Colombia and with its largest drainage basin in Brazil.

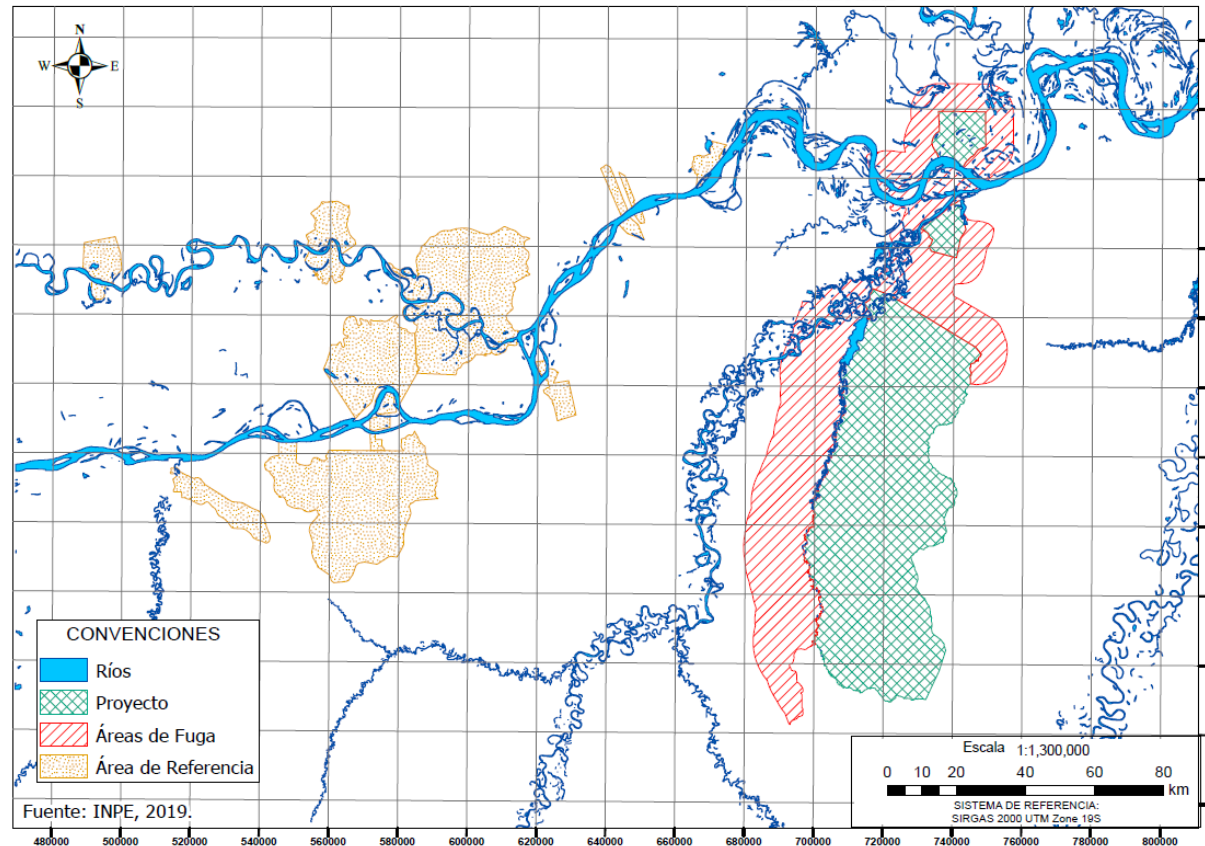
The Jutai river flows through the Juruá-Purus moist forests ecoregion. It runs northeast before reaching the southern bank of the Amazon River (Solimões section). It is west of the Juruá River and is roughly parallel to the lower Juruá. The indigenous territory of Acapuri de Cima is very rich in natural resources, because it is located at the mouth of the Jutai River.

Other minor water sources include tributaries of the Jutai River including: the Copatiana River that flows into this river about 20 km before the Jutai merges into the Amazon River; the Riozinho River that forms the eastern boundary of the Rio Jutai Extractive Reserve, the Biá River is a river that crosses the Cujubim Sustainable Development Reserve in the municipality of Jutai, and the Cuthuaia River.

The Riozinho and the Mineruazinho river, have a south/north direction and have black waters and originate in the Amazon basin. The forest cladding takes these rivers even cleaner and poorer in suspended particles. Riozinho and Mineruazinho belong to two large watersheds of the Jutai and Juruá rivers, respectively.

Other water sources include lakes that spread over the lands of occupation of the indigenous communities (*Aldeias*), especially Lake Grande and other lakes like Aladino, Ama-go Grande, Amagozinho, Buzina, Chaga, Ferruginho, Comprido, Grande, Mafra, Matá-mata, Matá-matazinho, Pirizinho, Porção, Redondo and Tucunaré, Igarapé da Lagoa, Igarapé Jurema, Igarapé Preto,

MAPA DE HIDROLOGÍA DEL PROYECTO JUTAI, ESTADO AMAZONAS, BRASIL.



Map 7. Hydrology in Project boundaries.

Types of vegetation and Life zones

The project area is located in the Amazon biome, which is formed mainly by forest formations, with the occurrence of small enclaves of savanna and grassland formations. Considered the largest tropical forest in the world, the phytophysionomies of the Amazon store a large amount of carbon. This biome represents 49.5% of the area of the country (NREF Brazil , 2022).

Table 1. Forest types considered in the Amazonia biome

Description (IBGE Vegetation Typologies)	
Aa	Alluvial Open Humid Forest
Ab	Lowland Open Humid Forest
As	Sub-montane Open Humid Forest
Cb	Lowland Deciduous Seasonal Forest
Cs	Sub-montane Deciduous Seasonal Forest
Da	Alluvial Dense Humid Forest
Db	Lowland Dense Humid Forest
Dm	Montane Dense Humid Forest
Ds	Sub-montane Dense Humid Forest
Fa	Alluvial Semi-deciduous Seasonal Forest
Fb	Lowland Semi-deciduous Seasonal Forest
Fm	Montane Semi-deciduous Seasonal Forest
Fs	Sub-montane Semi-deciduous Seasonal Forest
La	Forested Campinarana
Ld	Wooded Campinarana
Pa	Vegetation with fluvial influence and/or lake
Pf	Forest Vegetation Fluviomarine influenced
Pm	Pioneer influenced Marine influenced
Sa	Wooded Savannah
Sd	Forested Savannah
Ta	Wooded Steppe Savannah
Td	Forested Steppe Savannah

Source (Ministry of the Environment, 2014)

The Amazonas state is almost entirely covered by the Amazon Rainforest, and it is divided into three types of habitats:

- Igapos: permanently flooded land, roots of vegetation always submerged.
- Varzeas: Higher than igapos, land is only submerged when rivers are at their highest during the wet season.
- low plateau: higher still, never submerged.

The Amazon represents over half of the planet's remaining rainforests and comprises the largest and most species-rich tract of tropical rainforest in the world. Most of the vegetation corresponds to natural vegetation, followed by secondary vegetation, that consists of those forest areas that have been deforested and later abandoned for natural regeneration. Based on the relief forms, belonging to the Amazonian Planície, an area of accumulation of quaternary river sediments that occur in the form of bands along the rivers.

Within the municipality of Jutai important reserve areas are found:

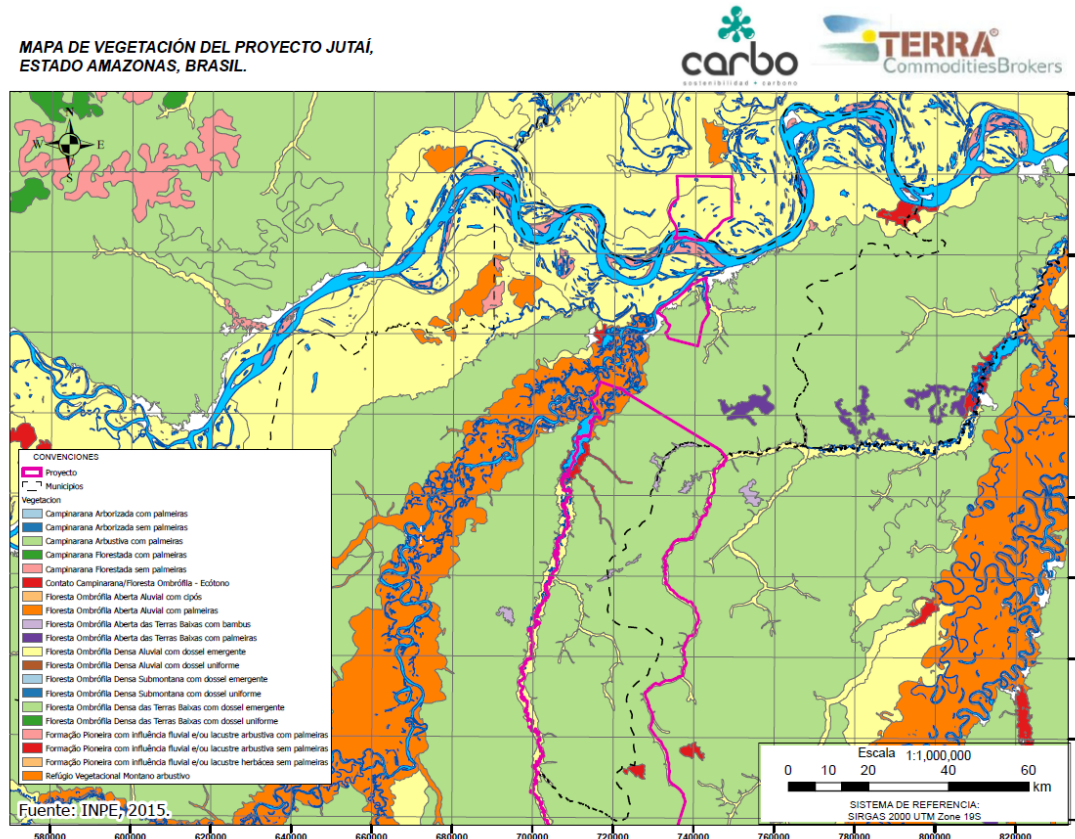
- The Cujubim Sustainable Development Reserve: one of the largest protected areas for sustainable use in Brazil, located within the Central Amazon Corridor in the watershed of the Jutai River. It is bordered on the Southwest by the Vale do Javari Indigenous Reserve and on the North by the Rio Biá Indigenous Reserve, it is part of one of the largest contiguous blocks of protected areas in tropical forest. It extends over 2,450,380 hectares and was established in 2003 (Conservation International; Government of Amazonas; British Embassy, s.f.)
- Rio Jutai Extractive Reserve: formed by the river downstream boundary with 275,533 hectares was created in 2002, to the southeast and the Jutai-Solimões Ecological Station to the northwest

On the other hand, in the municipality of Fonte Boa, reserve areas include:

- Mamirauá Sustainable Development Reserve: with an area of 1,124,000 ha, considered to be the first Brazilian conservation unit to try to reconcile the conservation of biodiversity with the sustainable development of resident human populations.

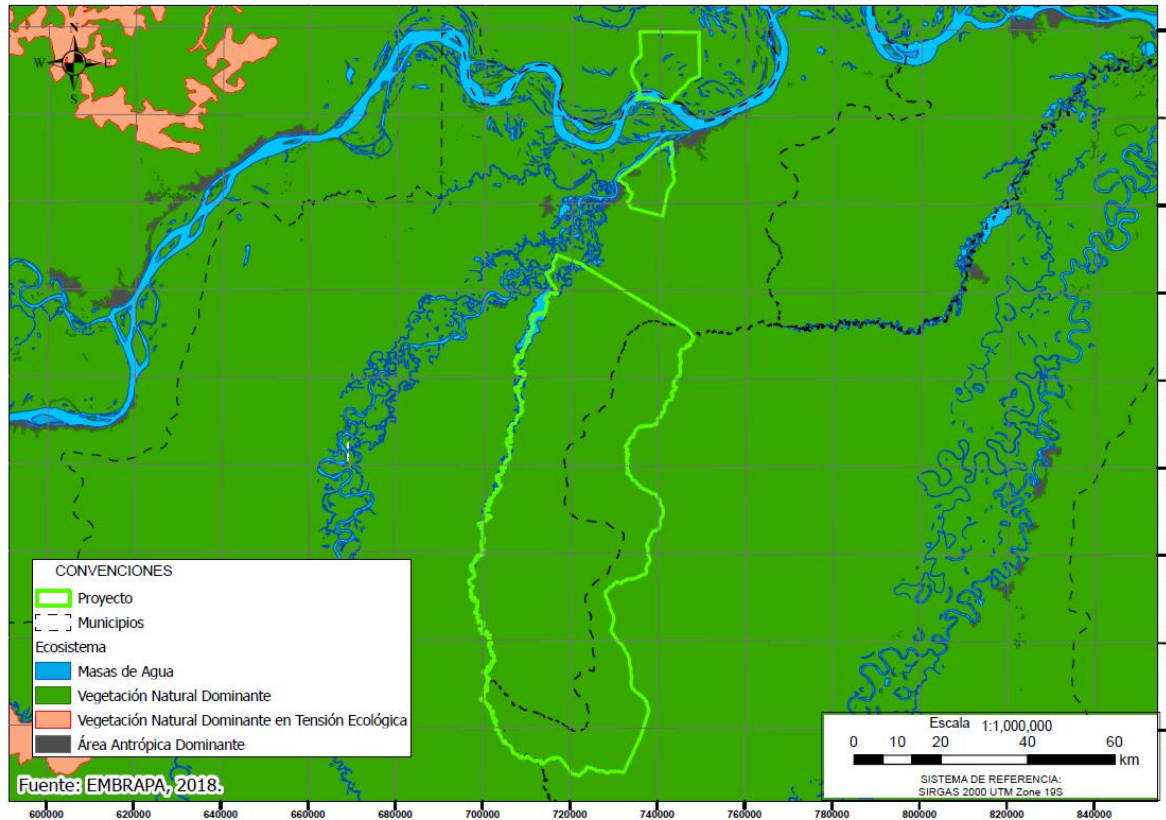
The vegetation can be classified as follows: i) chavascais, consisting of extensive areas, of low vegetation, shrubby, common between lakes, rivers and canals, subject to annual flooding of six to eight months: the characteristic plants of the chavascais are the bushes, vines, and species such as embaúba, munguba, carauçuzeiro and apuí; ii) low restinga, which is characterized as a transition between the chavascal and the forest areas of várzea, subject to annual flooding of four to six months: among the most abundant species are the abiorana and the matamatá; ii) restinga alta,

subject to annual flooding of two to four months: the main types of trees found there are the itaúba, the jacareúba (used for the construction of canoes), mamuí laurel, cedar, mulateiro (used as fire).



Map 8. Vegetation in Project boundaries.

MAPA DE UNIDADES ECOSISTÉMICAS DEL PROYECTO JUTAI,
ESTADO AMAZONAS, BRASIL.



Map 9. Ecosystems in project zone.

Land Use:

The municipality of Jutai has a primary forest that represents 98% of the total tree cover in the municipality. The total tree cover in each land cover corresponds to:

Land cover	% of tree cover	Area
Forest	98%	6.78 Mha
Grassland	<0.1%	1.22 kha
Settlement	<0.1%	517 ha
Wetland	0.28%	19.3 kha
Other	0.57%	39.2 kha

Source: (Global Forest Watch , 2023)

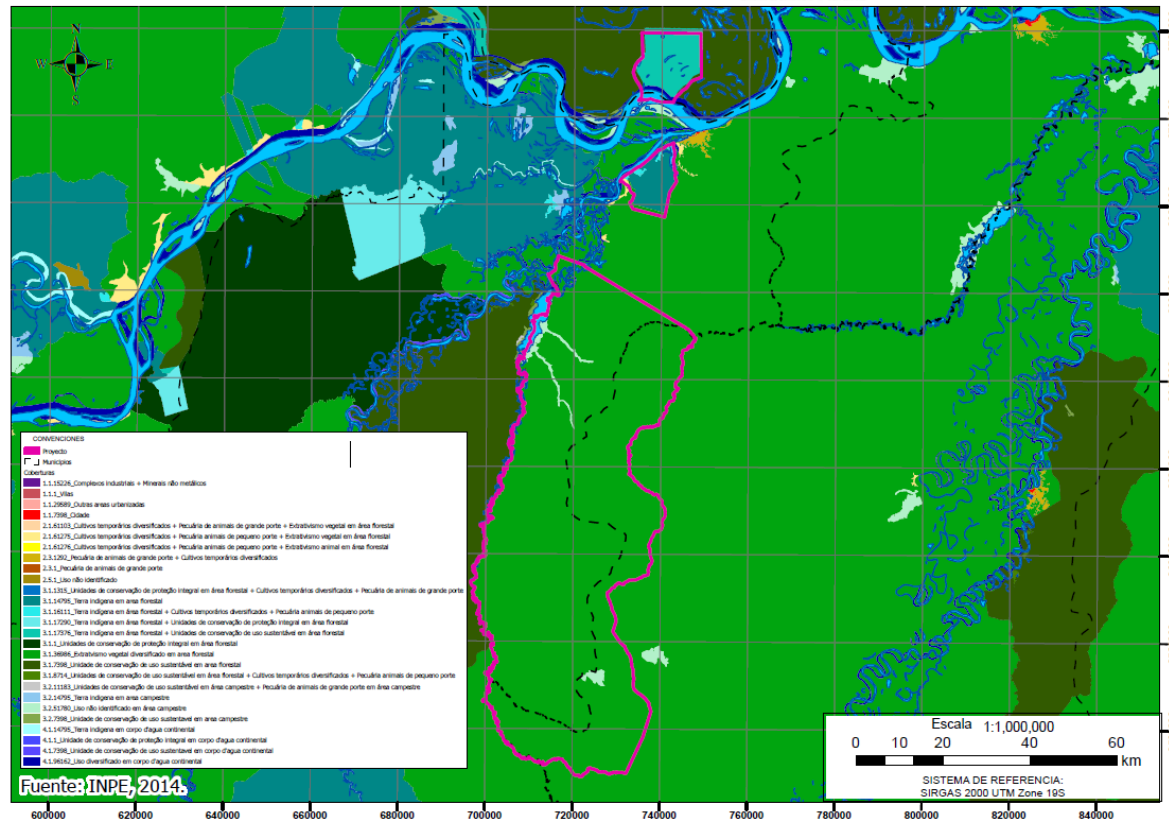
Regarding the municipality of Fonte Boa, the tree cover in each land cover corresponds to:

Land cover	% of tree cover	Area
Forest	98%	1.10 Mha
Grassland	<0.1%	642 ha
Settlement	<0.1%	381 ha
Wetland	1.2%	13.7 kha
Other	1.2%	13.2 kha

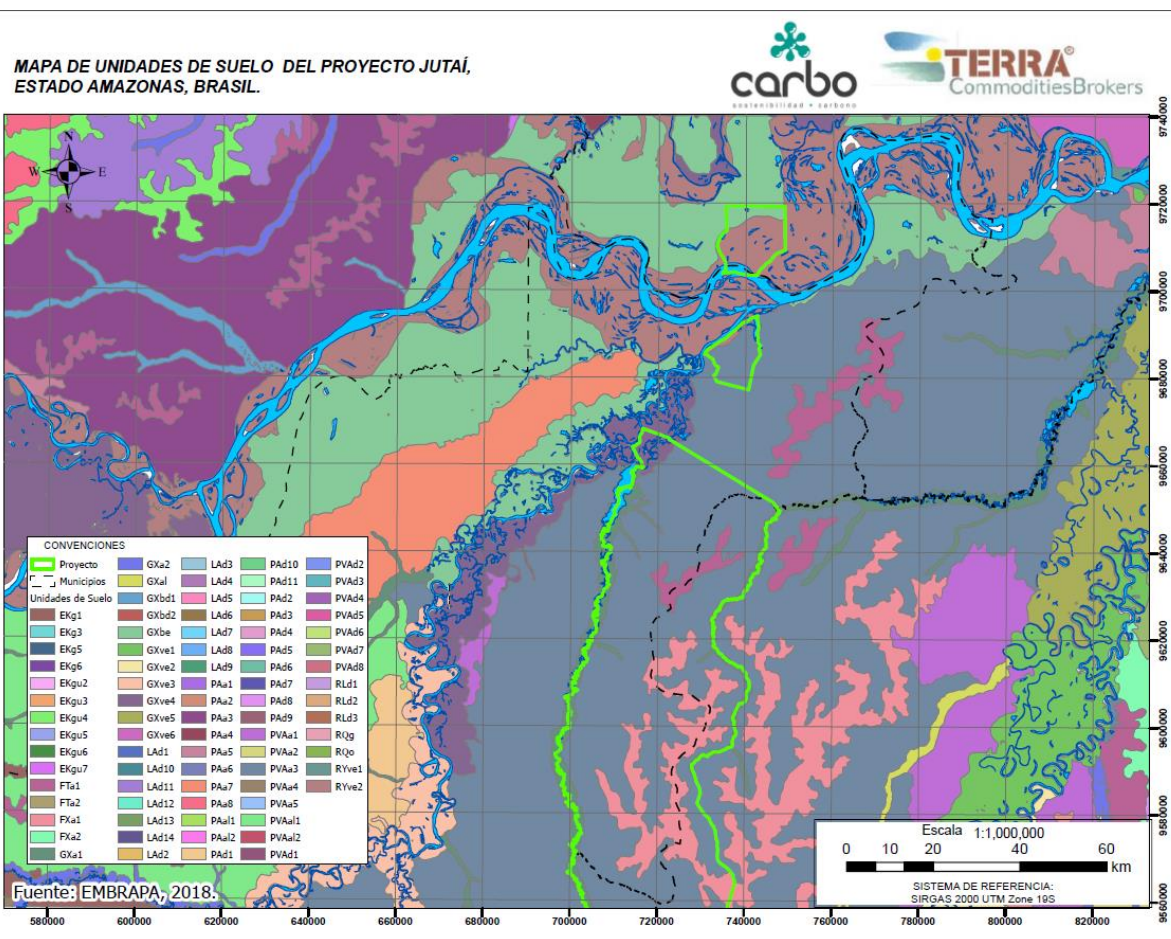
Source: (Global Forest Watch, 2023)

Fonte Boa had 1.15 Mha of tree cover, extending over 95% of its land area.

**MAPA DE COBERTURAS DE LA TIERRA DEL PROYECTO JUTAI,
ESTADO AMAZONAS, BRASIL.**



Map 10. Land use in project boundaries during 2018.



Map 11. Types of soils in the project boundaries.

2.1.15 Social Parameters (VCS, 3.18; CCB, G1.3)

Main Settlements

The municipalities of Fonte Boa, Jurua and Jutai are divided in urban and rural areas. The urban area is important in the strengthening and construction of associative dynamics with other municipalities. The municipality of Jutai counts with a population of 25.172, Fonte Boa of 25.871 and Jurua of 15.106. Moreover, political, cultural and economic relations between different communities are established within the urban area. The communities included in the project area are related in Table 2

Table 2. Indigenous communities in the project area

Indigenous Territory (<i>Terra Indígena</i>)	Indigenous communities (<i>Aldeias</i>)	# of people	Indigenous peoples (<i>Povos</i>)
Estrela da Paz	Bugaio	620	Kokama / Tikuna/ Kambeba
	Boa Vista	210	
	Inglaterra	56	
	Cajual	50	
	Curupiru	21	
	Escondido	30	
	Nova Esperança	25	
Acupuri de Cima	Acupuri de Cima	208	Kokama
Riozinho	Sao Luis	100	Kokama
	Novo Paraíso	80	
	Sampaio	150	
	Ariramba (de Baixo e da Cima)	28	
	Juruema	22	

In Acupuri de Cima families are settled in predominance of the floodplain ecosystem, which in turn unfolds into high floodplain and low floodplain. Houses built follow a linear planning along the left bank of the Solimões river, facilitating the access of each family unit to its own residence. Dwellings are built with paxiúba trunk and carana straw in the style of the riverside population, made of wood and zinc cover.

Due to their settlement in the high floodplain, the houses are built as stilts based on stakes (bar-rotas) fixed on the ground at a height of 1.50 m, making it possible to stay on site during the period of flood of the waters of the Solimões River.

In the case of Riozinho, families are located on the banks of the Solimões River and are usually bordered as well by a stream that provides greater access into the land with the houses arranged in lines, arranged close to each other. There is not a constant number of residences in each community. The communities are essentially made up of groups of relatives: each nuclear family occupies a residence and the vicinal groups usually

coincide with the establishment of kinship relationships: the new houses are being built next to the parents' house or the oldest member of the community.

Land Use and economic activities

Considered a local center of low influence in neighboring municipalities, the municipality of Jutai is close to the city of Tefé, Amazonas. Jutai is the eighth most populous municipality in the small region of Tefé. Its GDP is around R\$ 252.8 million reais, with 42.7% of added value coming from public administration, followed by the participation of agriculture (32.3%), services (42.7 %) and industry (4.8%). With this structure, Jutai's GDP per capita is R\$ 18.2 thousand, lower than the state average (R\$ 27.6 thousand), but higher than that of the Tefé region (R\$ 10.7 thousand). (Caravela Dados y Estadísticas, 2023). In 2021, the average monthly salary was 1.4 minimum wages. The proportion of employed people in relation to the total population was 6.0%. Compared to other municipalities in the state, it ranked 57th out of 62 and 28th out of 62, respectively. Considering households with a monthly income of up to half a minimum wage per capita, 50% of the population was in these conditions, which placed it in 36th position among 62 cities in the state and in position 1,404 among 5,570 among Brazilian cities (IBGE, 2023).

On the other hand, for Fonte Boa, the average monthly salary in 2021 was 1.5 minimum wages. The proportion of employed people in relation to the total population was 12.8%. In comparison with other municipalities in the state, it occupied positions 47 out of 62 and 4 out of 62, respectively. In comparison with cities across the country, it was in position 5029 out of 5570 and 2741 out of 5570, respectively. Considering households with monthly income of up to half a minimum wage per person, 53.3% of the population had these conditions, which placed it in position 15 out of 62 among cities in the state and in position 733 out of 5570 among cities in Brazil (IBGE, 2023).

The base of the economy of the project zone is supported in the use of renewable and non-renewable natural resources, such as forestry or selective logging, mining; agriculture for self-consumption mainly, with cassava products, banana, mandioca, acai, coco, cocoa, lemon tree and to a lesser extent livestock, especially fish farming and poultry. Agriculture schemes in this region are behind schedule and there are no incentives for strengthening and improving the related activities, and production is intended for the subsistence of the families of the Aldeias.

In the areas where floods occur, the communities plant perennial and semi-perennial crops such as cocoa, guava tree, coconut tree, açazeiro, lemon tree, banana. While in the low floodplain area, which is periodically submerged with the rise of the waters of the Solimões River, they plant short-cycle crops such as watermelon, corn, beans and pumpkin, whose production is commonly destined for domestic consumption and the surplus to commercialization in the regional market.

Tuberous crops (macaxeira, cassava, yam and sweet potato), which are planted in both types of floodplains. On average, the communities use less than ½ ha to open new gardens. The work system relating to the preparation of the soil and planting of crops is divided into two stages: in the first, the indigenous families agglutinate collectively in the "ajuri (joint work) for the cleaning of the land; in the next stage the agricultural work is carried out by the components of each family unit, then proceeding to the planting, cultural treatments, harvesting and "beneficiation" (production of flour).

Fishing is the fundamental economic activity for obtaining proteins. It is also a relevant source of monetary income through the sale of fish in the regional market. The instruments used for fishing are espinhel, caniço, harpoon, zagaia, malhadeira, flecha and curico, and açacú milk being used as timbó. This productive activity is exclusively carried out by men, but children participate when invited by their parents. The most appreciated and consumed species are the pirarucu, the tambaqui, the curimatã, the pacu, the matrinxã, the pirapitinga, the sardine, the piranha, the Surubim, the carauaçu and the peacock bass. Acupuri de Cima created the "Fisheries Committee" later transformed into the "Preservation Committee", whose objectives were no longer restricted only in the prohibition of predatory fishing, but also began to encompass the protection of flora and fauna in general as an attempt to conserve this primary resource of the indigenous habitat.

Hunting represents a supplementary source of protein, especially in the period of floods in the Solimões River, when fish becomes scarce. They hunt birds such as the mutum, nambu, maguari and mergulhão, and animals such as alligator and black, nail and smelling monkeys. Currently, the obtaining of these animals is carried out predominantly with firearms (guns).

In addition, they have developed the breeding of domestic animals such as ducks, chickens and pigs for consumption, and dogs that help in the search for hunting when necessary. The collection of fruits and roots, as well as woods, straws, vines used in the construction of houses and other artifacts.

Over the years, the gardens have been opened in places increasingly distant from the houses; the oldest (and close to the houses) are the current sites, managed areas where there are many fruit trees. Thus, close to the houses there is a mosaic consisting of sites, capoeiras and new gardens, which can configure a type of landscape unit recognized by the indigenous people. In addition to this space closer to the houses, there are two other sets of landscape units: the humid areas, where fish and birds are obtained, and the high areas, where the collection of fruits and resources are carried out for the construction and handicrafts and the hunting of larger animals.

Most of the commercialization of products by the Indigenous people takes place in the city of Jutai and with regattas that pass through the Solimões River.

The rubber economy has also taken place specially within the Indigenous Territory of Riozinho, developed under the aviamento system, had decisive effects on the current configuration of the population that traditionally occupies the Riozinho. Although the rubber economy has not been as intense in the upper and middle courses of Solimões as in other parts of the Amazon, it was decisive in the demographic and spatial reconfiguration of this entire region.

These Indigenous Territories (*Terra Indígena*) maintained the traditional form of occupation of space, based on the collective appropriation of natural resources, fallow (itinerant) agriculture and collective work, the "ajuri". Men and women participate in obtaining the basic resources of daily food, and the organization for work has a family basis (family unit of production). While men go further, entering the lakes that occur in the territory,

women fish on the edges of the streams closest to the houses. If, in the opening of new gardens, men participate with more intensity, in planting and collecting women have a fundamental role. It is also women who maintain the plantings of medicinal herbs, natural dyes and spices in beds (elevated ground canoes and jiraus are used for this purpose) in the yards of the communities. The landscape units recognized by the indigenous people preliminarily show their considerable territorial mobility: the availability of resources used in the food and construction of housing and canoes is subject to seasonality (winter/summer). the resources are not available throughout the year, as well as not all are concentrated in the same place. In winter (October to June). flood season, the predominant activities are hunting, fishing at less intensity, and care for the gardens, burned in the summertime from July to September). Flour production and collection, on the other hand, take place throughout the year.

Land Use	Occupation (%)
Dense high forest	72.56%
Fragmented forest with crops	5.45%
Mining zones	5.24%
Fragmented forest with secondary vegetation	3.15%
Rivers	2.39%
Dense high forest that can be flooded	2.18%
Grass and natural spaces mosaic	2.15%
Cropland, grass and natural spaces mosaic	1.96%
Low secondary vegetation	1.59%
High secondary vegetation	1.52%
Crops and grass mosaic	0.87%
Cropland and natural spaces mosaic	0.42%
Clean grasses	0.19%
Grass with undergrowth	0.13%
Crops mosaic	0.09%
Discontinuous urban areas	0.06%
Wetland	0.02%
Dense shrubs that can be flooded	0.01%
Grass with scarced trees	0.01%
Continental Acuaculture ponds	0.01%
Continuous urban areas	0.01%
Industrial and commercial areas	0.0049%

Socio – cultural information

This region is characterized for the lack of investment in social matters, poor coverage of education, health and sanitation systems. There is poor connection to communication services (telephone and internet access) and the community considers itself isolated as there are few transportation ways.

- Public services

For the municipality of Jutai 0.9% of households have adequate sanitary sewage, 1.8% of urban households on public roads with trees and 7.6% of urban households on public roads with adequate urbanization (presence of culvert, sidewalk, paving and curb). When compared to other municipalities in the state, it ranks 61st out of 62, 60th out of 62 and 25th out of 62, respectively. When compared to other cities in Brazil, its position is 5440 out of 5570, 5523 out of 5570 and 3069 out of 5570, respectively (IBGE, 2023).

The municipality of Fonte Boa presents 2% of households with adequate sanitary sewage, 48.5% of urban households on public roads with trees and 0.3% of urban households on public roads with adequate urbanization (presence of culvert, sidewalk, paving and curb). When compared to other municipalities in the state, it ranks 57 out of 62, 25 out of 62 and 52 out of 62, respectively. When compared to other cities in Brazil, its position is 5216 out of 5570, 4181 out of 5570 and 4686 out of 5570, respectively (IBGE, 2023)

- Health

The average infant mortality rate in Fonte Boa is 14.33 per 1,000 live births. Hospitalizations due to diarrhea are 0.3 per 1,000 inhabitants. Compared to all municipalities in the state, it ranks 29 out of 62 and 46 out of 62, respectively. The municipality counts with 5 health facilities (IBGE, 2023).

On the other hand, Jutai presents an average infant mortality rate of 18.35 per 1,000 live births. Hospitalizations due to diarrhea are 6.2 per 1,000 inhabitants. Compared to all municipalities in the state, it ranks 18 out of 62 and 3 out of 62, respectively. The municipality counts with 6 health facilities (IBGE, 2023)

- Education

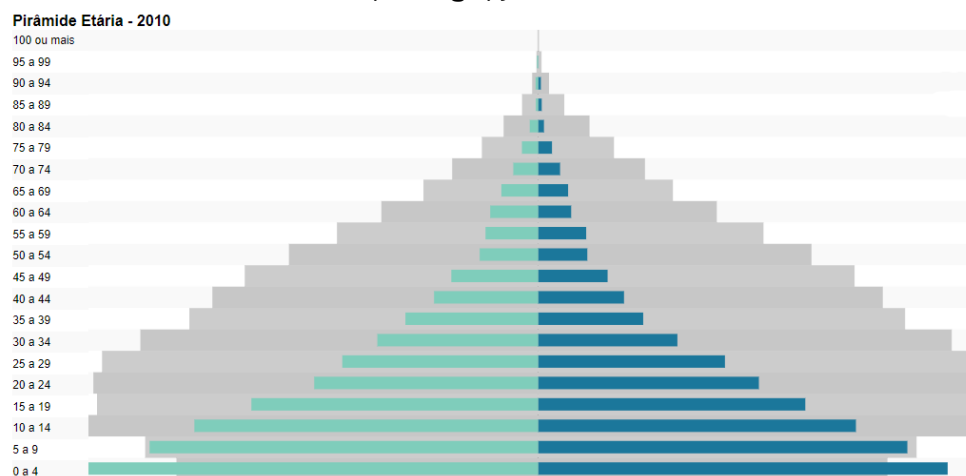
Education coverage is considered low regarding other municipalities and states of the country.

	Jutai	Fonte Boa
Schooling rate from 6 to 14 years old	88.4%	84.8%
IDEB - Early years of elementary school (Public network)	3.9	3.7

IDEA - Final years of elementary school (public school)	3.4	3.5
Number of elementary education establishments	88	54
Number of high school establishments	4	3
Schooling rate compared to other municipalities	5489 out of 5570	5527 out of 5570

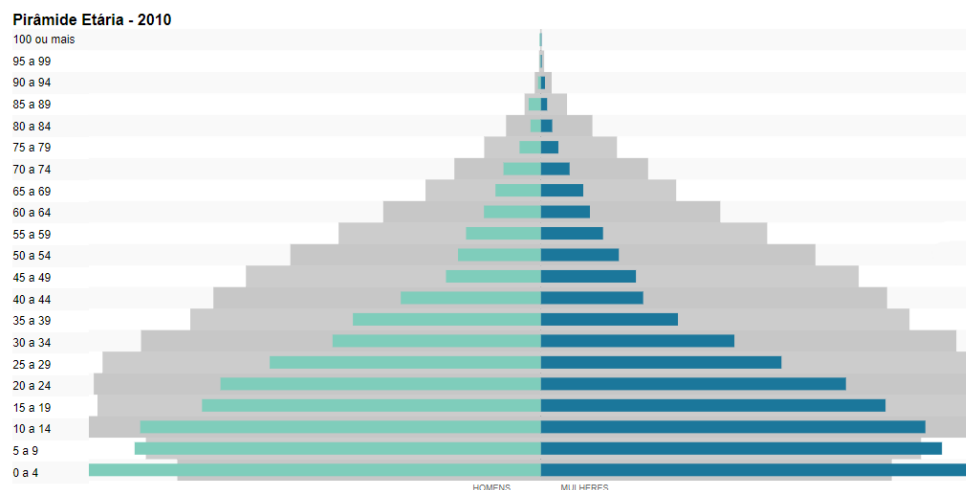
- Gender

Map 12 Age pyramid Jutai



Source: (IBGE, 2023)

Map 13 Age pyramid Fonte Boa



Source: (IBGE, 2023)

Relevant Historic Conditions

The primitive inhabitants of the region, which today includes the municipality of Jutai, were the Catuquinas, Marauás, Ariaceus and others. The origins of the municipality go back to the 17th century when the Jesuit Samuel Fritz founded the town of Tefé. The region was later object of dispute between the Spanish and the Portuguese, winning the latter at the end of the 18th century. Elevated to a municipality, Tefé in the mid-19th century reached an area of 500,000 km². Subsequently, several divisions took place, giving rise to several municipalities, among them, in 1891, Fonte Boa. In 1928, this municipality became the municipal seat and elevated to the category of municipality and district with the name of Jutai, by State Law no. 96, of 12-19-1955, dismembered from the municipality of Fonte Boa, and having its name in honor of the Jutai river. Headquarters in the current district of Jutai (former town of Boa Vista) (IBGE, 2023).

Likewise, the municipality of Fonte Boa, has its origins on the Taracotéua or Taracuariba, a village of the Omáguas Indigenous, who were the first nucleus of settlement of the current city of Fonte Boa. In that town, at the end of the 17th century, the Jesuit Samuel Fritz founded the religious mission called Nossa Senhora de Guadalupe. A few years after the foundation of the mission, the villa was destroyed by the Spanish and later restored by the Portuguese.

The European conquest of the Upper Amazon, which in Brazil is called Solimões, was made not only upstream, by the Portuguese enslavers and missionaries, but also in the opposite direction, by Spanish missionaries coming from the rivers that descended from the Ecuadorian Andes, and also by the Huallaga and Ucayali (COPIJU, 2011).

The religious missions stimulated, in the Amazon region, the extraction of products of animal and vegetable origin, which for two and a half centuries was done routinely using the forced labor of the indigenous people, albeit with gratification established by law. From the middle of the 19th century, the life of Solimões underwent important changes: steam navigation released the indigenous rowers, the river was opened to international navigation and the demand for rubber began to intensify, increasingly requiring more manpower and stimulating northeastern immigration. The Indigenous began to be summoned to produce rubber and, in the Putumayo basin, forced labor, violence, transfers of populations reached their peak. Perhaps it was where the indigenous population decreased the fastest and at high rates.

Although, after 1912, the area entered a new period of stagnation because of the fall in rubber prices, the system of shed or avia mento - from advance of goods to high prices in exchange for forest products at very low prices, generating an unsalable debt - continued to be in force, markedly marking interethnic relations. This region, at the beginning of the 20th century, represented a challenge both in terms of scientific knowledge and from the point of view of territorial occupation by the border national states and the new economic order of market integration. The history of some societies in the area in the present century has developed in order to overcome the heritage of the period of the rubber race, which includes many interethnic marriages and Lands inhabited by several Peoples (COPIJU, 2011).

The contact of the Kokama with non-indigenous society dates back to the first decades of colonization. The forced villages and displacements, imposed first by the missions and then by the extractive fronts, ended up creating such an adverse context of physical and cultural reproduction of these groups, that it aroused the denial of indigenous identity for many decades. Since the 1980s, however, the Kokama identity has been increasingly valued in the context of its political struggles - which include other indigenous peoples of Solimões - for land and access to differentiated programs of health, education and economic alternatives. Later studies indicated that its origin would be linked to the various migrations of Tupi groups from Brazil to Peruvian regions in pre-contact times. It is assumed that it is a language product of the interaction of some indigenous groups in the region of the upper Marañón, near the Huallaga, Napo and Ucayali rivers, and the Tupinambá language would have been the main linguistic source of the Kokama language, with approximately 60% of the vocabulary. In Brazil, there are few who have fluency and there are no narrated cases of communities that use Kokama, even if as its second language (COPIJU, 2011).

The Tikuna constitute the most numerous indigenous people in the Brazilian Amazon. With a history marked by the violent entry of rubber tappers, fishermen and loggers into the Solimões River region, it was only in the years 1990 that the Tikuna achieved the official recognition of most of their lands. Today they face the challenge of ensuring their economic and environmental sustainability, as well as qualifying the relations with the surrounding society keeping alive their very rich culture. These people lived at the top of the tributaries of the left bank of the Solimões River, in the stretch in which it enters Brazilian lands to the Içá/Putumayo River. It was with the growth of the indigenous movement from the 1980s, particularly with the recognition of indigenous rights by the 1988 Constitution and the multiplication of indigenous organizations, that the Kambeba began to

assert themselves as indigenous again and fighting for indigenous causes. Since then, they have taken a prominent position in the region for their great capacity for negotiation and political articulation with other indigenous groups and with governmental and non-governmental agencies, religious and secular, of the surrounding society. The Kambeba of the Upper Solimões were incorporated into the Tikuna on the occasion of the demarcation of the Tikuna indigenous lands, but today they are starting a process of ethnic affirmation (COPIJU, 2011).

The Katukina do Biá live in the tributary of the Jutai River where the massive invasion by rubber was not so violent due to the low quality of the latex. Throughout the 20th century, the Katukina have always been more numerous, but they lived with Brazilians and Peruvians in Biá, whether living or traveling in search of labor.

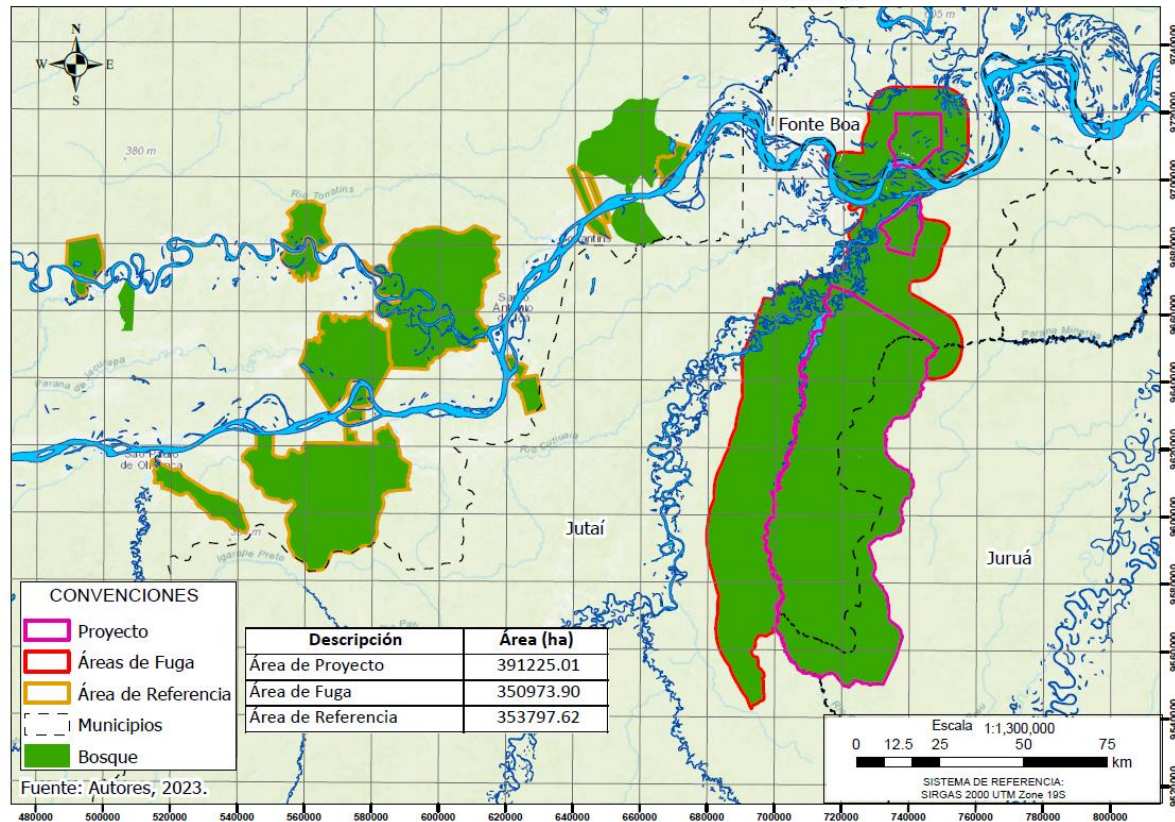
2.1.16 Project Zone Map and Project Location (VCS, 3.11, 3.18; CCB, G1.4-7, G1.13, CM1.2, B1.2)

The Estrela Acapuzinho REDD+ Project seeks the conservation of forests located in the indigenous territories of Estrela da Paz, Riozinho and Acapuri de Cima, located within the municipalities of Juruá, Jutai and Fonte Boa, in the Amazon State of Brazil, which has a total area of 391,225 ha defined as the project area and correspond to tropical rain forest that has been present for more than 10 years before the project start date.

The geographic coordinates of the indigenous territories are:

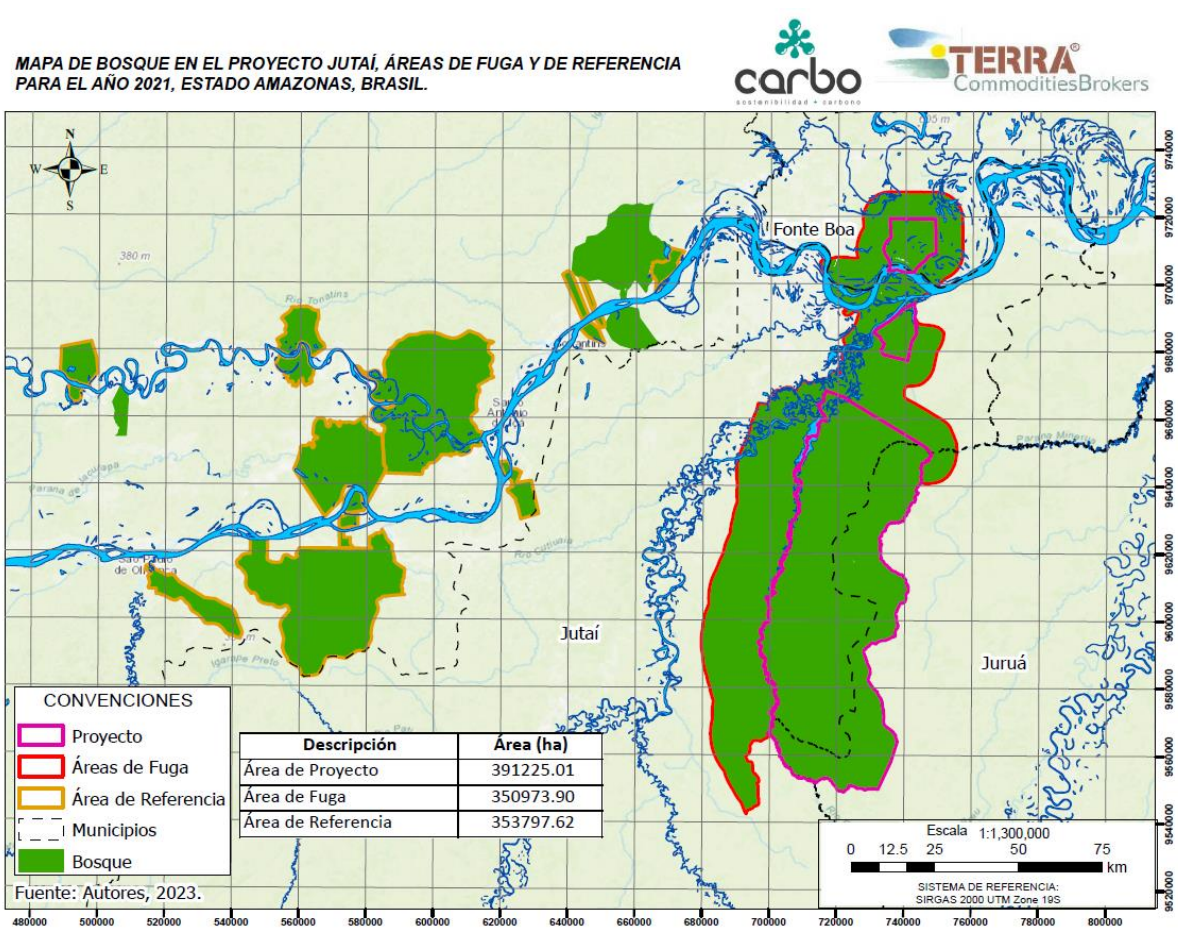
Indigenous territory	North	South	West	East
Acapuri da Cima	02°32'15.075984"S 66°53'0.928651"WGr	02°37'42.076374"S 66°45'38.018211"WGr	02°32'16.684942"S 66°48'9.636686"WGr	02°39'12.644736"S 66°53'10.924560"WGr
Estrela da Paz	02°50'40.744"S 66°55'37.946"WGr	02°54'51.970"S 66°50'58.972"WGr	02°52'17.859"S 66°53'47.912"WGr	02°49'28.311"S 66°48'32.132"WGr
Riozinho	03°01'05,9"S 66°59'30,0"WGr	03°59'31,3"S 66°53'07,9"WGr	03°04'45,0"S 67°03'18,7"WGr	03°10'10,3"S 66°46'28,9"WGr

MAPA DE BOSQUE EN EL PROYECTO JUTAI, ÁREAS DE FUGA Y DE REFERENCIA PARA EL AÑO 2021, ESTADO AMAZONAS, BRASIL.



Map 14. Project area, leakage belt and reference region in 2021

This area has experienced pressure due to the drivers of deforestation action in the last years. No forest within the community territory was converted to generate any carbon credits.



Map 14. Project area, leakage belt and reference region in 2021

The understanding of indigenous communities about the project consists of that through forest conservation there is a possibility to receive payments to benefit the community. This understanding, instead of representing a threat or risk that could result in increased deforestation to generate carbon credits, offer a social basis to promote forest conservation as a mean to access REDD+ opportunities. Their understanding reflects that the more forest they have, more benefits will be delivered. In this sense, no deforestation has taken place to take advantage of the mechanism. In the last 20 years, deforestation in these areas has been a national concern due to increasing rates and associated social and biodiversity impacts.

Local communities are located near rivers and scattered along the project zone within the three indigenous territories. High conservation values as the jaguar (*Panthera Onca*), tapir, anteater, and anaconda are found in dense forest landscape.

One of the REDD+ components involve sustainable productive systems. According to workshop #2 (see Taller 2 files), most of the community wants to participate and benefit from agricultural interventions. All the increased productive areas are going to be developed in already intervened areas, as well as improvement in existent productive areas. The extent of the activities depends on the available budget. Nevertheless, it is expected to benefit most of the families of the indigenous communities. The intervention scheme that offers the biggest benefit, i.e., new hectares under production, improve production practices, intensify production systems, is going to be define and implemented.

2.1.17 Project Activities and Theory of Change (VCS, 3.6; CCB, G1.8)

The theory of change aims to properly canalize investment resources to obtain the expected long-, medium- and short-term results, specifically in terms of controlling deforestation, preservation of biodiversity and sustainable community development (as presented in Appendix 2). Project costs are mainly associated with the activities described below and corresponds to those that would allow the VCU's to be generated.

Project strategy aims to address drivers of deforestation and to implement initiatives and activities within the framework of governance, productive activities, social investment and monitoring. Activities are described below:

1. REDD + strategies and activities

The causes or drivers of deforestation of the project are essentially of economic origin and illegal activities carried out by local and external agents in the indigenous territories. Local communities are motivated by the need to preserve their culture and ecosystems and to improve somehow their living conditions and economic status. The lack of technical capacity and the absence of the State which generally leads to the expansion of illegal activities (mining and logging), determine the poor sustainability of the productive activities in the area.

The REDD+ project aims to reduce existing pressures on the forest through the implementation of forest governance activities, profitable, sustainable and efficient production systems, local administrative strengthening and the monitoring of biodiversity. These activities, promote the protection and restoration of forests and biodiversity conservation and the improvement of social and economic development of the region. They address the main causes related with forest loss and represent an opportunity to consolidate social bonds and cultural identity.

The project activities are developed in four main components:

Productive activities:

Develop economic alternatives compatible with community well-being and nature conservation. This will be achieved through the implementation of the following activities:

- i) Design and prioritize productive activities aiming to promote the well-being of the community and nature conservation (i.e., cassava, banana, pineapple, corn, watermelon, pig farming, poultry farming (eggs and meat), aquaculture, others).
- ii) Strengthen technical y operational capacities of the communities to manage prioritized production systems.
- iii) Implement the prioritized productive systems.
- iv) Maintain, improve and monitor the productive systems implemented.

These activities are aimed to secure community food needs and promote opportunities and develop productive systems that generate incomes. The project will encourage and improve women enrollment in productive and managing activities. By generating economic income, the need for the people to participate of illegal activities will be reduced, as well as their related pressures that impact forest ecosystems, so this will contribute to biodiversity protection and community welfare.

According to the community needs manifested during workshop 2 (see folders Annex 7. Workshops, subfolder Workshop 2), and assuming a good carbon credit sales and revenues, at least 1 hectare per family of production systems may be implemented or improved in order to respond to the overall objective of the project that consists in improving community livelihoods and local biodiversity. Nevertheless, an to maintain a conservative approach, it is estimated that each family is going to improve the management of their traditional crops at least once during the project (1 ha/family). Assuming that population is near 1,600 people, near 390 hectares with improved management have been defined as a goal that could be achieved during the lifetime of the project.

This component includes training and new job positions. Considering the available resources, at least 20% of the people (320 persons) is expected to participate in trainings and 10% to improve incomes (that may benefit their families) during the project lifetime (160 persons). Men and women are going to be encouraged to participate on an equal basis.

Social Investment

This activity aims to improve the living conditions of the communities that live in indigenous territories and include but not limited to:

- i) Identify and prioritize the needs of the communities in matters of social investment.
- ii) Improve facilities and scholarships for community members to access formal education. This might include adaptation of classrooms, libraries, access to higher education, training, communication means. Also considers to strength their own indigenous differentiated educational initiative that is under development.

- iii) Improve access and health services for the inhabitants of the indigenous territories (i.e., improve or build a health centre, have medical supplies and health professionals in the territory, boat ambulance, among others).
- iv) Improve basic sanitation conditions and public services in the indigenous territories (i.e., drinking water, waste management, electricity, communication).
- v) Community infrastructure (sports, community house, transportation)

As stated during project designing and possibilities (see folder Annex 7. Workshops), it is expected to reach most of the community through this component and offer direct benefits to most of them, either by improvement in education (480 persons), health services (320 persons), transportation (320) or housing and basic services (320). Considering the available resources, at least 80% of the population is expected to benefit from at least one of these activities (1280 persons), keeping gender frequency as equal as possible. In regard of educational opportunities, through project activities is expected to reach young people during the lifetime of the project, offering equal opportunities to girls and boys and enhance their professional or technical skills that may result in better local development opportunities.

Governance and cultural protection

Strengthen community prospective, land use planning, own government and cultural identity are fundamental to guarantee their permanence in their territory and affirm social bonds. Protection of indigenous identity, traditional language and medicine, dances and rituals and integrity of the community will deliver biodiversity protection and cultural consolidation. To develop this project component five main strategies must be carried out:

- i) Strengthen the community's life plans.
- ii) Improve land use planning and implement related activities.
- iii) Promote and protect indigenous identity and traditional knowledge.
- iv) Strengthen capacities to manage and control their territory, business, legal and project matters.
- v) Implement activities defined in the life plans.

Local governance committee (REDD+ Committee) are to be defined by the community to develop and evaluate project activities. This group of people will be defined for each indigenous territory and will be in charge of the project implementation and monitoring specific activities. They will also administer the Requests, Complaints and Claims system.

Monitoring and biodiversity protection

This is a key activity that aims to contribute to biodiversity monitoring and conservation within the area of the indigenous territories and includes:

- i) Strengthening and building capacities to fulfil the functions of forest rangers and patrols and surveillance over the territory including control the entry of heavy machinery, control of mining permits, control of forest use, management of natural resources, management of equipment and techniques for monitoring biodiversity and conflict resolution.
- ii) Provide tools that contribute to the improvement of management capacities by the group of forest rangers. This might include acquisition of a boat for controls and supervision tours, equipment, location equipment, computer equipment.
- iii) Biodiversity follow-up and monitoring within the indigenous territories.

Through the implementation of activities related with these project components, it is expected to fulfill the climate, social and biodiversity objectives, because they address strategic aspects that surround the observed direct and indirect causes of land use changes. In this regard, the following is highlighted:

- a. Improve land management and maintain food security helps to protect forest integrity. Sustainable productive activities, planned and executed with appropriated and technical knowledge are an alternative to improve household income allowing the community to generate profits and reduce the high dependence on activities that require deforestation or forest degradation such as logging and mining. This will reduce deforestation and forest conservation is increased.
- b. Social investment is a key REDD+ strategy to be implemented. Since community needs are not properly addressed, this strategy allows the life quality and well-being to be improved and facilitate community participation in the project. In order to maintain the project benefits overtime and guarantee the access to health, education, among others, overtime the community control may be reinforce to avoid any activity that threatens the reception of the observed benefits derived from forest preservation, thus, contributing to maintain climate benefits.
- c. Improve governance will allow the reduction of deforestation and forest degradation, since life plans, land use plans, management capacity and other structural elements of the community are oriented towards territorial and cultural protection and permanence. Activities that relate with this component are fundamental to community wellbeing and identity, and they contribute directly to REDD+ objectives.
- d. Monitoring land cover will allow greater control over the forest resource, acknowledgment of the territory and identification of possible threats. Biodiversity studying, protection and management are basic elements to promote its conservation. These also enables impact assessment of the project activities over the land and evaluate the effectiveness of implemented measures, which is project's primary climate objective.

The REDD+ project is not located within a jurisdiction covered by a jurisdictional REDD+ program.

2.1.18 Sustainable Development Contributions (VCS, 3.17)

The project contributes to the fulfillment of the Sustainable Development Goals (SDGs) since it seeks to improve the income of the community and promote sustainable social and economic development alternatives, in addition to strengthening territorial environmental management and promote climate change adaptation and mitigation. The specific SDGs to which the project contributes are:

- SDG3 – Good health and well-being: part of the social investment is focused on improving the provision of health services for the community.
 - SDG4 – Quality education: with the project implementation, access to education is strengthened since it is planned to improve the available infrastructure, develop scholarship programs, training programs, among others.
 - SDG5 – Gender equality: the project promotes the participation of women in the activities to be implemented and their participation in decision-making instances.
 - SDG7 – Affordable and clean energy: the project promotes access to clean energy and electrical interconnection of homes.
 - SDG8 – Decent work and economic growth: the project promotes the development of sustainable economic activities that increase the income of families in the community.
 - SDG11 – Sustainable cities and communities: the project promotes sustainable communities since it seeks to conserve vegetation and biodiversity in the territory, the development of productive activities compatible with nature, improve economic income and strengthen community well-being.
 - SDG13 – Climate action: the project seeks to reduce GHG emissions from deforestation and forest degradation.
 - SDG15 – Life on land: the project seeks to reduce deforestation and degradation, thus contributing to the conservation of biodiversity.
 - SDG17 – Partnerships to achieve the goal: the project promotes alliances between community organizations, public sector and private sector.
- Similarly, project design is articulated with the action plans of public institutions at the local and regional level.

In addition, Brazilian Government has developed a series of strategies that aim to achieve an environmental management of climate change (involving both adaptation and mitigation) and to contribute to the international agenda of sustainable development 2030 (Sustainable Development Goals), and specially SDG 13 – Climate Action. Moreover, the country has developed a series of strategies that sought to identify and prioritize municipalities with high deforestation rates for the development of initiatives with the aim of preventing, controlling and combating illegal deforestation, to strengthen REDD+ activities, and conservation of forest areas specially in the Brazilian legal Amazonas.

The project to be developed in the Indigenous territories constitutes an important contribution for these National policies primarily in the Agriculture, Forestry and Other Land Use (AFOLU) sector, due to the deployment of mitigation measures such as deforestation reduction, development of economic alternatives compatible with community well-being and nature conservation, strengthening land use planning and mechanisms to guarantee food security for the communities and contributing to biodiversity monitoring and conservation. In addition, the project supports the National REDD+ Strategy as it seeks to protect existing natural resources while promoting the regeneration of degraded tropical forests and the reduction of global emissions in a high mega-diverse region of Brazil and the world.

According to the Paris Agreement and taking into account the principle of common but differentiated responsibilities, Brazil define it's Intended Nationally Determined Contributions (NDC) as a commitment to reduce its greenhouse gas emissions in 2025 by 37%, compared with 2005. Additionally, Brazil commits to reduce its emissions in 2030 by 50%, compared with 2005. Brazil´s commitments also include a long-term objective to achieve climate neutrality by 2050. Likewise, adaptation actions aim to reduce vulnerability in terms of water, energy, food, social and environmental security. Based on the second cycle of the National Adaptation Plan (NAP), adaptation measures in Brazil focus in strengthening the management of water resources, the diversification of energy sources, adaptation plans for the urban landscape, the development of adaptation strategies in the agricultural sector with a view to ensuring food security, to ensure the resilience of the population and infrastructure.

Therefore, the implementation of the Estrela Acapuzihno REDD+ Project represents a contribution for Brazil to achieve the proposed goals and a low carbon and resilient rural development.

2.1.19 Implementation Schedule (CCB, G1.9)

Date	Milestone(s) in the project's development and implementation
08/11/2021	Start date
2022 – 2024	Investment for the development of REDD+ activities. Activities monitoring and performance assessment
07/11/2024	Validation and verification of first project monitoring period
2025 – 2026	Investment for REDD+ activities development. Activities monitoring and performance assessment
06/2026 – 06/2027	Verification
2027 – 2029	Investment for REDD+ activities development. Activities monitoring and performance assessment

Date	Milestone(s) in the project's development and implementation
01/2029 – 06/2029	Baseline reassessment and REDD+ strategy updating
06/2029 – 12/2029	Validation and verification

2.1.20 Risks to the Project (CCB, G1.10)

Projects risks are presented in Appendix 3 Project risks table. Assessed risks are expected to be mitigated by either the project proponent's management activities or proper project activities.

Non-risk permanence appraisal was performed according to the guidelines subscribed in the VCS AFOLU Non-Permanence Risk Tool v4.0. For this, risks were classified into three categories: i) internal risks, ii) external risks, and iii) natural risks. An evaluation was carried out to determine the risk factors in the tool and estimate the risk rating per category and project.

Risk Category		Rating
a)	Internal risk	12,00
b)	External risk	0,00
c)	Natural Risk	7,25
Overall risk rating (a + b + c)		20

Note: Minimum risk rating equals 10%

Identified Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
Climate Risk	The community faces potential natural climate risks, such as an increase in the severity of fires and that they must deal	Project activities include developing of productive systems that contribute to food security, income generation and productive

Identified Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
	with natural cyclic climatic variations and derived weather conditions, especially droughts and floods that may alter the moisture exchange between forests and the atmosphere. Since communities tend to be located near water bodies, flooding has been historically managed through housing distribution out of the natural flooding areas. Average temperature presents low variations and annual precipitation usually exceed 2,500 mm. The average relative humidity is generally maintained above 85%, both in the period of higher rainfall and in the period of lower rainfall. Climate change is not expected to alter temperatures in the short-term (five years). However, in the long-term (more than 10 years), they may alter the presence of fires affecting forests and biodiversity.	diversification. The approach of diversification of activities, help to build resilience in the productive systems under conditions of weather variations and mitigate potential impacts regarding climate change. According to the appraisal significant natural risks were related to extreme weather. There is a community awareness that the risks like flooding, fires, disease outbreaks, water pollution, extreme weather and the increase of hot spots and landslides events are present. Through the protection of natural forests and reducing deforestation and ecosystem degradation, these risks are mitigated, and their impact is reduced as a result of natural water regulation, soil protection and tree integrity that the forest offers as part of the ecosystem services.
Community risks	Considering the context of the indigenous communities (Aldeias), the main community risks are related to human induced risks, weak governance capacity, economic matters, and the absence of community engagement. Other possible risks are related to net income from the project being below the	To mitigate these risks, a capacity building process has been integrated in a participatory manner as part of the project. Local workshops and social surveys were conducted to identify the characteristics and define the intervention model to address local and community context

Identified Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
	expected if proposed activities are not sufficient in comparison to former deforestation activities (agriculture, mining), leading to the development of alternative activities that may result in deforestation. Potential mid and long-term community-induced risks include change in the economic conditions in the municipalities (i.e.: economic activities which compete with large forest) or potential lack of forest carbon market to cover opportunity costs.	through the REDD+ strategy (see folder Annex 7. Workshops). Distribution of the project benefits and investment guidelines were discussed to ensure that all members of the community will benefit as a result of the project implementation. All the four components of the project strategy (sustainable productive systems, social investment, governance improvement, and biodiversity monitoring), include both social and training elements, as well as a general guideline to promote community enrollment, active participation, genre equality and equal individual opportunities integrated as part of the social safeguards. Also, a program for requests, complaints, and claims (grievances) management was established, where continuous community feedback will help adapt the project implementation. Governance weakness will be mitigated through the implementation of the components of project strategy that address governance and monitoring strengthen, including access to formal education and technical formation in administrative topics, business management and development, as well as defining and implementing the indigenous ethno-developing plan, land use plan and biodiversity monitoring plan are part of the elements that increase governance capacity of the indigenous communities (Aldeias) and that will be implemented through specific project activities.

Identified Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
		The REDD+ strategy includes governance improvement and sustainable production. The project is going to generate income opportunities, implement sustainable productive activities, create jobs, deliver social investment (education, infrastructure, sanitation, electricity, food security, health systems), as well as build capacities to improve territory governance and facilitate access to education programs. These activities address possible socioeconomic dynamics that result in forest loss, so it is expected to mitigate this risk through the explained integrated approach. In addition, community engaging, and development of participatory processes help to ensure that community members involve in an active way and the expected outcomes of the project in the short, middle and long-term are fulfilled. Finally, permanent socialization of the activities to keep people aware of the social and economic benefits related to the project implementation represent an action that contributes to mitigate risks linked to community commitment.
Biodiversity Risks	The theory of change implementation and the accurate development of project activities will deliver the biodiversity benefits.	The project activities are focused on reducing deforestation as well as avoiding any potential activity shifting derived from the project that could impact biodiversity.

Identified Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
	Biodiversity loss or affectation is not expected to occur.	Moreover, the project is going to monitor forest cover and sustainable productive activities in already cleared areas, biodiversity risks are mitigated through good productive practices adoption and forest protection. Likewise, to reduce leakage that may generate pressures over biodiversity out of the project area, community engagement and delivering income opportunities for the people, helps to control deforestation activity migration and reduce the risk over biodiversity out of the project area

2.1.21 Benefit Permanence (CCB, G1.11)

Capacity building, strong governance and good environmental management and education are key to maintain benefits beyond the project lifetime. Proper engagement and constant monitoring will help the community to internalize in their culture the improved forest management, the economic alternatives, and the activities to be developed by the project. It is expected as the benefits of the project are perceived by the community that the interest in maintaining the activities is kept in the long term.

2.1.22 Financial Sustainability (CCB, G1.12)

Estimated net carbon revenues from the project will be enough to cover all estimated project costs and net benefits to the communities. These additional revenues will be used by the communities to finance priority activities for social investments, not necessarily linked to drivers of deforestation, but important to improve livelihoods and achieve effective community internal control for keeping community members activities aligned with the REDD+ project over the long term. The financial assessment shows a positive EBITDA IRR resulting from the project cashflow over the implementation period, and also good investor IRR which are adequate returns for the type of investments and associated risks; it is expected

that the project reaches financial stability during the fourth year of implementation. Investments are mostly focused to social infrastructure, followed by livelihood and governance strengthening. Projected revenues consider a decreasing trend as the deforestation risk reduces over time.

Project budgets for the entire implementation period of the project (i.e. 30 years) have been estimated and account all project activities and interventions costs considered necessary to achieved project objectives. Comparing these costs and investments with the expected revenues derived from carbon sales, the financial model shows that there is adequate and positive cash flow that will enable to carry on with the REDD+ strategy and achieve the anticipated climate, community and biodiversity benefits. These confidential financial projections are available to the validator.

Additionally, the benefits sharing mechanism and the administration protocols developed with the community will ensure that the Indigenous Ethno-development Plan and the other REDD+ activities, within the theory of change framework, will receive financial support derived from carbon credit sales and reaffirm community long term commitment with project development.

2.2 Without-project Land Use Scenario and Additionality

2.2.1 Conditions Prior to Project Initiation and Land Use Scenarios without the Project (VCS, 3.13; CCB, G2.1)

The indigenous territories (*Terra Indígena*) of Acapuri de Cima, Estrela da Paz and Riozinho are located within the municipalities of Fonte Boa, Jutai and Juruá, in the Amazon state of Brazil, in a geographical and natural area rich in biodiversity, however, it's a region with social conflicts, a historic abandonment of the government and few sustainable and legal economic alternatives, which has led to an increased pressure on natural resources.

Current land use corresponds to agriculture and mining activities, activities that have led to environmental problems, such as significant reduction of natural vegetation cover and has increased this region vulnerability to climate events, such as forest fires, floodings and landslides, mainly.

Moreover, these indigenous communities face other socio-environmental problems regarding basic sanitation, water pollution and mismanagement of solid wastes.

Major threats to the community, the forests and biodiversity:

- Illegal Mining: Gold mining activities mainly.
- Illegal fishing and hunting: for commercial purposes represent a major threat for the biodiversity and food security of the communities.
- Logging firewood carried out by the community and for commercial purposes in some cases.
- Illegal logging
- Fire burns
- Agriculture

Designed REDD+ project activities are designed with the intention of reducing CO₂ emissions from unplanned deforestation and degradation compared to baseline levels. Land in the project area consists of contiguous and discrete areas covered by forests meeting the definition of forest as defined by the government of Brazil¹³. These areas were forests for a minimum of 10 years before the project start date. The project area would be degraded or deforested in the absence of the REDD+ project activity and the deforested and degraded areas are mining areas and mosaic areas.

Moreover, project activities are aligned with national and regional initiatives that seek to coordinate actions aimed at reducing deforestation in the legal Amazon as well as promoting conservation with the joint participation of the civil society and other stakeholders from the territories but that have so far been insufficient (González, 2018).

2.2.2 Most-Likely Scenario Justification (CCB, G2.1)

Current land use and activities will continue as they have been historically developed. There is no investment in the territory and there are no economic alternatives for the communities to improve their living conditions. The state absence continues to be a major problem for the territory governance despite the national initiatives to control deforestation in the Amazon region.

Post-deforestation Land Use after a 11-year-period is presented and shows how despite having a significant forest coverage, deforestation drivers (mining, agriculture, logging) are increasing.

Post-deforestation Land Use after a 10 year period (based on historical data of the region)	% of land use
Clean grass	20.5%
Mining zones	10,21%
Discontinuous Urban areas	4,25%
Wetland	30,62%
Cropland mosaic	34,42%

Through geographic information system (GIS) analysis, historical data of the region was assessed in order to determine the most likely scenario without the project. Information from years 2010, 2013, 2016, 2019, June 2021 and December 2021 has been reviewed and compared. The

¹³Forest is a minimum area of land of 0.5 hectares with tree crown cover (or equivalent stocking level) of more than 10-30 percent with trees with the potential to reach a minimum height of 5 metres at maturity in situ. FAO sets the minimum canopy coverage at 10%.

analyzed data shows that that major coverage loss corresponds to dense and open forest, fragmented forest with pastures and crops while pastures and crops and mining coverage increase. These shows that without the project deforestation patterns will continue.

2.2.3 Community and Biodiversity Additionality (CCB, G2.2)

All the indigenous communities in the three indigenous territories of Acapuri de Cima, Estrela da Paz and Riozinho are legally conferred with administrative autonomy over the land and natural resources and are free to decide how to manage natural resources and land use within their territories, thus they can decide whether to log their forests or preserve them.

Potential forest clearing in mid to long term in the absence of the project is perceptible through the historical analysis carried out of deforestation trends both in the project and reference area. Taking into account the local community's needs, the historical land use change analysis, the economic situation and the presence of income generating activities that lead to land use changes (variables that were considered for the determination of the baseline scenario and the barrier analysis, presented in section 3.1.4); and the absence of institutional readiness and government resources to invest in regional growth and development of these communities indicate that deforestation drivers may persist and no interventions are expected to change the historical land use dynamics.

There is a historical lack of law enforcement within this territory that has led to uncontrolled logging and mining by the local or national government. The community also lacks the means to deal effectively with this situation, mainly due to the close relationship between the need to meet economic requirements and the income-generating opportunity that these activities represent in these remote areas.

The REDD+ mechanism is seen by the indigenous communities as a unique opportunity to address deforestation drivers and natural resources depletion and deterioration observed in the territory. Obtaining economic resources through forest conservation activities represent the best opportunity to promote forest protection, legal activities and offer a path to transit from damaging land use practices to sustainable productive ones. In the absence of this project and the expected carbon revenues, the investments required to prevent forest loss and offer economic alternatives to implement a comprehensive strategy defined by the indigenous territories to protect their natural resources would not occur, mainly due to lack of funding opportunities and limited government investment for these specific areas and group of people. In addition, it is important to highlight that institutional barriers also contribute to assume that historical practices may continue to occur. Despite forest conservation is conceived as a government objective, lack of law enforcement capacity and reduced local investment resources have been a historical constant in the territory, which makes it difficult for the government to tackle deforestation drivers in a systematic and comprehensive way.

Additionality: Community

Development of economic alternatives compatible with the indigenous communities' well-being and nature conservation that contribute not only to improve living conditions in the three indigenous territories by supporting or improving community infrastructure, access to education, health, communication means and basic sanitation services, but to strengthen land use planning and mechanisms to guarantee food security is one of the key components of the REDD+ strategy and thus, the project delivers unique community benefits that would not have occurred in the absence of the project.

There are strong incentives in the REDD+ project to prevent land use changes as investments from the project revenues are directed to cover the communities' main needs and generate alternative sources of income to those coming from current deforestation drivers such as mining and agriculture, usually done on an unsustainable manner to secure quick financial gains. This is described in Section 4 through the definition of community conditions that would have occurred in the absence of the project, and in Section 4.2 and 2.1.17 where net-positive impacts are identified as result of the project's theory of change.

Additionality: Biodiversity

Strengthening communities' capacities to fulfill functions of forest rangers and surveillance over the territory, providing tools and knowledge that contribute to improve forest management and agricultural systems, biodiversity will be impacted positively and receive direct benefits that would not occur in absence of the project.

Forest protection, reducing pollution and good agricultural practices favor fauna and flora species. Historical land use practices have depleted natural resources and forest cover and polluted water streams. Since continuation of the historical practices is the most plausible alternative scenario for the project, biodiversity benefits under project scenario are additional to those expected under the current scenario. Details regarding biodiversity conditions in the absence of the project are described in Section 5; net-positive impacts to biodiversity are included in Section 5.2 and in the project's theory of change.

2.2.4 Benefits to be used as Offsets (CCB, G2.2)

No biodiversity or community benefits are intended to be used as offsets of any kind.

2.3 Safeguards and Stakeholder Engagement

2.3.1 Stakeholder Identification (VCS, 3.18, 3.19; CCB G1.5)

The process of stakeholder identification was initiated with the communities in the area and that are the owners of the land (*Terras Indígena*). This territory is inhabited by near 1,578 people. This is the people that owns the right over the territory of the project area and are in charge of the decisions. Also, there are some settlers that have been dwelling in the territory on a discrete basis, not being part of the indigenous communities, but interacting with the locals for commercial purposes. Finally, there are a few external agents that have economic interests inside the territory. This people are related basically with mining activities and local commerce.

The stakeholder identification process has respected the community's autonomy to stand for their own decisions regarding their engagement in REDD+ projects, local dynamics and regional institutional relation. During this process potential risks and benefits of participating in the project have been exposed and discussed, ensuring a transparent process. The interaction among community members and their interests regarding REDD+ strategy and activity portfolio has been identified through participative workshops (see folder *Annex 7. Workshops*, subfolder *Workshop 1, Workshop 2, Workshop 3*).

Moreover, some institutions were identified considering they should be articulated during the implementation phase, not as stakeholders, but as key partners to facilitate and contribute to project implementation and integration with the regional context and initiatives. Among these are the National Foundation of the Indian (FUNAI), Public Ministry, Jutai Municipality Government, Amazonas Departmental Government, Secretary of the Environment of Jutai and the Brazilian Institute of Environment and Renewable Natural Resources (IBAMA).

2.3.2 Stakeholder Descriptions (VCS, 3.18, 3.19; CCB, G1.6, G1.13)

Stakeholders were identified by the proponents of the project accordingly to workshops (see folder *Annex 7. Workshops*, subfolder *Workshop 1, Workshop 2, Workshop 3*) and consultations in the region. Following is a list of main stakeholders:

- Indigenous communities conforming the indigenous territories (Terra indígena) of Acupuri de Cima, Riozinho and Estrela da Paz: communities dwelling in the area, and legally owning the land.
- External agents with economic activities in the project area: exert pressure on the territory. People involved in illegal mining and logging activities.

Appendix 1 contains the full list of identified stakeholder's description.

2.3.3 Stakeholder Access to Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

Participation of local actors:

Since the beginning of the project, a properly designed consultation and participation process was established to ensure that all community members that live in indigenous territories of Estrela da Paz, Acapuri da Cima and Riozinho conforming the Estrela Acapuzihno REDD+ project could participate and have access to project information and documentation. This, to engage the communities' active participation during planning and implementation stages and increase project appropriation and community commitment with the REDD+ strategy.

Access to documentation:

Different groups and stakeholders have access to project documents and information. The project has the following mechanisms to facilitate communications and ensure information availability:

- E-mail: The project description and other documents have been sent to the Caciques of the communities and stakeholders physically. In this sense, as other documents become available like validated project document, monitoring reports, summaries, booklets, among others, were also deliver directly to specific stakeholders and community members.
- Printed Documentation: documents as the project description, validated project document, monitoring reports and others are going to be available physically in the offices of the Council of Indigenous Peoples of Jutai (Conselho dos Povos Indigenas de Jutai – COPIJU) and other facilities that the project defines. For this, booklets and billboards have been developed, as it can be seen in the evidence presented in folder Annex 7. Workshops, and folder Annex 9. Activities executed, subfolder Cartillas REDD+ Traducidas. Due to the poor internet connections and digital resources that the community members have, this mechanism facilitates delivering information to community members and stakeholders.
- Documents distribution: Physical material was produced to facilitate the project information distribution throughout the indigenous communities. A booklet was handed and socialized among the people during various workshops that took place during May 2023 (see folder Annex 7. Workshops, subfolder Workshop 2, and folder Annex 9. Activities executed, subfolder Cartillas REDD+ Traducidas). Further material, designed to transmit information to all community members, is going to be distributed among community members and stakeholders. Also, handling material in settlements is an appropriate strategy to reach many members that doesn't have the possibility to attend project workshops and meetings. This mechanism also contributes to deliver accurate information of the project to communities and different stakeholders.
- Online publications: Documents related to the project are available on the VCS and CCB websites. Additional documents will be posted on the website in accordance with the audits carried out, and the instructions of the VCS and CCB.

- Oral communication: This has been a main mechanism of communication with the communities (Aldeias) that are part of the Indigenous Territories (Terra Indigenas) part of the project. Project design has been defined through workshops, general assemblies and other community spaces that relied in oral communication. In addition, to guarantee the access of all the community members to project information, it has been agreed that the leaders and members of the indigenous communities should disseminate and transmit to the communities they represent all the information and relevant information of the project.
Among other stakeholders, such as government institutions, the representatives of the Communities have presented and socialized the main components of the project. During monitoring periods, further meetings will be arranged to present activities' progress, coordinate actions and identify common interests and articulation opportunities.

2.3.4 Dissemination of Summary Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

Summaries of project description and monitoring results have been presented to community members during workshops and project meetings and deliver physically to community leaders. All means of communication described in section 2.3.3. have been used to disseminate project summaries. Once the verification process is finished, reviewed and audited summary documents will also be posted in the VCS and CCB websites in accordance with each program's procedures. Summary documents are prepared in local language to ensure that community understands.

Project monitoring results are to be summarized and made available through all existing communication channels. The Management Mechanism will have to report on a permanent basis to the Assemblies and communities' meetings on the project's progress and results from state of forest monitoring.

According to the different stakeholders' groups, some specific communication channels are preferred to disseminate summary documents:

- Project proponents: project brochures and printed documents have been delivered to community members during workshops and are available on the COPIJU headquarters. Some members also participated in workshops in which summary documents were presented. Further project summaries are printed and sent to headquarters, while digital files are sent via email.
- Community members: information related to the project was printed and made available among community members. Sometimes, whatsapp are also helpful to ensure information dissemination. Project monitoring summaries are to be presented by the REDD+ Committee at the Assemblies.
- Other stakeholders: specific meetings (in person and virtual), emails, letters, phone calls and printed information are used as dissemination means. If required, printed document may be delivered, or summaries of project documents may be sent via email according to specific stakeholder characteristics.

2.3.5 Informational Meetings with Stakeholders (VCS, 3.18, 3.19; CCB, G3.1)

Publication and calling for meetings

Due to limited access to internet, calling for meetings and workshops within the territory is done through directly delivering letters to community leaders, who afterwards, are in charge of informing their respective community members via voice to voice contacting and direct conversation. This allows to keep record and officialized territory activities.

Descriptions of the meetings

Informational meetings with communities and local stakeholders are carried out in an active participatory way. Planning and implementation activities are based upon continuous and direct community's engagement. Social and environmental professionals provide technical assistance and help engage beneficiaries on the identification of problems related to deforestation and forest degradation, and on how to address drivers and set priorities.

During project design, workshops with all the representatives of the indigenous communities (*Aldeias*) were carried out. These workshops aimed to i) identify and analyze deforestation and degradation drivers, ii) identify solutions and strategies to guide investment resources of the REDD+ project, iii) define REDD+ strategy components and investment lines to achieve forest loss reduction (governance, productive initiatives, social investment and monitoring).

For project implementation, community leaders help set priorities and develop the budget and implementation plan for REDD+ strategies and tasks to take place. The Management Mechanism, in close coordination with the project developers, follows up disbursements and payments so that project's objectives are met. All important decisions regarding REDD+ strategy development take place with governing boards of the Indigenous Territories (*Terra Indígena*) and other community designated members that are to be appointed to assist implementation of specific topics and project components.

Below is a list of the meetings and workshops held with the members of the community (the evidence of these meetings is presented in the folder *Annex 7. Workshops*).

Workshops	Date(s)	Indigenous territory/Place	Attendees	Topic	Procedure
Workshop 1	04/11/2022 05/11/2022	Estrela da Paz	Indigenous Communities (<i>Aldeia</i>) Bugaio, representatives and members of the community Traditional Authorities CARBO-TERRA YAUTO Concepto Carbono COPIJU	REDD+ project presentation (objectives, interventions) Deforestation drivers Problem tree and solutions tree development Productive strategies	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA. The problem tree, solution tree, social cartography, deforestation drivers, governance strategies and productive strategies were carried out.
Workshop 1	15/02/2023	Riozinho	Indigenous Communities (<i>Aldeias</i>) Sampaio, Novo Paraíso, São Luís, Ariramba representatives and members of the community CARBO-TERRA	REDD+ project presentation (objectives, interventions) Deforestation drivers Problem tree and solutions tree development Productive strategies Governance strategies	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA. The problem tree, solution tree, social cartography, deforestation drivers, governance strategies and productive strategies were carried out.
Workshop 1	17/02/2023	Estrela da Paz	Indigenous Communities (<i>Aldeias</i>) Bugaio and Boa Vista representatives and members of the community Traditional Authorities CARBO-TERRA COPIJU	REDD+ project presentation Deforestation drivers Productive strategies Governance strategies	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA. Social cartography, deforestation drivers, governance strategies and productive strategies were carried out.
Workshop 1	23/02/2023	Estrela da Paz	Indigenous Communities (<i>Aldeias</i>) Cajual and Nova Esperança representatives	REDD+ project presentation (objectives, interventions) Deforestation drivers	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA.

Workshops	Date(s)	Indigenous territory/Place	Attendees	Topic	Procedure
			and members of the community Traditional Authorities CARBO-TERRA COPIJU	Problem tree and solutions tree development Productive strategies Governance strategies	The problem tree, solution tree, social cartography, deforestation drivers, governance strategies and productive strategies were carried out.
Workshop 1	24/02/2023	Acapuri de Cima	Representatives and members of the indigenous community Traditional Authorities CARBO-TERRA YAUTO COPIJU	REDD+ project presentation (objectives, interventions) Deforestation drivers Problem tree and solutions tree development Productive strategies	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA. The problem tree, solution tree, social cartography, deforestation drivers, governance strategies and productive strategies were carried out.
Workshop 1	18/05/2023	Estrela da Paz	Indigenous Community (<i>Aldeia</i>) Inglaterra representatives and members of the community Traditional Authorities CARBO-TERRA COPIJU	REDD+ project presentation (objectives, interventions) Deforestation drivers Problem tree and solutions tree development Productive strategies	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA. The problem tree, solution tree, social cartography, deforestation drivers, governance strategies and productive strategies were carried out.
Workshop 2	11/05/2023 12/05/2023	Riozinho	Indigenous Communities (<i>Aldeias</i> Sampaio, Novo Paraíso, São Luís, Ariramba representatives and members of the community Traditional Authorities	REDD+ project presentation Prioritization of productive systems, social investment Identification of mitigation actions Possible leakage and reference areas mapping	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA. Crops, social and economic investments in the area, mitigation

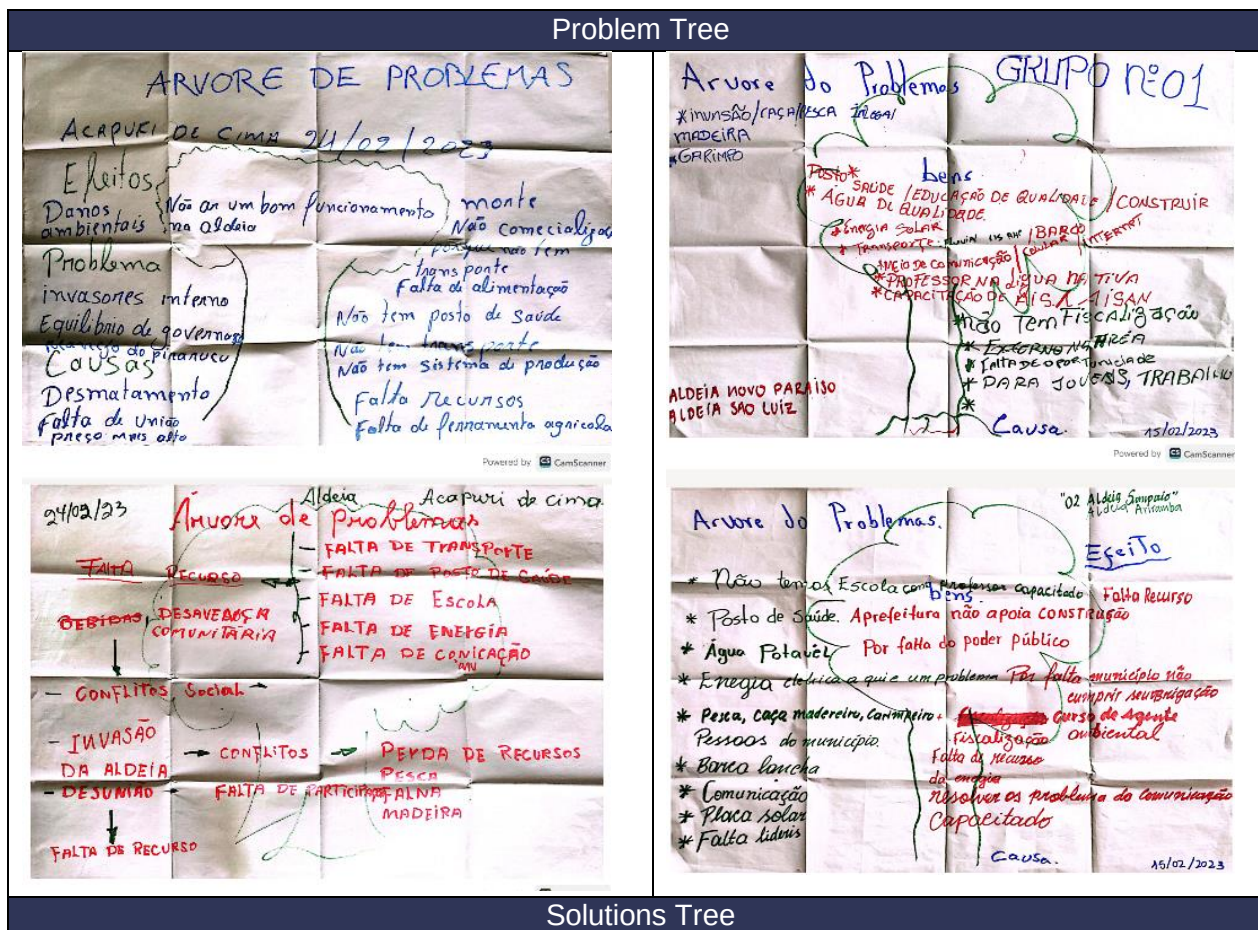
Workshops	Date(s)	Indigenous territory/Place	Attendees	Topic	Procedure
			CARBO-TERRA COPIJU UIDPA	Governance strategies	actions and steps to follow in the REDD+ project implementation process were defined.
Workshop 2	15/05/2023	Estrela da Paz	Indigenous Communities (<i>Aldeias</i>) Bugaio and Boa Vista representatives and members of the community Traditional Authorities CARBO-TERRA COPIJU UIDPA	REDD+ project presentation Prioritization of productive systems, social investment Identification of mitigation actions Possible leakage and reference areas mapping Governance strategies	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA. Crops, social and economic investments in the area, mitigation actions and steps to follow in the REDD+ project implementation process were defined.
Workshop 2	15/05/2023	Estrela da Paz	Indigenous Communities (<i>Aldeias</i>) Cajual, Escondido, Curupira and Nova Esperanza representatives and members of the community Traditional Authorities CARBO-TERRA COPIJU UIDPA	REDD+ project presentation Prioritization of productive systems, social investment Identification of mitigation actions Possible leakage and reference areas mapping Governance strategies	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA. Crops, social and economic investments in the area, mitigation actions and steps to follow in the REDD+ project implementation process were defined.
Workshop 2	17/05/2023	Acapuri de Cima	Community representatives and members of the indigenous community Traditional Authorities CARBO-TERRA COPIJU	REDD+ project presentation Prioritization of productive systems, social investment Identification of mitigation actions Possible leakage and reference areas mapping Governance strategies	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA. Crops, social and economic investments in the area, mitigation actions and steps to follow in the

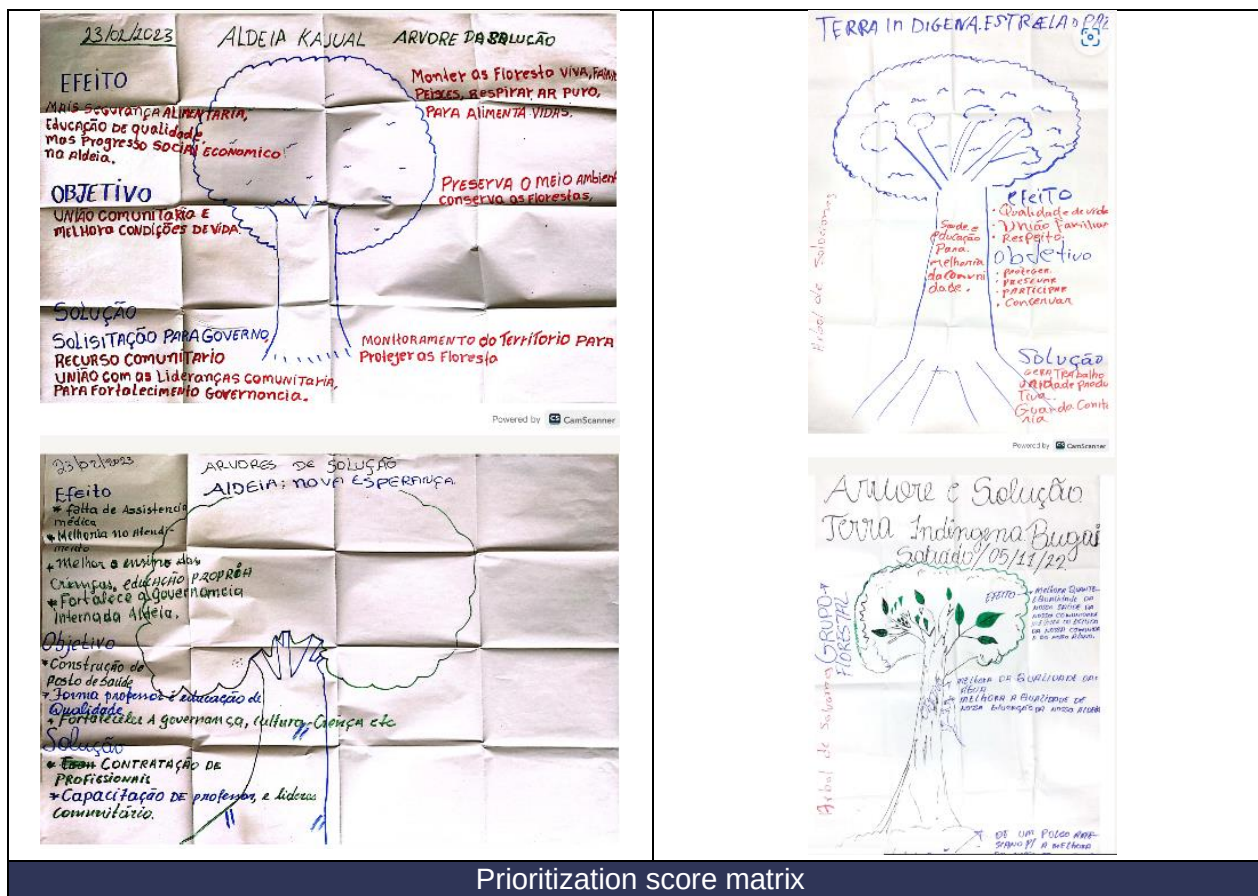
Workshops	Date(s)	Indigenous territory/Place	Attendees	Topic	Procedure
					REDD+ project implementation process were defined.
Workshop 3	06/07/2023 07/07/2023	Riozinho	Indigenous Communities (Aldeias Sampaio, Novo Paraíso, São Luís, Ariramba representatives and members of the community Traditional Authorities CARBO-TERRA COPIJU FUNAI	REDD+ project presentation REDD+ safeguards Activity prioritization Budget distribution per component REDD+ knowledge Benefit distribution	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA. General socialization of the project was presented, an analysis of each safeguard and its compliance to guarantee community rights was conducted, a prioritization matrix was developed in order to define the determine which activities are to be executed in the short-term, the distribution of the budget per component was defined.
Workshop 3	08/07/2023	Acapuri de Cima	Community representatives and members of the indigenous community Traditional Authorities CARBO-TERRA COPIJU	REDD+ project presentation REDD+ safeguards Activity prioritization Budget distribution per component REDD+ knowledge Benefit distribution	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA. General socialization of the project was presented, an analysis of each safeguard and its compliance to guarantee community rights was conducted, a prioritization matrix was developed in order to define the determine which activities are to be executed in the short-term, the distribution of the budget per component was defined.
Workshop 3	09/07/2023	Estrela da Paz	Indigenous Communities (Aldeias) Cajual,	REDD+ project presentation REDD+ safeguards	The workshops were carried out in a participatory way with the

Workshops	Date(s)	Indigenous territory/Place	Attendees	Topic	Procedure
			Escondido, Curupira and Nova Esperanza representatives and members of the community Traditional Authorities CARBO-TERRA COPIJU	Activity prioritization Budget distribution per component REDD+ knowledge Benefit distribution	methodological guide developed by CARBO-TERRA. General socialization of the project was presented, an analysis of each safeguard and its compliance to guarantee community rights was conducted, a prioritization matrix was developed to define the determine which activities are to be executed in the short-term, the distribution of the budget per component was defined.
Workshop 3	12/07/2023	Estrela da Paz	Indigenous Communities (<i>Aldeias</i>) Boa Vista, Bugaio and Inglaterra representatives and members of the community Traditional Authorities CARBO-TERRA COPIJU FUNAI	REDD+ project presentation REDD+ safeguards Activity prioritization Budget distribution per component REDD+ knowledge Benefit distribution	The workshops were carried out in a participatory way with the methodological guide developed by CARBO-TERRA. General socialization of the project was presented, an analysis of each safeguard and its compliance to guarantee community rights was conducted, a prioritization matrix was developed to define the determine which activities are to be executed in the short-term, the distribution of the budget per component was defined.
General Assembly	3/11/2023 4/11/2023 6/11/2023	Estrela da Paz Acapuri de Cima Riozinho	Indigenous Community (<i>Aldeia</i>) representatives and members of the community CARBO-TERRA	PDD socialization and approval	The workshops were carried out in a participatory way. CARBO-TERRA presented the Project Design Document, then an autonomous space was given for the community to discuss. Afterwards, the

Workshops	Date(s)	Indigenous territory/Place	Attendees	Topic	Procedure
					Indigenous representatives approved the PDD.

Some of the results obtained during these workshops are presented below (further information can be found in folder Annex 7. Workshops)





2.3.6 Community Costs, Risks, and Benefits (CCB, G3.2)

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The process with the community for the identification of costs, risks and benefits related with REDD+ mechanism has been done through a series of participatory workshops and information disclosing. The first community spaces were used to informed what is a REDD+ project and then analyzed the communities' situation. During workshop 1, the communities' developed the tree problem which findings are:

Causes	Problem	Consequences
Mining Low household income generation Weak food security Weak social investment Weak presence of the Government Lack of economic alternatives and commercialization means. Weak access to basic services (health, education, among others) Poor territorial control expansion of settlement areas Illegal mining, fishing incidence Wood extraction Forest burns	Increasing deforestation within the territories of the indigenous territories of Estrela da Paz, Acapuri de Cma and Riozinho in the municipalities of Jurua, Jutai and Fonte Boa.	Pollution Climate change Vulnerability to natural disaster events Poverty Ecosystem and biodiversity loss Poor territory control

It was identified that mining activities, governmental neglect, lack of economic resources in the region threatens their stability and future (see folder *Annex 7. Workshops*, subfolder *Workshop 1*). In this way, the participants described deforestation trends and identification of deforestation drivers (problem tree).

In a second instance of the first workshop proposed solutions were identified with the elaboration of a solution tree, making emphasis in alternatives or actions that lead to a change in current deforestation trends and what activities can be implemented to offer livelihood improvement and stability (see folder *Annex 7. Workshops*, subfolder *Workshop 1*). In a second workshop the members of the community established in what activities the carbon credit sales resources should be invested and how the distribution of the benefits should take place, according with the needs and expectations of the hole community (see *Annex 7. Workshops*, subfolder *Workshop Taller 2*). During a third workshop, the prioritization of places for productive systems development, timeline of the implementation of the project activities and budget distribution were defined with the members and leaders of the Indigenous Territories (see folder *Annex 7. Workshops*, subfolder *Workshop 3*).

For educational purposes, a REDD+ booklet was design and given to the community members where they can find the outcome of all the workshops and all the principal aspects regarding their REDD+ project.

By carrying out this socialization and project designing process, the community has received relevant and transparent information about REDD+ project development, and they agree to participate in a voluntarily and determined fashion (see folder *Annex 7. Workshops*).

2.3.7 Information to Stakeholders on Validation and Verification Process (VCS, 3.18.6, 3.19; CCB, G3.3)

Measures taken include diffusion of the REDD+ educational booklet that includes the description of the project, activities, purpose, benefits, values subject to conservation.

Direct contact with the legal representatives of the Indigenous Territories (*Terra Indigena*) is done and through this channel information is sent and the leaders of each community are contacted. It is reported when visits to the territory are going to be made including when the audit is going to be carried out so that the announcement is made among the community members.

The legal representative defines the places to carry out these conferences with the representatives of the communities and the project developers and defines the agenda and places to visit.

Visits to the territory and participatory workshops are carried out and the topics of the REDD+ project are addressed, and the educational booklet is socialized to reinforce concepts and commitments. During the visits and meetings held, the communities have had the space to express doubts associated with the project, as well as aspects that they consider should be strengthened within the framework of the project. Similarly, meetings have been held between the proponents of the project to resolve the concerns of the stakeholders.

Further details regarding communication with stakeholders on validation and verification process is related with other mechanisms defined in the Project Management Mechanism (see file *Management Mechanism_Estrela Acapuzihno REDD+ Project.pdf* in folder *Annex 15. Management mechanism*).

2.3.8 Site Visit Information and Opportunities to Communicate with Auditor (VCS, 3.18.6; CCB, G3.3)

The communities have been informed that there will be a visit by an auditor at least fifteen days in advance. They are informed through the community leader who receives the information from the legal representative.

During the visits and meetings of the audit, members of the community attend, on behalf of their partners, who will be available to directly answer the questions of the auditors in the spaces that the auditor requires.

2.3.9 Stakeholder Consultations (VCS, 3.18; CCB, G3.4)

Consultations with stakeholders have been carried out accordingly to the community's ways and reality. A set of workshops has been conducted to discuss and build the REDD+ Project on a participatory manner.

The PDD was presented during the last Assembly for discussion and approval (November 2023). Once approved, a Portuguese translation of this project description was given to the cacique of each indigenous territory (March 2024), for further consultation among the community members. As with all the project design process and information dissemination, this action contributed to ensure access for any community members interested in reading the document of the project.

Different strategies have been implemented throughout the socialization process, including presentations, documents delivery and voice to voice scenarios. These activities aimed to make emphasis on the development of appropriate community level engagement processes and materials to ensure complex issues are presented in a way that communities can understand. This included identification of the project zone, participatory mapping of project zone, identification of risks to the communities, prioritization of project activities, and role-playing to better understand potential financing and implications for the project.

The scheduling of meetings and workshops has been done to ensure: 1) formal approval from community assemblies is undertaken in an objective and timely manner; 2) communities understand and can take coherent decisions regarding the negotiation of any term sheet with investors; 3) the communities are able to understand and commit to any final contract that emerges because of these activities.

During the consultation activities the community members have stated their interest in the different stages of the project, the productive activities being the most preferred activity, followed by monitoring and supervision ones and finally governance activities. A detailed schedule of stakeholder consultations is presented in section 2.3.5 Informational Meetings with Stakeholders.

Further details regarding communication with stakeholders during any stage of the project is related with other mechanisms defined in the Project Management Mechanism (see file *Management Mechanism_Estrela Acapuzihno REDD+ Project.pdf* in folder *Annex 15. Management mechanism*).

Another important stakeholder interaction has been with the Fundacion Nacional del Indio (FUNAI). This institution, among other functions, protects and supervises indigenous communities' relationships with external actors. Since the first meeting between the communities and the other proponent companies the Technical Local Coordinator of Jutai of FUNAI was informed and socialized about the community REDD+ project (see Annex 5, *Lista asistencia Jutai REDD+ FUNAI_20072022.pdf*). FUNAI was also aware of the implementation of the activities to protect the forest and the traditional cultures that began since 2020 regarding the environmental amazon indigenous guard. The Project structure and activities

implementation was also socialized with FUNAI (Annex 5, *Carta para FUNAI de UIDPA_22082022.pdf*; *Carta_Caciques_Copiju_Funai 27032024.pdf*; *v2_Presentación REDD+ Estrela Acapuzinho FUNAI_30112023.pdf*; *Carta radicada FUNAI_072023 y remisión Brasilia.pdf*), and they were also participants of a indigenous leaders commission from Jutai to Bogota, which purpose was to know the companies ad planned further project activities (see Annex 5, *Invitación a Bogota FUNAI_20032023.pdf*; *Acta Comisión Jutai a Bogotá 26032023.pdf*). Until the end of project documentation and basic agreements definition, the Technical Local Coordinator of Jutai of FUNAI was accompanying the meetings and other advances of the project (see Annex 5, *Asistencia Firma_Acuerdo Comercial_JUTAI_112023.pdf*).

Demonstrate in the table below how due account of all and any comments received during the consultation and public comment period has been taken. Include details on any updates to the project design or demonstrate the insignificance or irrelevance of comments.

Summary of comment received	When comment was received	Actions taken
Provide a summary of each comment received (1 per row)		Provide a summary of actions taken and any project design updates. If no actions were taken, justify the insignificance or irrelevance of the comment.
...		

2.3.10 Continued Consultation and Adaptive Management (VCS, 3.18; CCB, G3.4)

The project has decision-making bodies that respect the indigenous traditional government and their territorial management. There is a REDD+ committee that includes the leaders of the community and the developers of the project. This instance oversees making budget, investments and project activities development, ensuring that the project's safeguards are respected, as well as the lines of action prioritized by the communities during general assembly's meetings. This committee must meet at least once a year to make the investment plan. In this committee exist a representant for each topic and project component, thus explained in detailed in section 2.4.1. The committee is composed of community members that addresses specific issues and is constituted as support for monitoring the implementation of activities. There is also a member that fulfills a secretariat function and supports recording and documentation of the project activities. And lastly, there is also a grievance attendance mechanism. It is defined as the requests, complaints and claims attention system and is manage by a specific member of the REDD+ committee, which oversees receiving, reviewing and responding every notification in relation with this mechanism.

The specific details regarding the administration of the project are described in the annex 15, *Management Mechanism REDD + Jutai Project.pdf*

2.3.11 Stakeholder Consultation Channels (CCB, G3.5)

As mentioned in section 2.3.10, consultation channels include:

- Workshops
- Presentations
- Document Delivery
- Community assemblies

2.3.12 Stakeholder Participation in Decision-Making and Implementation (VCS, 3.18, 3.19; CCB, G3.6)

Within stakeholder consultation and participation processes for the project's design and implementation, participatory workshops, including women engagement, have been carried out for the communities to express their perceptions, knowledge, and expectations (see folder Annex 7. Workshops).

Participation in decision-making include:

1. Development of a problem tree
 - Identification of mining and logging as the principal deforestation drivers
 - Social Needs (transportation, health and education systems, sources of economic incomes)
2. Mapping of forest areas and areas with the highest deforestation rates (social cartography)
3. Development of a solutions tree
 - Generate a profitable and sustainable productive alternative
 - Social empowerment and capacity building
 - Clean crops
 - Forest, soil and biodiversity recovery
 - Commercialization of goods
4. Construction of governance strategies
 - Control in logging and mining activities
 - Control over the activities carried out in the territory
 - Punishments for the ones who does not obey community laws
 - Community controls
 - Surveillance committee
 - Construction of the indigenous ethno-development plan

5. Brainstorming on productive sustainable activities
 - pisciculture, poultry and pork farming, cassava, banana, pineapple, acai, copoazu
6. Brainstorming on monitoring strategies
 - Forest ranger families
 - Community monitoring
 - Pedagogy
7. Definition of project activities and benefit distribution

2.3.13 Anti-Discrimination Assurance (VCS 3.19; CCB, G3.7)

In order to comply with The Climate, Community & Biodiversity (CCB) Standards, an Implementation Framework Agreement will be executed with the project investor as defined in the Project Management Mechanism which has the measures needed and taken in order to ensure that the project proponent (Communities) and all other entities involved in Estrela Acapuzihno REDD+ Project design and implementation, are not involved in harassment or discrimination, including discrimination based on gender, race, religion, sexual orientation or other habits.

2.3.14 Feedback and Grievance Redress Procedure (VCS, 3.18.4; CCB, G3.8)

Development process	The organizational structure of the Grievance and Redress Mechanism will be as follows: The project has a Grievance and Redress responsible as part of the REDD+ committee functions, which is at least one member of the community and may include traditional leader's accompaniment. These people are in charge of what is call as Requests, complaints and claims system.
Grievance redress procedure	The committee oversees receiving, reviewing, and responding to any grievance, suggestions or comments made by any member of the communities or interested part. The community may communicate

at any moment suggestions or grievance to their community leader through a verbal or written manifestation.

The leader of each community must register the manifestation and should take it to the grievance and redress committee and should follow up the process of response.

The details of the operation of this committee are described in the Project Management Mechanism Document (see file *Management Mechanism_Estrela Acapuzihno REDD+ Project.pdf* in folder Annex 15. Management mechanism). The templates and formats to be used for this purpose are available in folder Annex 15. Management mechanism.

2.3.15 Accessibility of the Feedback and Grievance Redress Procedure (VCS, 3.19; CCB, G3.8)

The mechanism for feedback and grievance redress accessibility is described in the Project Management Mechanism. The project has a Grievances and Redress mechanism stating clear procedures and responsible parties for recording, processing, and replying to potential information requests, suggestions, complaints and claims posed by community members related to the implementation of the REDD+ project.

Written responses will be delivered to the community and properly documented. A registry of responses will be taken and when applicable they will be available for consultation. In addition, follow-ups are included in the correspondent formats; this activity is carried out by the Grievance and Redress committee.

Further details regarding accessibility of the Feedback and Grievance Redress Procedure are detailed in the Project Management Mechanism (see file *Management Mechanism_Estrela Acapuzihno REDD+ Project.pdf* in folder Annex 15. Management mechanism).

2.3.16 Worker Training (VCS, 3.19; CCB, G3.9)

Each of the four main components conforming the REDD+ strategies include training and capacity building activities that seek to promote adequate and useful skills and knowledge focused in increasing and strengthening local participation in the project implementation and all related activities procuring for a sustainable development of the community and the benefits permanence.

The methodology for trainings includes a previous selection of people in the proposed activities. Technical and more specific training will be focused on the following topics: (i) sustainable production systems, (ii) administration and business management, (iii) territory governance, cultural practices, tradition protection (iv) forest conservation and protection, (v) biodiversity monitoring. Additional to this training, there is a REDD+ activity focused

in offering and ensuring access to formal education for members of the communities, according to budgets and availability, and is conceived to reach as many members of the communities as possible.

Tools and strategies to be implemented are designed to be culturally appropriate.

2.3.17 Community Employment Opportunities (VCS, 3.19.13; CCB, G3.10)

There is no employment discrimination within the indigenous communities based on gender, age, beliefs, religion, ethnicity, among other characteristics.

The project implementation will represent employment opportunities in the different activities regarding productive activities, forest governance and monitoring. It is expected from the indigenous communities a transparent and objective hiring process, which ensures competitive selection process for equal opportunities among recipients.

2.3.18 Occupational Safety Assessment (VCS, 3.19; CCB, G3.12)

Brazil has a well-developed system for occupational safety, that is required by the consolidation of labor laws and regulatory standards known as NR (Brazilian NR, 2023). The occupational safety assessment is carried out in every specific case according to the implementation process of REDD+ activities. Each of the activities get analyzed and the occupational risks identified, so that every participant receives the information and training sessions needed in order to ensure and minimize any identified risk. All occupational safety regulations related with these activities should be complied and all required insurance, personal protection elements, trainings and risk awareness need to be provided.

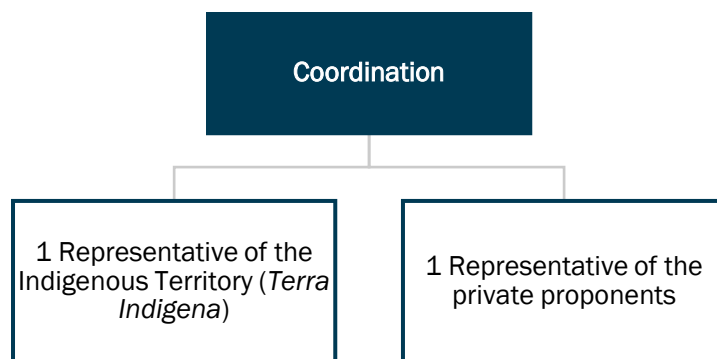
When activities are implemented by external entities, all national regulations related with occupational safety must be complied and evidence of compliance will be required.

2.4 Management Capacity

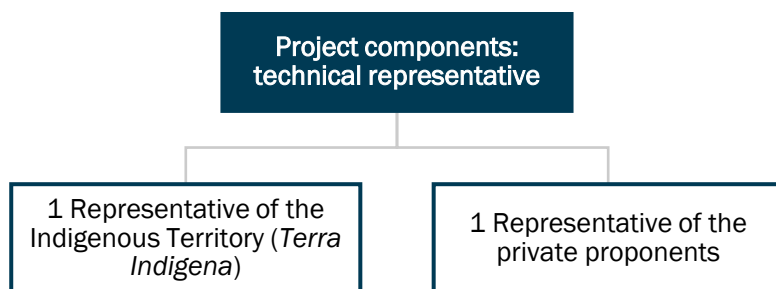
2.4.1 Project Governance Structures (CCB, G4.1)

The project has a REDD+ Committee which facilitates the governance of the project and is composed by members of the communities and the companies. It has a defined structure, along with the roles and responsibilities of the participants (see file *Management Mechanism_Estrela Acapuzihno REDD+ Project.pdf* in folder *Annex 15. Management mechanism*). The main components of the REDD+ Committee were presented during meetings and specific members will be define after the first verification period (see Annex 7, subfolder Workshop 4, archives in subfolders *Presentación_Lectura_PDD_RIOZINHO.pptx*, *Presentación_Lectura_PDD_Estrela_ESTRELA_DA_PAZ.pptx* and *Presentación_PDD_Cartilla_ACAPURÍ DE CIMA.pptx*). The general structure is as follows:

Coordination:



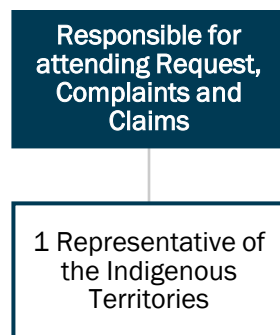
Technical components: may be constituted per topics, i.e., productive systems, governance, monitoring activities, social investments, among others.



Secretary: this person is in charge of administrative matters and evidence of the project implementation.



Requests, complaints and claims system:



2.4.2 Required Technical Skills (VCS, 3.19; CCB, G4.2)

The project proponents have a team of professionals that includes environmental engineers, social workers, biologists, cadastral engineers, and economists with experience in field and community development projects, who have the responsibility for supporting project design and implementation, as well as for producing monitoring reports. This team, along with the community leaders, follow up all project activities, the indicators regarding climate, community and biodiversity impacts generated by the project and participatory mechanisms. This team is also involved during validation and verification processes and participate in field visits.

Changes in carbon stocks are monitored according to description of data and parameters found in section 3 and are assessed by SIG engineers. In relation to the community leaders, appointed representatives of the Indigenous Territories (*Terra Indigena*) are experienced community leaders or have local expertise to address different project components. The local participants also help to ensure that community interests and ideas are incorporated in project planning and implementation processes.

The project activities are grouped in four main components and each of these need specific trainings. The main topics identified will cover the following: (i) sustainable production systems, (ii) administration and business management, (iii) territory governance, cultural practices, tradition protection (iv) forest conservation and protection, (v) biodiversity monitoring. During project implementation, according to needs and capacity or skills requirements, specific trainings are provided to people that participates. Follow-up training activities are included in the monitoring plan to ensure that capacity building is carried out as expected. These trainings are part of a major objective because they are conceived as a vehicle to improve development and increase community commitment with their territory protection.

Training activities are developed through strategic alliances with institutions such as Federal Institutes of Education, Science and Technology, Jutai and Fonte Boa city halls, or private or government educational programs, among others. In this way, appropriate technical assistance and strengthening capacities are addressed properly and contribute to achieve the expected short-, medium and long-term results.

Additional to these topic specific trainings, the social investment component of the project will deliver new opportunities and improve access to formal education that will benefit community members and in the future increase employment opportunities.

2.4.3 Management Team Experience (VCS, 3.19; CCB, G4.2)

The developers of the project have vast experience implementing community-based conservation projects in the amazon region (national and international). For more than ten years they have participated in the design and implementation of sustainable productive schemes and community strengthening. Moreover, they have also developed carbon projects in the Colombian amazon regions with ethnic groups and have actively participate in public policy making regarding climate change.

The legal representatives and *caciques* of the Indigenous Territories (*Terra Indigenas*) are elected by the communities according to their experience and management capabilities to guide and represent the will and needs of the communities. They have demonstrated experience in community affairs and relations management. Normally the elected person as representative has had experience in projects implementation within the community and has administration knowledge. These people are the promoters of the REDD+ project. Thanks to their active participation the project gains direct access to all community members and through their vision and knowledge of the territory the REDD+ activities can be oriented in an effective way.

2.4.4 Project Management Partnerships and Team Development (VCS, 3.19; CCB, G4.2)

The implementation of the project requires specific skills which are gathered through the participation of different entities. To enhance the social relationships and access to community members the project has included the company Yauto SAS which has worked for more than five years in the amazon region and has developed different projects with the communities.

For activities related with training and education the project has considered participation of educational institutions that have presence in this area and are known for developing specific training programs for communities. It is also expected to involve universities to offer formal education grants for community members.

For the social investment component, it is defined the need to involve the municipality and stataal government to articulate related activities regarding public health, housing, transportation, and education. The relationships with these entities are to be established when the first process of verification takes place, and the economic resources are identified.

2.4.5 Financial Health of Implementing Organization(s) (CCB, G4.3)

For readiness and project registration, validation and verification, the developers have designated all the economic resources needed. These resources come from private funding. All the carbon monitoring activities and validation and verification processes over the project lifetime are to be covered by the developers. Investments related to these activities are detailed in the investment plan.

Regarding other activities, the project relies completely on the carbon credits sales. Once there is an income from carbon sales the resources are going to be transferred to a bank trust. As stated in the Project Management Plan, each year an implementation and investment plan should be defined and approved by the coordinator committee and the bank is in charge of the money management. This bank trust offers different financial services which includes investment reports, balance sheets and economic analysis.

Regarding implementation, implementation is to be led by the community according to the agreed benefit distribution and administration mechanisms agreed. Funds will be kept at a financial institution through a trust where specific rules are set to ensure that all project funding proceeds are used according to the project´s objective and within the theory of change framework. During implementation, the community members in charge of preparing projects and procurement terms and documentation will be trained to build capacity. Training will include topics such as corporate associative schemes, business model development, financial management, procurement, and monitoring and reporting. Thus, the economic resources derived from carbon credits sales will be deposited in a bank trust and all the investments are going to be regulated trough a community-based decision process. All payments are going to be supported in the annual investment plan, which is limited by the available budget.

To diversify the possible incomes for the project implementation, the sustainable production systems component is oriented towards reaching economic independence and real sustainable initiatives. It is expected to develop profitable business during the first years of the REDD+ project to reduce financial risks associated to carbon markets and try to secure food safety and livelihood opportunities for community members.

2.4.6 Avoidance of Corruption and Other Unethical Behavior (VCS, 3.19; CCB, G4.3)

All project developers are searched within authorities and control entities databases, to find if there are any bad practices or law-breaking reports and bad financial behavior alerts and make decisions according to the case.

With respect to REDD implementation, in the Project Management Mechanism it is stated that all hiring processes respond to a legal and ethical selection process carried out by the coordinator committee. Each investment decision and supplier or contractor selection goes through a transparent process that the community coordinator committee conducts. The proposals and offers are analyzed and selected according to a cost-benefit assessment, experience of the proponent and the objectives that the community has in each case. Decisions are made only if the majority of the committee members approve the selection. This Mechanism helps to reduce corruption, bribery, and other bad practices regarding money management.

2.4.7 Commercially Sensitive Information (VCS, 3.5.2 – 3.5.4; CCB Rules, 3.5.13 – 3.5.14)

Sensitive commercial information includes the Commercial and Development Agreement between the project developers and the community and the project's financial model.

2.5 Legal Status and Property Rights

2.5.1 National and Local Laws (VCS, 3.1, 3.6, 3.7, 3.14, 3.18, 3.19; CCB, G5.6)

The project proponents are committed to comply with all applicable laws, statutes, property rights and other regulatory frameworks. The extensive stakeholder consultation process and implementation mechanism of the project ensure that compliance is achieved. A legal compliance matrix is included as part of the project monitoring, in which it is described how the regulatory framework is followed. Listed below are the Brazilian laws and regulations, referent to the project:

- National Constitution: includes property rights for indigenous territories and natural resources related regulations, establishing the right to an ecologically balanced environment, which is an asset of common use and essential to a healthy quality of life, and both the Government and the community shall have the duty to defend and preserve it for present and future generations.
- Environmental Policy Act: establishes the National Environmental Policy, its aims, formulation and enforcement mechanisms; the National Environmental System; the National Environmental Council and the Federal Technical Register of the Activities and Instruments of Environmental Protection. The National Environmental Policy has the aim of preserving, improving and recovering the environmental quality conducive of a healthy life, with a view to ensuring socio-economic development, the interests of national security and the protection of human life. This Act establishes the National Environmental Council NAENVCO, its composition, organization, responsibilities and mode of operation shall be regulated by the Executive Power. It establishes penalties as well defined by the federal, state and municipal legislation,

the non-compliance with the measures necessary for the prevention and correction of the damages caused by the degradation of the environmental quality.

- Forest Code (Law 12,651): The law established regulates the protection, and sustainable use and exploitation of native forests and other indigenous plants. The Law aims to promote the economic development of Brazilian selected areas by the following activities: recognizing the existing native forests and other indigenous plants as Brazilian wealth; confirming the preservation of National forests, habitats, biodiversity, soil conservation and water resources for the well-being of the future generations; recognizing the strategic function of rural production in recovering and protecting natural forests and the sustainable agroforestry production; including private forests on the same level as public; sustainable use of forest resources in line with the Environmental, Water and Land National Policies. This law was amended by Law 12.727.
- Law 12.727: Amendments establish general norms related to plant protection, permanent preservation areas and Legal Reserve areas; forestry exploitation requirements, control over the forestry origin products and prevention of forest fires, prescribing economic and financial instruments to reach the objectives.
- Brazilian Federal Decree No. 10,144/2019: Established CONAREDD+, The National REDD+ Commission as an execution and advisory body to the States, the Federal District, and the Ministry of the Environment, destined to formulate guidelines and issue resolutions related to REDD mechanism.
- Decree 11548 of 2023 Establishes the National Commission for Reducing Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conservation of Forest Carbon Stocks, Sustainable Management of Forests, and Increase of Forest Carbon Stocks - REDD+. The National Commission for REDD+ aims to coordinate, accompany, monitor and review the National Strategy for REDD+ - ENREDD+ and coordinate the development of requirements for access to payments for REDD+ results in the country, recognized by the United Nations Framework Convention on Climate Change.
- Decree 11.075/2022: creates the National System for the Reduction of Greenhouse Gas Emissions (SINARE), which has the purpose of serving as a single center for recording emissions, removals, reductions, and offsets of greenhouse gases and acts of trade, transfers, and transactions of certified emission reduction credits.
- Decree as of July 3rd, 2003, creates a Permanent Inter-Ministry Working Group, in order to control high deforestation rate. The measures proposed by that Working Group were gathered in the Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm)
- Law No. 12.187/2009: National Policy on Climate Change outlines the objectives and guidelines for addressing climate change in Brazil. Moreover, the National Policy and the National Plan on Climate Change intend to promote measures to reduce the adverse effects of climate change and the vulnerability of environmental, social and economic systems, thus contributing to adaptation. The coordination, supervision and impact monitoring of the National Policy and the National Plan are the responsibility of the Interministerial Committee on Climate Change (CIM) and its Executive Group on Climate Change (GEx), in accordance with Decree No. 6.263/2007 (Ministry of the Environment, 2016)
- Law No. 9.605/1998: Provides for criminal and administrative sanctions applicable to conducts and activities harmful to the environment.
- Law No. 9.985/2000: Institutes the National System of Nature Conservation Units (SNUC)

- Law no. 10.650/2003 the Law on Data and Information of the National Environment System
- Ordinance No. 365/2015: established the Brazilian Biomes Environmental Monitoring Programme to improve environmental monitoring at the national level.
- Decree No. 5.902/2004: establish the Priority Areas for Conservation, Sustainable Use and Biodiversity Benefits Sharing
- Decree No. 7.747/2012: The National Policy for Environmental and Territorial Management of Indigenous Lands. It aims at guaranteeing and promoting the protection, recovery, conservation, and sustainable use of the natural resources of indigenous lands and territories, ensuring the integrity of indigenous heritage and all the necessary conditions for the physical and cultural reproduction of current and future generations of indigenous peoples, respecting their sociocultural autonomy and improving the quality of life.
- Decree No. 8.505/2015: establishes the Amazon Region Protected Areas Programme – ARPA
- Public Forest Management Law, 2016: introduced several mechanisms for regulating public forest use, created the Brazilian Forest Service (BFS) and established a national forest development fund. Under the law, the BFS has registered almost 200 million hectares of public forest, of which 123 million hectares are destined for community use. The Public Forest Management Law also allowed government bodies to manage national, state and municipal forests directly, and for the first time gave state agencies a role in monitoring compliance with regulations.

In addition to the above-mentioned national laws and regulations, Brazil has adopted the following international legislation:

- Convention for the Protection of the World Cultural and Natural Heritage. Paris, 1972.
- Convention on International Trade in Endangered Species: wild fauna and flora. Washington, 1973.
- Convention on Biological Diversity. Rio de Janeiro, 1992.

Other relevant national laws and regulations include:

- National REDD + Strategy defines the REDD + policies and measures that will reduce GHG emissions associated with the forestry sector. It draws up the "roadmap" that establishes the activities that can be carried out, how they can be carried out and the financial resources required.
- The National Plan for the Control of Illegal Deforestation and Recovery of Native Vegetation provides guidelines for combating deforestation based on three cross-cutting themes: business environment, innovation and technological solutions, and financing for sustainable practices. The Plan proposes a goal zero deforestation, meaning the elimination of illegal deforestation and ending compensation for the legal suppression of native vegetation. It aims to achieve this by strengthening the implementation of forestry legislation and recovering and increasing the stock of native vegetation through economic incentives for conservation and sustainable forest management.

- Brazilian NDC: Commitment to reduce its greenhouse gas emissions in 2025 by 37%, compared with 2005. Additionally, Brazil commits to reduce its emissions in 2030 by 50%, compared with 2005. Brazil's commitments also include a long-term objective to achieve climate neutrality by 2050.

2.5.2 Relevant Laws and Regulations Related to Worker's Rights (VCS, 3.18.2; CCB, G3.11)

Taking into account Brazilian Labor Regulation and Laws, the project is committed to comply with all the requirements which include worker rights and safety and employers' duties. All the staff will be notified about their rights and duties under any contract established.

The most important regulation and laws in Brazil applied to the employment components including types of workers, working hours, employment terms, statutory contributions and to the Severance Pay Fund (FGTS), Social Security and Pensions throughout project development, among others, correspond to Federal Constitution, Brazilian labor legislation, labor code, collective bargaining, and collective agreements. Other regulations including maternity and family leave rights include Law 11.770/2008 and Articles 137 to 142 of Decree 10854/2021.

2.5.3 Human Rights (VCS, 3.19)

All the project activities are structured through participatory scenarios with the community and are designed accordingly to national regulations and international human rights laws as described in section 2.3, 2.5.1 and 2.5.2. Project design and implementation, is not involved in harassment or discrimination, including discrimination based on gender, race, religion, sexual orientation or other conditions.

The project recognizes, respects, and promotes the protection of the rights of indigenous peoples, local communities and customary rights holders in line with applicable international human rights law, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples. The project development scheme and implementation processes, along with agreements between project proponents are part of the evidence that allow to demonstrate that project activities recognize, respect, and promote the protection of the indigenous communities' rights according to international conventions and national laws.

Also, the proponent's LATAM Code of Ethics ("Code") – which applies to all its employees, directors, advisors, service providers, agents, and suppliers, regardless of hierarchical level – has a main concern on the Principle of Inviolability of Personal Information and Human Dignity, which involves:

- respecting individuals, therefore, not tolerating any abuse of human rights and preserving the inviolability of information and personal data of clients, suppliers, employees, visitors, and any individuals that the Company may have access to in the course of its activities.

- not engaging in or being involved in any business that goes against human dignity.
- not discriminating against any individual based on race, color, sex, religion, or nationality.
- not engaging in activities that exploit child labor, human trafficking, or any other form of forced or involuntary labor.
- not tolerating any form of harassment.
- not dealing with any individuals engaged or involved in any violations of human rights.

2.5.4 Indigenous Peoples and Cultural Heritage (VCS, 3.18, 3.19)

Project design and all the related activities are developed considering the communities' ways and cultural identity, practices and heritage. The project was designed with the communities' participation, and general intervention strategy was analysed to ensure that all social safeguards were considered to guarantee community rights respect and recognition. Project activities and high conservation values are related with subsistence dynamics and cultural identity. While delivering new economic opportunities for the community, it is expected that the communities' gain stability for the people in the territory so opportunities to strengthen their cultural identity will come with the project activities.

The project preserves and protects cultural heritage as part of objectives. The proponents consider direct engagement with the traditional communities in the region as essential to project conception and management. The indigenous populations are the owner of the project. The REDD+ strategy has a cultural approach in all the components. Nevertheless, some activities are directed to improve specific cultural elements: strengthening traditional productive agriculture, improve indigenous languages education, promote traditional medicine, strengthen traditional government and social structure, dances and rituals. All forms of communication between proponents are guided on ethical principles and cultural practices, in a respectful manner, without infringing upon any history, ancestral heritage, or intrinsic culture of a peoples.

2.5.5 Statutory and Customary Property Rights (VCS, 3.18, 3.19; CCB, G5.1)

The land that contains the project area belongs to the local communities. Brazil's Constitution recognizes and protect indigenous rights including their right for "the lands they traditionally occupy," without any time limits or arbitrary cutoff dates, and states that it is incumbent upon the federal government to demarcate Indigenous territories and to protect them (Human Rights Watch, 2023). Prior, to the constitution, the Statute of the Indian (Law 6,001 of 1973) set the regulatory legislation regarding indigenous rights in the country, followed by law and decree on the demarcation of indigenous lands.

A set of documents presented are an important reference for the understanding of the main procedures for land regularization of indigenous lands and, indirectly, to contribute to the protection and management of indigenous territories. This include: Funai (National Foundation of Indigenous Peoples) recognizes and approves the studies of identification of indigenous lands and the Declaratory Ordinance is signed by the Minister of Justice, after analysis and approval of the documentation sent by the Funai, declaring the limits of the land, the permanent possession of the indigenous group and determining the demarcation. Then, Homologation Decrees are signed by the President of the Republic to confirm the physical demarcation of the land, carried out by Funai. Then the Registration in the Registry Office of Real Estate of the District of the situation of the indigenous land and the Registration in the Secretariat of Heritage of the Union of the Ministry of Finance and the map of the demarcated indigenous land may take place.

In 1997, a technical group was formed through Ordinance No. 1280/PRES/97 to carry out the identification and delimitation of the indigenous territory of Acupuri de Cima, and the work was complemented in compliance with the Ordinance No. 227/PRES/98. The Ministry of Justice through Ordinance No. 287 of April 13, 2000, declare permanent possession of the Kokama indigenous group, the Indigenous Territory of Acapuri de Cima, with an approximate surface of 19,400 ha (Nineteen thousand four hundred hectares), and approximate perimeter of 57 km.

For Estrela da Paz, Decree of July 3, 1995 from the president of the republic of Brazil approved the administrative demarcation by the National Foundation of the Indigenous (Funai), of the Indigenous Territory, located in the Municipality of Jutai, State of Amazonas, characterized by traditional and permanent possession of the Indigenous Tikuna Group, with an area of 12,876 ha and a perimeter of 53,56 km. Later Certificate 14/1997 contains the registration of this Indigenous Territory.

The technical Group constituted through Ordinance No. 651, of May 17, 2012 in continuity with Ordinance No. 1257/PRES, of December 5, 2002 started demarcation of the Indigenous Territory of Riozinho. In 2015, the Interim President of FUNAI, in accordance with Decree 1775/96, in view of the Process FUNAI/BSB No. 086020.042690/2013-10, and considering the Detailed Report of Identification and Delimitation, approved the conclusions of the identification and delimitation studies of the Indigenous Territory of Riozinho, of traditional occupation of the Kokama and Tikuna indigenous peoples, State of Amazonas, municipality of Jutai and Juruá, with an approximate area of 362,495 hectares and an approximate perimeter of 461 km. Ordinance 485 of April 22 of 2016 declares the Indigenous Territory.

2.5.6 Recognition of Property Rights (VCS, 3.7, 3.18, 3.19; CCB, G5.1)

In accordance with the provisions of chapter VIII of the 1988 Constitution of the Federative Republic of Brazil, in its articles 231 and 232; Decree No. 1775, of January 8, 1996; Ordinance MJ No. 14, of January 9, 1996, the lands of the Indigenous Territories (*Terra Indigenas*) that are given have the legal character of collective lands of indigenous groups, they are of collective property and not alienable, also imprescriptible and unattachable.

Consequently, areas that are assigned to a family group may be alienated, due to the dissolution of that group or for any other cause indicated by the internal regulations approved by the governance structure of the Indigenous community (*Aldeia*). In any case, the exercise of the preferential right of occupation or acquisition may only fall on another member of the respective community or, failing that, on other members of the ethnic group to which the beneficiary indigenous community belongs.

2.5.7 Free, Prior and Informed Consent (VCS, 3.18; CCB, G5.2)

Since 2020, indigenous people have started the creation of the Indigenous Environmental Guard of Amazonas, and in the statutes of its constitution, in May 2022, the carbon market is defined as part of its strategy to finance its conservation and cultural protection activities (see *Annex 9*, subfolder *Guardia Indígena*, document *Estatuto_Guardia_Indigena_Estrella_Paz.pdf* article 52 part IV).

Before the beginning of the project documentation and commercial relationship with private companies, the indigenous communities of Jutai decided to undertake the path of territorial protection and education as part of the strategy to contribute to their sustainable development and take advantage of the international financing mechanism REDD+. This initiative gave rise to the legal constitution of the Amazon Indigenous Environmental Guard (GAIAM) and the Differentiated Indigenous University of the Amazon Peoples (UIDPA) as part of the educational component. In order to develop their REDD+ project, the leaders sought the support of developing companies. Among various, they contacted Concepto Carbono and convened meetings to receive an explanation and present the work plans (including the distribution of benefits and scope of the project) (see *Annex 5, Carta Autorización ingreso Jutai_02072022.pdf*).

After this meeting, the communities held other autonomous spaces to socialize the partnership intention and conditions to develop their project. In this way the communities reviewed the initiative and took their decisions in a free, prior and informed consent to move forward with the project, and also to define the partner to develop it. After holding these spaces, the indigenous people convened an assembly (see *Annex 5, Carta Presupuesto Asamblea 102022_Jutai sin costo.pdf*) and invited the company to a new meeting (see *Annex 5, Carta Autorización ingreso Jutai_112022.pdf*), where the caciques and the members of the communities informed their will to work with this company and to start the collection and monitoring of their activities to participate in the carbon market with their REDD+ project. At that time they signed an agreement with the company to meet this objective, which was an expression of intention to work together and allows communities to decide to give up their intention without any penalties or fines (see *Annex 1*, letters of intention to develop the REDD+ project).

Under this agreement, multiple community workshops have been held in the aldeias (community settlements) of each indigenous territory, where members have identified the livelihoods, pressures, solutions and most important aspects to structure the REDD+ strategy in each territory, which will help to implement their life plans and improve the conditions of their inhabitants and reduce the risks of deforestation that have occurred historically.

This free, prior and informed consent is mentioned in the minutes of the meeting of the indigenous leaders (caciques) of 27 March 2024, paragraph 2, 5, 6, 9 and 10 (see Annex 5, *Ata_Caciques_Copiju_Journalistas 27032024.pdf*). This information can also be confirmed in the communication issued by the president of GAIAM (see Annex 5, *Comunicado_UIPA_GAIAM_REDD+.pdf* in paragraph 5), the document generated by the president of the Council of Indigenous Peoples of Jutai - COPIJU (see annex *Carta aberta COPIJU sobre REDD+_30102023.pdf*, in paragraph 2) and also in information presented to FUNAI (see Annex 5, *Carta COPIJU a FUNAI sobre REDD+_072023.pdf*, *v2_Presentación REDD+ Estrela Acapuzinho FUNAI_30112023.pdf* and *Carta_Caciques_Copiju_Funai 27032024.pdf*).

Description of process for obtaining consent	Summary: The community began the REDD+ project since 2021, with own resources and activities to protect their culture and territory. In July 2022 the communities invited Concepto Carbono SAS to explain the conditions to help them develop their project. After discussing among the members, in November 2022 the community invited again the company and agreed the conditions of a partnership to receive the technical and economic support to carry on with the project and look for its registration. Since then, more than 16 community meetings have taken place to design and document the project advances. In November 2023 a formal commercial contract was defined and approved during general assemblies in each territory. Afterwards, other events and activities have been held to keep going with the community initiative and ratified their consent to move forward in their project.
Outcome of FPIC process	The project proponents are the communities themselves who have been granted property rights on their lands by means of an FUNAI and the Ministry of Justice Ordinance. The community plan is to live in their territory and improve cultural an environmental protection. Thus, the Indigenous Territories (Terra Indigena) Acapuri de Cima, Estrela da Paz and Riozinho are freely pursuing and developing the ESTRELA ACAPUZHNO REDD+ project opportunities. Due to this, the Free Prior Informed Consent (FPIC) criteria are met precisely by the free and unfettered participation of the communities in the REDD+ project, which is based on the community intention,

activities and efforts. The project development has followed a thorough process of internal consultation and community approval, reflecting the community's own institutional governance structure and by-laws.

2.5.8 Benefit Sharing Mechanisms (VCS, 3.18, 3.19;)

Process used to design the benefit sharing plan

A benefit sharing mechanism has been designed through participatory workshops oriented to define the overall allocation of funds resulting from the commercialization of VCUs, always within the theory of change framework, so that the project objectives can be attained and sustained over time.

Since the first meeting between the proponents of the project in July 2022, a proposal of benefit sharing mechanism was presented to the community members and leaders, for analysis and comments (see Annex 5, documents *Carta Autorización ingreso Jutai_02072022.pdf* and *Proyecto REDD+ COPIJU_20072022.docx*). After this presentation, the communities discussed the proposal, and in November 2022, during a general assembly, they approved the overall benefit sharing scheme (see Annex 1, letters of intention to develop the REDD+ project). After this meeting, more than 12 community workshops were carried to define the project strategy and investment priorities (see Annex 7). All community members were invited to these spaces. The decision-making followed the own government structure, where the general assembly is highest instance to make any decision that relates with their community and the caciques are the chiefs of each *aldeia* within the territory. During the workshops the people had the responsibility to describe their own participation and governance structure, and these were used to define the project activities and benefit sharing scheme (see Annex 7, subfolder Workshop 2, archives *Taller_2_Gobernanza*; subfolder Workshop 3, *Taller_3_Distribucion_Presupuesto* and *Taller_3_Cronograma Actividades*).

Summary of the benefit sharing plan	The bases for the benefit distribution mechanism are the existing communities' guidelines defined during the workshops. Within the REDD+ framework, the benefits distribution mechanism is designed to address the communities' priorities on how REDD+ revenues will be distributed. This process has been developed considering the protection and benefits to the most marginalized and vulnerable members of the communities', especially women, children, and elderly. Each community designed the project interventions and the distribution scheme of the benefits, as well as its prioritization (see Annex 7, subfolders <i>Workshop 2</i>, <i>Workshop 3</i> and <i>Workshop 4</i>),
Approval and dissemination of benefit sharing plan	The benefit sharing plan was revised, approved and ratified during community meetings. During the first meetings the general plan was presented. Afterwards, during another meeting an intention letter to formalize the proponent's commercial relationship was approved. Then 12 workshops with community members gathered their intentions and integrated their specifications and realities in the project design, and a general assembly in each territory was sustained to present and approve the final project design and ratified project benefits distribution (see Annex 7, subfolder <i>Workshop 4</i>). According to the benefit sharing plan, detailed in the commercial contracts (see Annex 2), all project investments and carbon credit sales will be managed in a transparent manner, through bank entities and all finance information is going to be presented and the community leaders and members will have access to it.

2.5.9 Property Rights Protection (VCS, 3.18, 3.19; CCB, G5.3)

The project proponents have agreed that the project boundaries will be protected and patrolled accordingly to approved governance and monitoring activities and in accordance to the indigenous ethno-development plan.

The project activities do not involve removal or relocation of property rights of the territories belonging to the Indigenous Territories (*Terra Indigenas*) Acapuri de Cima, Estrela da Paz and Riozinho.

2.5.10 Illegal Activity Identification (VCS, 3.19; CCB, G5.4)

Illegal activities identified correspond to:

- Illegal Mining: Generate new economic activities and strengthening of environmental management to reduce the need for mining activities.
- Illegal fishing and hunting: Carried out mainly by external actors entering the indigenous territories and by members of the community seeking for economic activities.
- Forestry and logging without any authorization and without any environmental management plans: Building capacities on environmental and forest management and governance are required. Strong monitoring on activities proposed by the project and community engagement to the project to avoid unauthorized logging activities within the project area are necessary.

2.5.11 Ongoing Disputes (VCS, 3.18, 3.19; CCB, G5.5)

Ongoing disputes over mineral resources are present in the territory of the Indigenous Territories. With the participation of local people and external agents, these resources are exploited and have demonstrated huge negative impacts over the natural resources, ecosystems, and social dynamics. Through the REDD+ strategy it is expected to increase economical and work possibilities and provide environmental education regarding traditional practices for the community members involved in this kind of activities. So, in the short and midterm, the objective is to reduce, in a progressive manner, illegal exploitation and bad production models of mineral resources within the territory.

Also, some people find in other illegal activities (logging, fishing and hunting) an opportunity to perceive economic resources, despite the ecosystem damage and degradation they generate. In this respect, the before mentioned REDD+ strategy to increase economical and work possibilities also address this issue and the expected outcome is to reduce illegal practices in the short and midterm.

2.5.12 Approvals (CCB, G5.7)

The project has been approved by the General Assembly and by the Governing Board of the Indigenous territories. No other permission by local or national authorities is required.

2.5.13 Double Counting and Participation under Other GHG Programs (VCS, 3.23; CCB G5.9)

2.5.13.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program, or any other form of community, social, or biodiversity unit or credit?

☐ Yes ☒ No

2.5.13.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

2.5.13.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

2.5.14 Double Claiming, Other Forms of Credit, and Scope 3 Emissions (VCS, 3.24)

2.5.14.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

2.5.14.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

2.5.14.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

2.6 Additional Information Relevant to the Project

2.6.1 Leakage Management (VCS, 3.11, 3.15)

The project is going to promote forest monitoring and sustainable productive activities in already cleared areas, as well as strengthen productive chains that contribute to regional development. This will help to reduce leakage since the members and general population may find an opportunity to integrate good productive practices. Community engagement and delivering income opportunities for the people, helps to control deforestation activity migration and reduce displacement of any land use change activity out of the project area.

2.6.2 Further Information

Not applicable.

3 CLIMATE

3.1 Application of Methodology

3.1.1 Title and Reference of Methodology (VCS, 3.1)

Project activities have been designed as part of the REDD+ project with the intention of reducing CO2 emissions from unplanned deforestation and degradation compared to baseline levels. As required by VM0007, the land in the project area consists of contiguous and discrete areas covered by forests meeting the definition of forest as defined by the government of Brazil. These areas were forests for a minimum of 10 years before the project start date. The project area would be degraded or deforested in the absence of the REDD+ project activity and the deforested and degraded areas are mining and mosaic agricultural areas.

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Methodology	VM0007	VM0007 REDD+ Methodology Framework (REDD-MF),	1.6

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Methodology	VMD0001	VMD0001 Estimation of carbon stocks in the above- and belowground biomass in live tree and non-tree pools (CP-AB)	1.1
Methodology	VMD0004	VMD0004 Estimation of stocks in the soil organic carbon pool (CP-S)	1.0
Methodology	VMD0007	VMD0007 Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation (BL-UP)	3.3
Methodology	VMD0010	VMD0010 Estimation of emissions from activity shifting for avoiding unplanned deforestation (LK-ASU)	1.2
Methodology	VMD0015	VMD0015 Methods for monitoring of greenhouse gas emissions and removals (M-MON)	2.2
Methodology	VMD0016	VMD0016 Methods for stratification of the project area (X-STR)	1.2
Methodology	VMD0017	VMD0017 Estimation of uncertainty for REDD+ project activities (X-UNC)	2.2
Tool	VT0001	VT0001 Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities	3.0
Tool		VCS AFOLU Non-Permanence Risk Tool (T-BAR)	4.2

3.1.2 Applicability of Methodology (VCS, 3.1)

Applicability conditions of methodology VM0007 and its associated modules are detailed in Table below.

Reference ID/Title	Applicability condition	Justification of conformance
VM0007 - REDD-MF – Section 4.3.1	Land in the project area has qualified as forest (following the definition used by VCS) at least 10 years before the project start date.	Condition satisfied. The project proponent has obtained satellite imagery from ten years before the project start date to demonstrate that the land in the project area qualified as forest in accordance with the Brazilian national definition of forest
VM0007 - REDD-MF – Section 4.3.1	If land within the project area is peatland and emissions from the soil carbon pool are deemed significant, the relevant WRC modules (see Table 3 must be applied alongside other relevant modules.	Condition satisfied. Project area is not peatland; thus, WRC must not be applied in this case. The project area includes natural forest considered to be important carbon sinks and the soil organic carbon which also in considered to be high.
VM0007 - REDD-MF – Section 4.3.1	Baseline deforestation and forest degradation in the project area fall within one or more of the following categories: • Unplanned deforestation (VCS category AUDD) • Planned deforestation/degradation (VCS category APD) • Degradation through extraction of wood for fuel (fuelwood and charcoal production) (VCS category AUDD)	Condition satisfied. Baseline deforestation and forest degradation fall in the category of unplanned deforestation.
VM0007 - REDD-MF – Section 4.3.1	Leakage avoidance activities must not include: • Agricultural lands that are flooded to increase production (eg, paddy rice); • Intensifying livestock production through use of feed-lots and/or manure lagoons.	Condition satisfied. The project does not include agriculture on flooded lands or intensification of livestock production.
VM0007 - REDD-MF – Section 4.3.2	Baseline agents of deforestation must: (i) clear the land for settlements, crop production (agriculturalist) or ranching or aquaculture, where such clearing for crop production or ranching does not amount to large scale industrial agriculture activities; (ii) have no documented and uncontested legal right to deforest the land for these purposes; and (iii) be either residents in the Reference Region for Deforestation	Condition satisfied. Agents of deforestation meet all the conditions required. See section 2 and 3

Reference ID/Title	Applicability condition	Justification of conformance
	or immigrants. Under any other condition this methodology must not be used.	
VM0007 - REDD-MF – Section 4.3.2	If, in the baseline scenario of avoiding unplanned deforestation project activities, post-deforestation land use constitutes reforestation, this methodology may not be used.	Condition satisfied. The project does not promote reforestation activities as post deforestation land use in the baseline.
VM0007 - REDD-MF – Section 4.3.3	Avoiding Planned Deforestation / Degradation activities are applicable where conversion of forest lands to a deforested condition must be legally permitted.	Condition not satisfied. The conversion of forest lands to a deforested land is not legally permitted.
VM0007 - REDD-MF – Section 4.3.4	Avoiding Forest Degradation (Fuelwood/Charcoal): If degradation is caused by either illegal or legal tree extraction for timber, this methodology cannot be used.	Condition satisfied. Degradation of forest is caused by legal and illegal tree extraction for timber.
VMD001 – CP-AB	This module is applicable to all forest types and age classes. Inclusion of the aboveground tree biomass pool as part of the project boundary is mandatory as per the framework module REDD-MF.	Condition satisfied. Aboveground biomass is included in project emissions estimations.
VMD001 – CP-AB	Non-tree aboveground biomass must be included as part of the project boundary if the following applicability criteria are met (per framework module REDD-MF): Stocks of non-tree aboveground biomass are greater in the baseline than in the project scenario, and Non-tree aboveground biomass is determined to be significant (using the T-SIG module).	NTAGB not included in project area, as stocks changes are not significant.
VMD001 – CP-AB	Belowground (tree and non-tree) biomass are not required for inclusion in the project boundary because omission is conservative.	Below ground biomass (trees) is included, as allowed by the methodology.

Reference ID/Title	Applicability condition	Justification of conformance
VMD002 – CP-D	Dead wood shall be included if stocks are greater in the baseline than in the project scenario (in conformance with REDD-MF)	Dead wood not included in the project area, as carbon stocks changes are not significant.
VMD002 – CP-D	Dead wood shall be included if determined to be significant	Dead wood not included in the project area, as carbon stocks changes are not significant.
VMD0007 – BL-UP	The module must be applied to all project activities where the baseline agents of deforestation: (i) clear the land for settlements, crop production (agriculturalist), ranching or aquaculture, where such clearing for crop production, ranching or aquaculture does not amount to large scale industrial agri/aquaculture activities; (ii) have no documented and uncontested legal right to deforest the land for these purposes; and (iii) are either resident in the region or immigrants.	Condition satisfied. Module is applied to all project activities where the baseline agents of deforestation correspond to crop production (agriculture) of low and/or medium scale. There are no large-scale producers.
VMD0010 – LK-ASU	Activities subject to potential displacement are conversion of forest land to grazing lands, crop lands, and other land uses.	Condition satisfied. Project activities are conversion of forest lands to pastures, crops and settlements. See Section 2.1.11
VMD0010 – LK-ASU	The module is mandatory if module BL-UP has been used to define the baseline and the applicability conditions in module BL-UP must be complied with in full.	Condition satisfied. Module BL-UP is used, so this leakage module is applied.
VMD0012 – LK-DFW	Where fuelwood is collected or charcoal is produced for sale in regional or national market the market effects leakage must be considered using Module LK-ME.	Condition not applicable: fuelwood sale in regional or national markets is not considered in the project.

Reference ID/Title	Applicability condition	Justification of conformance
VMD0013 – E-BPB	This module is applicable to avoiding unplanned deforestation or degradation (AUDD), avoiding planned deforestation (APD) and avoiding degradation project activities, whether or not situated on peatland.	Condition satisfied. Project aims at avoiding unplanned deforestation, not situated on peatlands. See condition 2 and 3 above.
VMD0015 – M-REDD	Strata as defined in the relevant baseline modules are fixed and may not be changed without baseline revision.	Condition satisfied.
VMD0015 – M-REDD	The module is always mandatory. Without application of this module the methodology shall not be used.	Condition is satisfied: the module is applied.
VMD0015 – M-REDD	Where selective logging is taking place in the project case: • Emissions from logging may be omitted if it can be demonstrated the emissions are de minimis using T-SIG. • If emissions from logging are not omitted as de minimis, logging may only take place within forest management areas that possess and maintain a Forest Stewardship Council (FSC) certificate for the years when the selective logging occurs. • Logging operations may only conduct selective logging that maintains a land cover that meets the definition of forest within the project boundary. • All trees cut for timber extraction during logging operations must have a DBH greater than 30 cm. • During logging operations, only the bole/log of the felled tree may be removed. The top/crown of the tree must remain within the forested area. • The logging practices cannot include the piling and/or burning of logging slash	Condition not applicable. The project does not involve selective logging.

Reference ID/Title	Applicability condition	Justification of conformance
	Volume of timber harvested must be measured and monitored.	
VMD0016 – X-STR	In case of REDD, above-ground biomass stratification is only used for pre-deforestation forest classes, and strata are the same in the baseline and the project scenario. Post-deforestation land uses are not stratified. Instead, average post-deforestation stock values (eg, simple or historical area-weighted approaches are used, as per module BL-UP).	Condition satisfied. The pre-deforestation forest class consists in only one strata (tropical rain forest). Carbon stock average for post-deforestation land uses was estimated. See application of this module in section 3.2.1.
VMD0016 – X-STR	For peatland rewetting and conservation project activities this module must be used to delineate non-peat versus peat and to stratify the peat according to peat depth and soil emission characteristics, unless it can be demonstrated that the expected emissions from the soil organic carbon pool or change in the soil organic carbon pool in the project scenario is de minimis.	Condition not applicable. See condition 2 above.
VMD0016 – X-STR	In the case of peatland rewetting and conservation project activities, the project boundary must be designed such that the negative effect of drainage activities that occur outside the project area on the project GHG benefits are minimized.	Condition not applicable. See condition 2 above.
VMD0017 – X-UNC	This module is mandatory when using methodology REDD-MF.	Condition satisfied. This module has been used throughout the project design to estimate uncertainties.
T-ADD	AFOLU activities the same or similar to the proposed project activity2on the land within the proposed project boundary performed with or without being registered as the VCS AFOLU project shall not lead to	Condition satisfied. Project activities do not violate any applicable laws. They all aligned with government guidelines.

Reference ID/Title	Applicability condition	Justification of conformance
	violation of any applicable law even if the law is not enforced;	
Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities” T-ADD	Forestation of the land within the proposed project boundary performed with or without being registered as the A/R CDM project activity shall not lead to violation of any applicable law even if the law is not enforced.	Condition satisfied. REDD activities do not violate any Brazilian applicable law.
Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities” T-ADD	This tool is not applicable to small - scale afforestation and reforestation project activities.	Condition satisfied. REDD activities are not classified as small scale.

3.1.3 Project Boundary (VCS, 3.12)

Definition of the project boundary and identification of the relevant GHG sources, sinks and reservoirs for the project and baseline scenarios (including leakage if applicable) is shown in the following table.

Carbon pool	Included?	Justification
Aboveground tree biomass	Yes	Major carbon pool affected by project activities.
Aboveground non-tree biomass	No	Excluded considering that the land cover under the baseline scenario is not perennial tree crop.
Belowground biomass	Yes	Major carbon pool affected by project activities.
Dead wood	Yes	Carbon pool affected by project activities.
Litter	Yes	Carbon pool affected by project activities.
Soil organic carbon	Yes	Carbon pool affected by project activities.
Wood products	No	Conservatively excluded

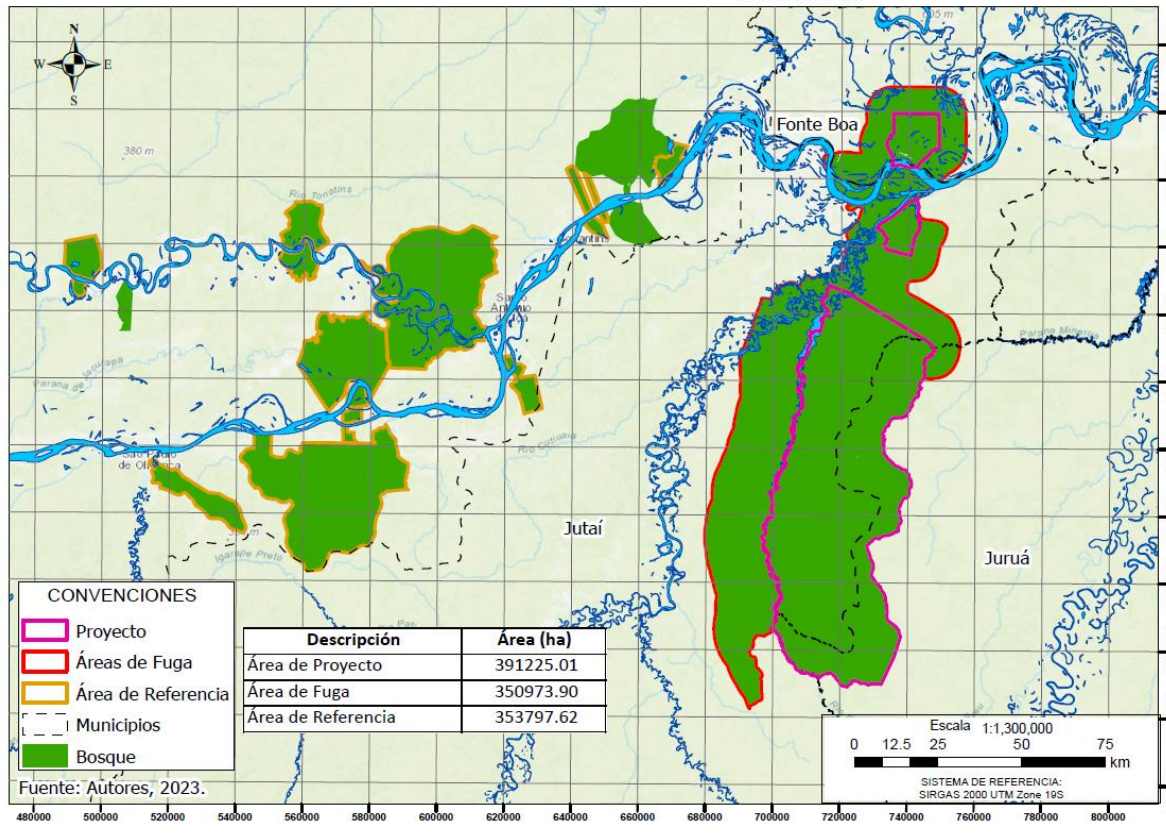
Source		Gas	Included?	Justification/explanation
Baseline	Deforestation	CO ₂	Yes	Aboveground and belowground biomass losses resulting from deforestation are included
		CH ₄	No	Not required for REDD projects per the VCS AFOLU requirements.
		N ₂ O	No	Not required for REDD projects per the VCS AFOLU requirements.
		Other	No	
	Biomass burning	CO ₂	No	Aboveground biomass losses as a result of fire are conservatively assumed zero
		CH ₄	No	Aboveground biomass losses as a result of fire are conservatively assumed zero
		N ₂ O	No	Aboveground biomass losses as a result of fire are conservatively assumed zero
		Other	No	Aboveground biomass losses as a result of fire are conservatively assumed zero
	Combustion of fossil fuels	CO ₂	No	Conservatively omitted
		CH ₄	No	Conservatively omitted
		N ₂ O	No	Conservatively omitted
	Use of fertilizers	CO ₂	No	Conservatively omitted
		CH ₄	No	Conservatively omitted
		N ₂ O	No	Conservatively omitted

Source		Gas	Included?	Justification/explanation
Project	Deforestation	CO ₂	Yes	Aboveground and belowground biomass losses resulting from deforestation are included
		CH ₄	No	Not required for REDD projects per the VCS AFOLU requirements.
		N ₂ O	No	Not required for REDD projects per the VCS AFOLU requirements.
		Other	No	
	Biomass burning	CO ₂	No	Aboveground biomass losses as a result of fire are conservatively assumed zero
		CH ₄	No	Aboveground biomass losses as a result of fire are conservatively assumed zero
		N ₂ O	No	Aboveground biomass losses as a result of fire are conservatively assumed zero
		Other	No	Aboveground biomass losses as a result of fire are conservatively assumed zero
	Combustion of fossil fuels	CO ₂	No	Conservatively omitted
		CH ₄	No	Conservatively omitted
		N ₂ O	No	Conservatively omitted
	Use of fertilizers	CO ₂	No	Conservatively omitted
		CH ₄	No	Conservatively omitted
		N ₂ O	No	Conservatively omitted

In each community settlement have been identified the productive lands and places that have been transformed before the beginning of the project. These areas are the priority zones to strengthen agricultural practices and implement improvements in production processes (see Annex 7, subfolder Workshop 3, archives *Taller_3_Acapuri_Cima Mapa_Actividades.pdf*, *Taller_3_Estrela_Da_Paz_Nova Esperanza_Mapa Actividades.pdf*, *Taller_3_Boa_Vista_Mapa_Actividades.pdf*, *Taller_3_Riozinho_Mapa Actividades.pdf*).

Regarding specific project boundaries, the following map presents detailed information:

MAPA DE BOSQUE EN EL PROYECTO JUTAI, ÁREAS DE FUGA Y DE REFERENCIA PARA EL AÑO 2021, ESTADO AMAZONAS, BRASIL.



Map 15 Project Boundaries, leakage belts, reference area

3.1.4 Baseline Scenario (VCS, 3.13)

The baseline scenario was selected in accordance with the procedure defined in the VM0007 methodology using the T-ADD. The application of the steps defined in the tool is presented in section 3.1.5 below. As a result, the baseline scenario corresponds to the continuation of the current practices.

Expansion of the agricultural frontiers is given by internal and external agents that clear primary forest areas for mining purposes and crops establishment. These practices have been ongoing in the project area and reference region over the last decade and are unlikely to cease without an effective and differentiate intervention. The proposed project activities require significant financial resources and long-term assistance in the project area. Without significant capital and high financial returns out of the proposed activities, the community will not be able to address the deforestation problem and maintain or implement the required actions to stablish and exercise adequate territorial governance and livelihoods. Based on historical evidence, local or regional government institutions are not able to allocate the technical support or economic resources to develop an integrated and midterm strategy that delivers productive opportunities to enhance communities' livelihoods and reduce deforestation.

3.1.5 Additionality (VCS, 3.14)

3.1.5.1 Regulatory Surplus (VCS, 3.14)

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

3.1.5.2 Additionality Methods (VCS, 3.14)

Additionality: Application of VCS Tool and CDM combine Tool

The project has used Tool VT0001 v.3.0 for Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) project activities and CDM Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities.

Step 0. Preliminary Screening base on the starting date of the A/R project activity

The project start date is November 08 of 2021. This is the date on which the community voluntary involved in the development of a REDD+ project through pedagogy activities.

Step 1: Identification of alternative land use scenarios to the proposed VCS AFOLU project activity.

Realistic and credible land use scenarios which would occur in the absence of the project were identified considering national policies and local circumstances, as well as the historic land uses that have occurred in the indigenous territories of Acapuri de Cima, Estrela da Paz and Riozinho. The communities have direct and indirect interactions with the forest present within their territory.

Sub-step 1a(a): Identify credible alternative land use scenarios to the proposed VCS AFOLU project activity

a. Scenario 1: Continuation of historical land use in the project area:

This scenario corresponds to the continuation of the conversion of forest to develop mining activities, expansion of the agricultural frontier and logging, which were recognized as part of the common practices identified in the municipalities of Fonte Boa, Juruá and Jutai (COPIJU, 2011). These activities were also identified during workshops held with the members of the Indigenous Territories (*Terra Indígena*) and communities (*Aldeias*), as stated in the problem tree assessment (see Section 2.3.4 and folder *Annex 7. Workshops*), and as a result of the cartographic analysis (section 2.2.2).

The communities establish subsistence crops and fishing practices as the main mechanism to guarantee food security for the territories (*Aldeias*) and to generate revenues from surpluses. Activities of illicit origin also represent significant changes in carbon content, mainly those related to mining without legal title and logging. Historical lack of government presence has shaped economic dynamics and cultural approach to productive activities, which facilitates land use continuation under current circumstances.

A multitemporal analysis was performed to identify the main changes in the land use within the boundaries of the Indigenous Territories. Land use activities have occurred historically and may continue in the future. As described in section 2.2.2, historical land use change since 2010 has demonstrated to be constant and the tendency reveals that new transformed areas appear each year and forest cover decreases. The results indicate that during 2010-2021 near 25% of the forest covers were transformed, mainly to mining extraction zones, mosaic of pasture crops and natural spaces, and mosaic of pastures with natural spaces which are related to the activities previously mentioned and which cause deforestation.

Logging and some of the mining activities that take place in the territory do not have permits issued by the environmental authorities (Secretary of the Environment of Jutai and the Brazilian Institute of Environment and Renewable Natural Resources IBAMA). and are considered illegal. The

economic dependence that the communities has towards these activities and the lack of government control limits the possibility to tackle deforestation causes in an appropriate approach.

b. Scenario 2. REDD+ project without certification of GHG emission reductions:

Considering that the Indigenous communities are responsible for the protection and proper management of the environmental resources within their territory (Federal Constitution, Decree 747/2012 and Law 12651). Therefore, the members of the communities should use their territory in a sustainable way and carry productive and land use activities that do not compromise the stability of the natural ecosystems or the provision of environmental services.

Under this scenario, members of the communities, with support of COPIJU and institutions and public funds, decide to contain deforestation drivers through initiatives that are compatible with nature and forest conservation. The community may exercise an effective territorial control and change productive activities that affect the environment with those that are compatible with ecosystems protection and sustainable use.

During the historic reference period, COPIJU has involve the community in different projects like the *Aldeias Project*, that seek to improve land use practices and local economies, such as a) implementation of a biodiversity and its environmental services information system, b) monitoring of the environmental resources, c) governance strengthening, d) cultural strengthening (see folder Annex 9. *Activities executed* and folder Annex 14. *Documents of interest*); in addition, national and regional policies for the Brazilian amazon have prioritized actions to implement alternative productive activities, conservation and restoration of degraded land as mechanisms to tackle deforestation trends and recover natural areas.

These projects seek to deliver new opportunities to improve land use practices and contribute to financial sustainability for local people. The general intervention schemes involve creating new opportunities to assist transition from unsustainable practices to legal and sustainable ones. Under this scenario the members of the Indigenous Territories (*Terra Indigena*) are not expecting or financing activities through revenue derived from VCU sales but relay on government assistance and financial support to start and maintain the activities.

Considering that the first limitation to promote these local changes is associated with economic resources availability, it is possible, though very unlikely, that the communities' members, COPIJU, and other local and regional government institutions articulate their efforts and gather the economic resources to support short- and medium-term investments that are required to support the interventions. Developing alternatives to generate income and change unsustainable land use practices, substitute illegal activities (mining and logging) and reduce deforestation in a significant way demands significant investments that usually have slow economic retroactivity. Any future intervention should offer economic and technical support during several years to obtain the expected results. Nevertheless, if the intervention strategy does not meet local economic and basic needs of people, the national experience has demonstrated that national policies have not been effective enough to tackle deforestation and achieve forest stability considering political influence (Sandy, s.f.)

- c. Scenario 3. Activities like the proposed project activities on at least part of the land within the project boundary resulting from legal requirements or observed similar activities in the region:

Before the start date of the project, Brazilian Amazon has been identified as a prior area that requires the implementation of programs and projects to improve community and biodiversity conditions. The Permanent Interministerial Working Group (GPTI) proposes and coordinate actions aimed at reducing deforestation in the Legal Amazon and the Action Plan for the Control of Deforestation in the Amazon (PPCDAm) intends that all conservation initiatives were backed by a large set of measures that had to be implemented and executed with the joint participation of the federal government, state governments, municipalities, networks of experts and civil society. Thus, the country has been taking a range of measures to tackle deforestation and improve governance in the Amazon, including legal reform, anti-corruption campaigns, improved remote sensing technologies and expansion of protected areas (IUCN, s.f.).

To implement the initiatives important economic resources are needed, which are limited for the government, which expects to be supported by international development financing institutions and non-governmental organizations to materialize these programs, and to allocate national resources. In this sense, the implementation of these initiatives is uncertain and represent voluntary actions, thus, they cannot deliver or benefit all the communities of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho considering the geographic scope and insufficient economic funds.

Sub-step 1a(b): Credibility of identified land use scenarios

Scenario 1: corresponds to the current land use scenario in the project area and prior to the project start date, thus is a credible scenario. Analysis, characterizations, and documentation of primary and secondary sources confirmed that the expected land uses are based on historical and common practices that take place in the project area and are likely to continue in absence of the project.

Scenario 2: is considered credible since some of the baseline activities are not in compliance with the national law and the proposed project activities offer a sustainable perspective for the people and their territory. It is also very important to recognize that the government has established priorities for the prevention, monitoring, control, and reduction of deforestation and forest degradation in the Amazon, as well as reduce forest conversion to other land use without legal permission and a sustainability approach.

Scenario 3 is considered credible because government institutions have established cooperation schemes like Amazon Fund with different agencies to implement regional strategies to reduce deforestation and forest fires, which will provide financial support to cities and territories that succeed in working to prevent, control, and reduce the rates of deforestation and degradation of the Amazon biome in their territories (Alves, 2023).

Sub-step 1a(c): List of credible alternative land use scenarios

1. Continuation of historical land use in the project area, see scenario 1.
2. Project activities without carbon credits revenue, see scenario 2.
3. Activities like the proposed project activities on at least part of the land within the project boundary resulting from legal requirements or observed similar activities in the region, see scenario 3.

Sub-step 1b(a): Consistency of credible land use scenarios with enforced mandatory applicable laws and regulations

Scenario 1: The indigenous communities own their territory and their resources and have administrative autonomy conferred by the national government. The communities can decide how natural resources are and must be used if regulations regarding their use are complied. Moreover, they must guarantee the environmental integrity of the territory and the conservation of the natural heritage.

The current scenario also involves activities that are not under national law. Due to the dynamics of this territory and the low presence of the government, activities such as illegal mining, illegal extraction of wood, have found conditions that favor their development, causing loss of forest and deterioration of the environmental conditions. Neither the communities nor the State have managed to stop the illegal use of natural resources. Communities face a scenario in which the main profitable activities are related to these economies and there are no legal alternatives that offer sufficient and stable income for families to satisfy their needs.

Scenario 2: This land use scenario is based on activities aiming to contribute to sustainable development, protect natural resources and limit practices that do not comply existing laws. The Indigenous Territories (*Terra Indigena*) have the responsibility to consider all regulations and legal requirements related with land and natural resources use in their planning instruments and local activities. In this sense, activities under this scenario meet all legal requirements and comply with mandatory regulations. Activities aimed at halting the environmental deterioration of the area and the loss of forests are of interest for the members of the communities and are aligned with the law and all development plans that government institutions have for the region.

Scenario 3: This land use scenario involves the action of government institutions, cooperants and other stakeholders which are required to operate under the law. Design and implementation of regional and local projects and programs is only feasible if they fulfill and addressed all legal requirements, while they also contribute to sustainable development objectives. This scenario also considers that people migrate from illegal practices to legal ones. According to these characteristics, this scenario also complies with mandatory laws and regulations.

Based on above, no land use scenarios have been removed.

Sub-step 1b(b): Outcome of Sub-step 1b

1. Continuation of current land use, see scenario 1.

2. REDD+ project without certification of emission GHG reduction, see scenario 2.
3. Implementation of activities like the proposed project activities on at least part of the land within the project boundary by regional programs and projects involving national institutions and cooperation agencies, see scenario 3.

Sub-step 1c: Selection of baseline scenario

The expansion of the agricultural and pasture frontiers is given by internal and external agents that clear primary forest areas for mining purposes and crops. This scenario has been ongoing in the project area and reference region over the last decade and is unlikely to cease without an effective and differentiate intervention.

The proposed project activities require significant financial resources and long-term assistance in the project area. Without significant capital and high financial returns out of the proposed activities, the communities will not be able to address the deforestation problem and maintain or implement the required actions to establish and exercise adequate territorial governance and livelihoods. Based on historical evidence, local or regional government institutions are not able to allocate the technical support or economic resources to develop an integrated and midterm strategy that delivers productive opportunities to enhance communities' livelihoods and reduce deforestation.

Also, non-governmental agencies or international cooperants programs and projects can't concentrate many resources in one location. These interventions involve a wider geographic region and investments at specific location are low so it is possible to increase the number of beneficiaries and a larger geographic scope (Brazilian amazon). The government has been promoting jurisdictional REDD+ projects in different areas of the country, where many national government institutions, cooperants and communities combine resources and efforts to develop an ambitious regional strategy to contain deforestation. Very low local impact and non-permanence of deforestation reductions were observed in the short term. But even this multistakeholder intervention had to rely on economic resources associated with payments based on carbon emissions reductions and participants could not allocate own investment resources to implement the activities to support local interventions. Moreover, national funds rely on political influence and international cooperation and have a wide scope. These important limitations regarding regional, national and other stakeholders support to implement REDD+ activities without carbon revenues, makes scenarios 2 and 3 unfeasible to be selected as baseline scenario.

Due to the lack of adequate governance and resources to limit historical land conversion within the project area, the most plausible baseline scenario is the continuation of previous and current land use (scenario 1).

Step 2: Investment Analysis

The VCS Additionality Tool requires that either step 2 (investment analysis) or step 3 (barrier analysis) be undertaken (or both). The barrier analysis was selected, and the analysis is completed below.

Step 3: Barrier analysis

Step 3a: Identify barriers that would prevent the implementation of the type of proposed project activity

Investment barriers

One of the components of the project (sustainable productive systems) seeks to generate surpluses for the beneficiaries of the project and thus contribute to the environmental and financial sustainability of the expected results with the project over time. Although these initiatives have a favorable economic potential in this region, the members of the indigenous communities do not have any facility to access credit mechanisms, nor do they have external investors that provide the resources to develop the initiatives. Collectively owned land cannot be used as guarantee for loans, due to legal requirements, and the communities do not have other relevant assets to establish liens on.

The establishment of production systems requires significant investments for the community and the lack of financing mechanisms is a limitation that has occurred historically in this region. Development or improvement of these productive activities could only be materialized with investment resources provided by the State or by national or international cooperants. There are no other financial sources available for communities to develop their production plans.

Activities that aim to strengthen governance, establish biodiversity monitoring systems, improve communities' infrastructure and basic sanitation will not generate any type of profitability and their implementation depends entirely on financial aid. Likewise, formulation, validation and verification of a REDD+ project have important cost, which could not be financed in the absence of income from the commercialization of VCUs.

The only financing mechanism identified to modify production practices with high environmental impact consists of framing the activities in a REDD+ project that allows the commercialization of VCUs, whose economic value allows to cover the required investments.

Social barriers

Market conditions and current practices represent a limitation for the substitution of practices that have high environmental impact. Historically, fishing, agricultural and mining production models have generated a progressive deterioration of natural areas. The general perception of the communities regarding these models is that are the only opportunity to generate income and there is resistance to immediate change because benefits related to other productive alternatives or more efficient land use models have not been perceived.

In the regional context, there is evidence of the implementation of financing models based on the commercialization of carbon certificates that developed production systems that displaced previous inefficient or high impact practices that compromised the stability of ecosystems (other

REDD+ Projects). This serves as evidence to demonstrate the viability of introducing changes in the productive arrangements of this region and thus, to modify the general perception that the community has regarding existing possibilities to achieve more sustainable models. Through the commercialization of the VCU derived from this project, economic funds will be available to strength the community decision on developing economic alternatives and enhance their governance, which makes it possible to reduce the population economic dependence on illegal activities and displace other unsustainable land use practices.

Step 3b: Show that the identified barriers would not prevent the implementation of at least one the alternative land use scenarios (except the proposed project activity):

Scenario	Investment Barrier	Social Barrier	Analysis and implementation barriers
1	No	No	None of these barriers prevent the continuation of the activities that have been historically developed in the territory
2	Yes	Yes	Investment: Without the availability of investment capital, there is no transition from current productive activities to those that do not affect forest cover and biodiversity. High economic investment is needed to address an integrated development strategy for Indigenous Territories. Social: Considering actual informal economic models and the absence of legal alternatives to guarantee sustainability, food security and improvement of living conditions for the communities, common practices that cause deforestation and natural resource depletion will continue. If the population does not have a legal financial mechanism, it is very unlikely that new land use practices can be establish and begin to offer employment opportunities and demonstrate short and midterm benefits so population can be progressively engaged. Resistance to change has taken root given the historical context and low technical assistance. Unsustainable production models are the most common and widespread, so it's difficult to change people production approach. However, there is a percentage of the population that really wants change and a REDD+ project without registration, although it may have resistance, can overcome the social barriers if examples of other successful REDD+ experiences in the region have introduced better land use practices and good results are observed.

Scenario	Investment Barrier	Social Barrier	Analysis and implementation barriers
3	Yes	Yes	<u>Investment:</u> The implementation of the Jurisdictional REDD+ Programs and policies to tackle deforestation depend on political commitment and they rely on international cooperation and national co-financing. In the scenario that these initiatives prosper, the investment resources directed to the Indigenous Territories conforming the Estrela Acapuzihno REDD+ Project would correspond to a small fraction these types of programs, as they can't concentrate investments in only one population, because they usually have a landscape approach or wide geographic boundaries to deliver fewer local benefits but to a greater number of communities. This would limit the participation of the Indigenous Territories of Acupuri de Cima, Estrela da Paz and Riozinho to a small part of the interventions. Also, regional and national government investments have been limited, but their planning instruments contained ambitious goals. Budget allocation has always been low for this area, so it's very unlikely to change in the near future. <u>Social:</u> may be overcome if local interventions offer an integrated approach and assistance. If continue support is offer to the community, new and improve productions models can be introduce and spread within the three indigenous territories.

The barriers identified above will not prevent the land use scenario defined as baseline scenario in Step 1, which consist of the continuation of land use conversion.

Step 4: Common practice analysis

Common land uses in the region include:

- Illegal Mining
- Illegal Logging
- Subsistence agriculture: cassava, banana, pineapple, corn, watermelon
- Pig farming
- Poultry farming (eggs and meat)

These activities may continue in the future, as they represent the main income source for the community. Not all the people have economic resources to develop their own productive systems, so they work in the economic activities that take place in the project zone, even if it involves deforestation

and ecosystem degradation activities. The economic benefits derived from the commercialization of the VCU of this project are a source of investment resources that allow the development of economic alternatives needed to counteract the activities that are deteriorating the forests and strength those that helps improve the community livelihoods.

The alternatives represent job opportunities and income generation, which reduces reliance on activities that cause deforestation in the territory. The region is comprised of lands titled in favor of Indigenous communities, who do not have the financial capacity to implement activities similar in scope to those presented here. Therefore, efforts to reduce deforestation through supporting local governance capacity, land-use planning and implementation and local capacity building, are not common practice in the region. Under the defined baseline scenario, the project area would be deforested at existing or historical rates in the absence of the project. This scenario would result in significant emissions of carbon dioxide compared to the expected mitigation from project activities under the with-project scenario.

No other planned or expected intervention will have the same impact on the Indigenous Territories of Acupuri de Cima, Estrela da Paz and Riozinho. Due to the lack of governmental and community resources, there are no similar development assistance projects or initiatives to reduce deforestation and forest degradation on this scale in the region except for other REDD+ Projects.

The conclusion of the step-by-step approach is the following: The region is predominately comprised of lands titled to Indigenous Territories, who do not have the financial capacity or government support to implement activities like those proposed in this project. Therefore, efforts to reduce deforestation and forest degradation through sustainable production systems, improving livelihoods conditions, supporting local governance capacity and land-use planning and monitoring the biodiversity on a community-based scheme are not a common practice in the region. As a result of this analysis, it is demonstrated that the project activities do not correspond to the baseline scenario, therefore, the project is additional.

Community and biodiversity additionality is stated in section 2.2.3 above.

3.1.6 Methodology Deviations (VCS, 3.20)

The project does not involve deviations from the methodology.

3.2 Quantification of Estimated GHG Emission Reductions and Removals

3.2.1 Baseline Emissions (VCS, 3.15)

The quantification of baseline emissions followed the VM0007 methodology modules BL-UP, X-STR, C-AB, E-BB. Following the module BL-UP the baseline deforestation rate was calculated from the Reference Region for Deforestation (RRD). Projected areas of deforestation are based on the simple historic approach and on the average deforestation rate observed in the reference region during the reference period, thus, the population driver approach was not used.

Definition of boundaries:

Spatial boundaries:

Reference Region for Deforestation (RRD)

A combination of different geospatial data was used to delineate the reference region. To define the minimum area of the RRD, the following equations were used:

$$\text{MREF} = \text{RAF} * \text{PA}$$

$$\text{RAF} = 7500 * \text{PA}^{-0.7}$$

At the beginning of the historical period, the project area consists in 393,409 ha, so the RRD should be at least 357,696 ha. The selected RRD had 357,893 ha of forest at the beginning of the reference period (year 2010), which meets the minimum required by the methodology.

The boundaries of the RDD were defined according to section 1.1.1.1 of the Module VM0007 BL-UP. For this, the following criteria was considered:

Main agents of deforestation: to identify the main agents of deforestation and establish a comparison between RRD and the PA, especially regarding agriculturalist proportion, a land-use approach was used. Considering the proportion of agricultural lands at the beginning of the reference period, it was possible to compare the magnitude of the extension that is dedicated to these economic activities in each area of interest. The total area dedicated to agriculture is almost distributed in the same proportions, so they represent similar threats respect future land use change. This was also confirmed through a post-deforestation land uses analysis in RRD (described in section 3.2.1.3) and the project zone (described in section 2.2.2), where mining activities, agriculture and wood extraction represent the main causes of deforestation.

Landscape factors: Forests in the PA and the RDD correspond to only one type of forest which is tropical rain forest, but with different spatial arrangement as can be seen in land use characterization using Corine Land Cover methodology level 4: High dense forest, floodable high dense forest, fragmented forest with pasture and cropland and fragmented forest with secondary vegetation. According to forest cover, at the beginning of the reference period, the project area represented 96% of the total project zone, and in the RRD it represented more than 95%, thus, corresponding to a 99% of similarity. In the following table are presented the comparison of soil type, slope classes and elevation classes between RRD and PA:

Criteria	Variable	Unit	RRD	PA	Comparison
Soil type	Hydrogenic clastic deposits (mixed alluvial) and organic deposits	% with respect to the total area	1.77%	0.00%	1.77%

Criteria	Variable	Unit	RRD	PA	Comparison
(suitable for the land-use practice used by the main agents of deforestation)	Hydrogenic clastic deposits, mixed alluvial	% with respect to the total area	4.63%	3.43%	1.20%
	Hydrogenic clastic deposits (clays and silts)	% with respect to the total area	1.70%	1.45%	0.25%
	Hydrogenic clastic deposits boulders, gravels and sands	% with respect to the total area	13.97%	3.99%	9.98%
	Hydrogravigenic clastic deposits (clays and silts)	% with respect to the total area	1.99%	1.25%	0.75%
	Hydrogenic clastic deposits (silts and sands)	% with respect to the total area	0.96%	2.96%	-2.01%
	Hydrogenic clastic deposits, mixed alluvial	% with respect to the total area	2.70%	0.00%	2.70%
Soil type Land-use practice	Agroforestry with permanent crops	% with respect to the total area	24.91%	11.62%	13.30%
	Agrosilvopastoral with permanent crops	% with respect to the total area	15.38%	1.95%	13.43%
	Semi-intensive transient crops in warm climates	% with respect to the total area	1.11%	0.01%	1.10%
Slope	Ratio slope <15%	% with respect to the total area	71.36%	57.05%	14.31%
	Ratio slope >15%	% with respect to the total area	28.64%	42.95%	-14.31%
Elevation classes	0 m - 100 m	% with respect to the total area	29.97%	12.85%	17.12%
	100 m - 200 m	% with respect to the total area	51.10%	32.35%	18.75%
	200 m - 500 m	% with respect to the total area	13.69%	33.23%	-19.54%
	500 m - 1000 m	% with respect to the total area	4.57%	11.81%	-7.24%
	1000 m - 1500 m	% with respect to the total area	0.62%	6.55%	-5.93%

Criteria	Variable	Unit	RRD	PA	Comparison
	> 1500 m	% with respect to the total area	0.05%	3.21%	-3.16%

Transportation networks and human infrastructure: the PA and RDD present similarities related to road density (64.35 m/km² and 72.08 m/km², respectively) and navigable river density, where PA and RDD have a stream density of 1,349.29 m/ km² and 1,224.48 m/ km², respectively.

As for the settlement density, the PA had 0.041 settlements/km² at the beginning of the reference period, while the RDD had 0.034 settlements/km², which corresponds to an 81% of similarity.

Social factors and policies: the project area is characterized for being inhabited by indigenous peoples who have settled alongside water bodies and considering social dynamics of the ethnic groups Tikuna, Kocama, Kambeba. The RRD and the PA fall within these municipalities. Also, all the areas are part of the RRD and the PA correspond to collective territories of indigenous peoples, which denotes similar ethnic composition. This also reflect the type of policies and regulations schemes that are found in the RRD and PA. The main defining criteria to define the boundaries of the reference region was to include Indigenous Territories (*Terra Indigena*), which correspond to the same collective land tenure of the project zone. By defining the reference region boundaries within this type of territories, it is ensured that similar systems of governance, regulations, social structure, and customs are the same to those in the project zone.

Exclusion of planned deforestation: areas of planned deforestation, such as mining titles, were excluded from the reference region boundaries.

Project Area

The project area corresponds to the area that is under threat of deforestation and where the project activities take place. It has a total area of 391,225 hectares of stable forest land at the start date of the project.

Leakage belt

The leakage belt was defined as an immediate vicinity of the project area. According to the project area the leakage belt must be at least 354,388 ha (90% of the project area). The leakage belt had 354,388 ha of forest at the beginning of the reference period (year 2010) and 350,973 ha at the beginning of the project, which meets the minimum required by the module VM0007, section 1.1.3.

The leakage belt was defined considering the accessibility and mobilization of the agents of deforestation identified in the project baseline, and considering the following criteria:

Landscape factors: Forests in the PA and the LK belt correspond to only one type of forest which is tropical rain forest, but with different spatial arrangement as can be seen in land use characterization using Corine Land Cover methodology level 4: High dense forest, floodable high dense forest, fragmented forest with pasture and cropland and fragmented forest with secondary vegetation. In the following table are presented the comparison of soil type, slope classes and elevation classes between RRD and PA:

Criteria	Variable	Unit	PA	LK	Comparison
Soil type (suitable for the land-use practice used by the main agents of deforestation)	Hydrogenic clastic deposits (mixed alluvial) and organic deposits	% with respect to the total area	0.00%	0.08%	-0.08%
	Hydrogenic clastic deposits, mixed alluvial	% with respect to the total area	3.43%	2.72%	0.71%
	Hydrogenic clastic deposits (clays and silts)	% with respect to the total area	1.45%	4.28%	-2.83%
	Hydrogenic clastic deposits boulders, gravels and sands	% with respect to the total area	3.99%	4.43%	-0.44%
	Hydrogravigenic clastic deposits (clays and silts)	% with respect to the total area	1.25%	0.00%	1.25%
	Hydrogenic clastic deposits (silts and sands)	% with respect to the total area	2.96%	2.60%	0.36%
Soil type Land-use practice	Agroforestry with permanent crops	% with respect to the total area	11.62%	8.25%	3.37%
	Agrosilvopastoral with permanent crops	% with respect to the total area	1.95%	1.85%	0.09%
	Semi-intensive transient crops in warm climates	% with respect to the total area	0.01%	1.27%	-1.26%
Slope	Ratio slope <15%	% with respect to the total area	57.05%	60.25%	-3.21%
	Ratio slope >15%	% with respect to the total area	42.95%	39.75%	3.21%
Elevation classes	0 m - 100 m	% with respect to the total area	1.53%	1.53%	-0.11%
	100 m - 200 m	% with respect to the total area	1.53%	1.53%	-4.17%
	200 m - 500 m	% with respect to the total area	1.53%	1.53%	4.44%
	500 m - 1000 m	% with respect to the total area	1.53%	1.53%	-2.46%
	1000 m - 1500 m	% with respect to the total area	1.53%	1.53%	0.62%
	> 1500 m	% with respect to the total area	1.53%	1.53%	1.68%

Transportation networks and human infrastructure: the PA and LK belt present similarities related to road density (64.35 m/km² and 75.02 m/km², respectively) and navigable river density, where PA and RDD have a stream density of 1,349.29 m/km² and 1,341.36 m/km², respectively.

As for the settlement density, the PA had 0.041 settlements/km², while the LK belt had 0.036 settlements/km², which corresponds to an 88% of similarity.

Social factors and policies: this region have been characterized by the presence of indigenous peoples. Being the indigenous territories of Acapuri de Cima, Estrela da Paz and Riozinho, located in the municipalities of Jutaí, Fonte Boa and Jurua, part of the amazon region of Brazil, where a major emphasis in controlling deforestation and forest fires is a priority. The PA and the Leakage belt are located within these municipalities. In addition, both areas present a similar social composition which is collective territories of indigenous people.

Temporal boundaries:

The following temporal boundaries were defined according to the section 1.2 of the module VM0007 ML-UP:

Start date and end of the historical reference period: the reference period started in 2011 and ended in 2021, which correspond to a 10-year-period prior to the project start date. In addition, to estimate the deforestation rate five points of time were analyzed.

Start date and end of the crediting period: The crediting period goes from November 08 of 2021, to November 07 of 2051, resulting in an expected lifetime of 30 years.

Date at which the project baseline will be revisited: the baseline will be renewed every 10 years after the start date of the project, in 2031 and 2041.

Analysis of historical deforestation:

The calculation of historical deforestation corresponds to the area of forest at the beginning and end of the historical reference period, and the number of hectares deforested for each interval of the historical reference period. The four intervals used for this analysis are: 2010-2013, 2013-2016, 2016-2019, 2019-2021. Gross deforestation was measured. Image interpretation was carried on according to the national guide for deforestation trends assessment, which corresponds to the National Reference Level methodology. All images used have less than 10% of clouds and the classification accuracy is 91%.

The estimation of annual areas of unplanned baseline deforestation in the RRD was identified through the historical average of annual deforestation during the historical reference period. No regression was applied because the acceptable forms that could be used didn't meet the significance

and r^2 expected values ($r^2 > 0.75$ and $p < 0.05$) (see folder *Annex 8. Emission reduction calculations*). So, the mean area deforested, in hectares per year, across the reference period was used and the same mean rate was used for each year of the baseline period. Annual average of forest lost during the reference period was 380 ha/year.

Year	Forest in RRD	Total Hectares loss during interval	Annual average of hectares loss
Forest 2010	357.893,7		
Forest 2013	357.344,9	549	183
Forest 2016	355.752,0	1593	531
Forest 2019	354.729,4	1023	341
Forest 2021	353.797,6	932	466

Deforestation in Project Area

To estimate the projected unplanned baseline deforestation in the project area the equation #5 of the BL-UP was used:

$$A_{BSL,PA,unplanned,t} = A_{BSL,RRD,unplanned,t} * P_{PA}$$

After this equation, the sum of the deforested areas was estimated using the equation #7 of the same tool:

$$A_{BSL,PA,unplanned} = \sum_{t=1}^{t^*} A_{BSL,PA,unplanned,t}$$

The result of these equations is that 415 hectares/year are expected to be deforested in the project area and during the project implementation (30 years) this number goes up to 12,117 hectares.

Deforestation in Leakage belt

To estimate the projected unplanned baseline deforestation in the leakage belt the equation #6 of the BL-UP was used:

$$A_{BSL,LB,unplanned,t} = A_{BSL,RRD,unplanned,t} * P_{LK}$$

After this equation, the sum of the deforested areas was estimated using the equation #8 of the same tool:

$$A_{BSL,LB,unplanned} = \sum_{t=1}^{t^*} A_{BSL,LB,unplanned,t}$$

The result of these equations is that 362 hectares/year are expected to be deforested in the leakage belt and during the project implementation (30 years) this number goes up to 10,872 hectares.

Pre- and post-deforestation carbon stocks

Pre-deforestation strata was identified according to module X-STR and for the total area subject to deforestation in the project area and leakage belt. The analysis regarding forest classification was developed using available spatial data such as a Digital Elevation Model (DEM), soil, land cover types, climate, life zone and topography. Through field survey along the project area, above ground biomass did not differ over 20% on average in respect to the mean value, thus indicating that the forest can be considered as one stratum, as it is stated on *X-STR Module*. These results showed that only one type of forest is identified in these areas, which corresponds to tropical rain forest. According to land use characterization using Corine Land Cover methodology level 4 (section 2.1.5 *Physical Parameters* and 2.1.6 *Social Parameters*), the project area presents High dense forest (87%), Floodable high dense forest (2,6%), Fragmented Forest with pasture and cropland (6,5%), Fragmented Forest with secondary vegetation (3,7%). In this sense, there is only one type of forest identified, but with different natural dynamics or different landscape arrangement. Nevertheless, these forests are tropical rain forests that share similar species composition and aboveground biomass volumes (as field survey demonstrates, found in file *Estudio Forestal*). Despite that there have been different degradation rates of the forest in the project area, due to fine wood extraction, selective logging or forest fragmentation, the forest remains in the category of tropical rain forest, according to the life zone and physical parameters of the area. Considering that the National Reference Level of Forest Emissions in Brazil uses a unique above ground biomass classification for all forests located within the Amazon Region, one forest strata for the project area that shares species, biomass and hydrometeorological parameters, is consistent and rightly supported.

In the project area and RRD, deforestation does not meet the mosaic configuration considering that there are forest areas that exceed 1,000 ha of coverage, so a location for future deforestation was carried. Thus, for the project scenario, considering that there is only one forest class in the project area, it is assumed that deforestation is to happen within this unique stratum. To estimate carbon stock changes and greenhouse gas emissions, step 4 of the BL-UP tool was followed.

Post-deforestation land uses were identified to estimate the long-term average carbon stocks based on current land uses in areas deforested during the historical reference period in the reference region. Carbon stock changes per strata was defined according to modules CP-AB and CP-S. Carbon pools excluded from the project have counted as zero (DW and L). Estimation of post-deforestation carbon stocks are assumed to be the long-term average stocks on the land following deforestation. The historical area-weighted average approach was used to determine the carbon stocks of these land uses. The area-weighted average of the mature carbon stock for each land use is calculated from the historical land-use change matrix and is assumed to represent all post-deforestation carbon stocks in that land use during the project term.

Due to the lack of local studies, post-deforestation carbon stocks of the identified land-use classes were obtained from IPCC (2006). The following table presents the results and the carbon stock average defined for post-deforestation land use during project implementation.

Post-deforestation Land Use in RR	% of land use post-deforestation	BA tree (tC/ha)	BA non tree (tC/ha)	BA (tCO2/ha)	BA weighted post-deforestation (tCO2)	BG (tCO2/ha)	BG weighted post-deforestation (tCO2)	Post-deforestation DW (tCO2/ha)	Post-deforestation Litter (tCO2/ha)	COS (tC/ha)	COS weighted post-deforestation (tC/ha)	COS weighted (tCO2/ha)
Clean grass	20,51%	0,0	3,1	11,37	2,33	18,19	3,73	0,00	0,00	63	12,93	47,42
Mining zones	10,21%	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0
Discontinuous Urban areas	4,25%	0,0	0,0	0,00	0,00	0,00	0,00	0,00	0,00	0	0,00	0,00
Wetland	30,62%	0,0	0,0	0,00	0,00	0,00	0,00	0,00	0,00	86	26,33	96,56
Cropland mosaic	34,42%	0,0	2,5	9,17	3,16	2,5	0,86	0,00	0,00	29	9,88	36,23
Total carbon content post-deforestation (tCO2/ha)					5,49		4,59	0,00	0,00			180,20

Above ground biomass estimations and emission factor

Above ground biomass of forest was estimated through a field survey, using Module CP-AB as reference. The field survey and data collection occurred in June 2021 using the Protocol for National and Subnational Biomass-Carbon Estimation in Brazil. Plot-based measurements were taken in forest areas corresponding to tropical rain forest. The protocol followed was: 40 circular plots of 15 m radius, located on a systematic random fashion over the project area, consisting of a seven series of L arrangements, with 6 plots located approximately every 50 m. All trees with DBH ≥ 10 cm were register. Heights for at least 40% of all plot trees with DBH was measured.

The final location of the forty sampling plots in the territory was the result of several exercises carried out using a geographic information system (GIS) and social and environmental conditions in the project zone.

Due to the complexity for determining wood density with direct field measurements, default values were used, for each species using different tropical wood densities databases reported by IPCC (2006, 2019), Chave *et al.* (2006) and (Zanne 2009), among others. The project applied the recommendations given by IDEAM (2011) to manage wood densities, which consists in the following procedure: first, a list of potential species in

the area was defined using regional species databases and local forest studies, where the scientific names and the common names are related. Then, during the field surveys, supported by local experts and agroforestry engineers' expertise, *in situ* identification of each species was carried out according to tree characteristics as wood color, appearance, smell, taste, tree flowers, fruits, resins and structure, among others. Afterwards, each identified species was found on the tropical wood density databases to obtain the targeted data. When no information for a specific species was found, or when the field identification was only possible to reach genus or family, the average density of the superior taxonomic level (genus or family) was used, according to the list of the potential species in the area. When a specimen could not be identified *in situ*, then the average wood density of all the species found at the same plot was used for this individual. In the case where several species or genus had the same common name (local name), the average density of each possible species or genus was used, so a single value was applied for carbon content estimations.

In order to estimate carbon stock changes due to deforestation, equations 15, 17 and 21 (BL-UP) were followed:

$$\Delta C_{ABtree,i} = C_{ABtree_{bsl},i} - C_{ABtree_{post},i}$$

$$\Delta C_{BBtree,i} = C_{BBtree_{bsl},i} - C_{BBtree_{post},i}$$

$$\Delta C_{SOC,i} = C_{SOC,bsl,i} - C_{SOC,PD-BSL,i}$$

The sum of baseline carbon stock changes is resumed in the next chart (equation 22 of Module BL-UP):

$$\begin{aligned} \Delta C_{BSL,i,t} = & A_{unplanned,i,t} * (\Delta C_{ABtree,i} - \Delta C_{WP,i} + \Delta C_{ABnon-tree,i} + \Delta C_{LI,i}) \\ & + \left(\sum_{t=10}^t A_{unplanned,i,t} \right) * (\Delta C_{BBtree,i} + \Delta C_{BBnon-tree,i} + \Delta C_{DW,i}) * \left(\frac{1}{10} \right) \\ & + \left(\sum_{t=20}^t A_{unplanned,i,t} \right) * (C_{WP100,i} + \Delta C_{SOC,i}) * \left(\frac{1}{20} \right) \end{aligned}$$

Forest SOC (tCO ₂ /ha)	Forest BA (tCO ₂ /ha)	Forest BB (tCO ₂ /ha) (*1/10)	BA + BB total tCO ₂ /ha	Post-deforestation SOC (tCO ₂ /ha)	Change in SOC stocks (1/20)	BA Post- deforestation tCO ₂ /ha	Total Carbon stock change post- deforestation (tCO ₂ /ha)
220,0	445	15,5	543	99,9	14,00	14,4	542,6

Calculation of net emissions is the result of the baseline unplanned deforested area times the average carbon stock changes within the project area, according to equation 25 and 26 (module BL-UP):

$$\Delta C_{BSL,unplanned} = \Delta C_{BSL,PA,unplanned} + GHG_{BSL,E}$$

$$\Delta C_{BSL,PA,unplanned} = \sum_{t=1}^{t^*} \sum_{i=1}^M \Delta C_{BSL,PA,i,t}$$

Baseline emissions from unplanned deforestation in the project area are estimated to be 612,697 tCO₂/year and remain constant during the accreditation period.

3.2.2 Project Emissions (VCS, 3.15)

Emissions that result from biomass burning, combustion of fossil fuels or nitrogen application (use of fertilizers) are not part of the project greenhouse gases sources as stated in section *¡Error! No se encuentra el origen de la referencia. ¡Error! No se encuentra el origen de la referencia..*

3.2.3 Leakage Emissions (VCS 2.5, 3.2, 3.6, 3.15, 4.3)

Ex-ante leakage estimation was done with tool *VMD0010 Estimation of emissions from activity shifting for avoiding unplanned deforestation (LK-ASU)*, which states that a conservative estimate must be produced to reflect potential GHG emissions in the leakage belt as a result of displaced deforestation agents.

In the case of the project, where no location of deforestation analysis has been conducted, the annual deforestation area in the leakage belt is given directly by *ABSL,LB,unplanned,t*. So, to estimate the projected unplanned baseline deforestation in the leakage belt, the equation #6 of the BL-UP was used:

$$ABSL,LB,unplanned,t = ABSL,RRD,unplanned,t * P_{LK}$$

The results are that 362 ha/year can be loss in the leakage belt during the accreditation of the project. This number is assumed to maintain stable during the implementation years.

After this equation, the sum of the deforested areas was estimated using the equation #8 of the same tool:

$$A_{BSL, LB, unplanned} = \sum_{t=1}^{t^*} A_{BSL, LB, unplanned, t}$$

After applying this equation, 10,872 ha are expected to be deforested in the baseline scenario during a 30-year period. Ex-ante Estimating Emissions from Leakage is done by multiplying the GHG emissions estimated for the baseline by a factor < 1.0 representing the % of deforestation expected to take place. The result is added to the estimated leakage emissions calculated following VMD0007.

In the project scenario, it has been determined that deforestation may increase 10% in the leakage belt due to displacement of deforestation activities. Calculation of net emissions is the result of the baseline unplanned deforested area in the leakage belt times the average carbon stock changes, according to equation 27 (module BL-UP).

This represents that an additional 37,3 ha/year may be lost in the leakage belt with the project implementation, and this corresponds to 54,975 tCO₂/year emissions, that are assumed to be constant during the whole project scenario, reaching 1,649,245 tCO₂ during a 30-year period.

Because activity shifting leakage is produced because of the same acting drivers of deforestation and degradation identified in the project area, the strategy for leakage management is consistent with the main project activities.

Leakage mitigation strategies for the project include the implementation of productive activities which improve socio-economic status, the strengthening of governance, and the establishment of forest patrols. The strengthening of local governance improves the ability of the indigenous communities to manage local participation in project activities that generate income (not from illegal activities), and to effectively distribute those economic benefits. Finally, the presence of forest patrols may discourage illegal activities in the leakage area that directly borders the project area boundaries. The predicted outcome of implementing these types of activities throughout the project zone is the successful mitigation of leakage by providing legal alternatives and incentives.

3.2.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals (VCS, 3.15, 4.1)

Describe the procedure for the quantification of estimated GHG emission reductions (reductions) and carbon dioxide removals (removals). Include all relevant equations.

For data and parameters monitored, use the estimated data/parameter values provided in Section 3.3.2 below. Document how each equation is applied in a manner that enables the reader to reproduce the calculations. Provide calculations for all key equations to allow the reader to reproduce the annual calculations for estimated reductions or removals. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all of the above in the emission reduction and removal calculation spreadsheet.

Complete the tables below by vintage period (calendar year). Note that the baseline or project emissions subtotals may be negative where sinks exceed emissions. Only specify the estimated VCUs for reductions and removals separately where the applied methodology provides procedures and equations to do so.

Provide the requested information using the table below:

State the non-permanence risk rating (%)	Example: 20%
Has the non-permanence risk report been attached as either an appendix or a separate document?	☐ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e, the Long-term Average (LTA).	
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☐ No <i>If no, provide justification.</i>
State, in tCO ₂ e, the expected total GHG benefit to date.	
Is the number of GHG credits issued below the LTA?	☐ Yes ☐ No <i>If no, provide justification.</i>

Uncertainty

A cumulative uncertainty of 13.6% was calculated using module X-UNC “Estimation of uncertainty for REDD project activities” (see folder *Cálculos emisiones*). This uncertainty results from the variability of data used to estimate the carbon pools for all strata. Uncertainty is expressed as the 95% confidence interval as a percentage of the mean of the carbon pools measurements. All carbon estimations uncertainty for the carbon pools (AGB, BGB, SOC) in each post-deforestation land use strata identified (i.e, tropical rain forest, cropland, grass) were assessed. As indicated by the tool, if the error is below 15%, then no further deduction should be made to the Net Emission Reductions adjusted by the buffer. This is the case for this project. The following equations were used:

$$U_{REDD-BSL,SS,i} = \frac{\sqrt{\sum_1^n (U_{REDD-BSL,SS,i,pool\#} \times E_{REDD-BSL,SS,i,pool\#})^2}}{\sum_1^n E_{REDD-BSL,SS,i,pool\#}} \quad (4)$$

Where:

$U_{REDD-BSL,SS,i}$	Percentage uncertainty in the combined carbon stocks and greenhouse gas sources in the REDD baseline scenario in stratum i (%)
$U_{REDD-BSL,SS,i,pool\#}$	Percentage uncertainty for carbon stocks and greenhouse gas sources in the REDD baseline scenario in stratum i (%)
$E_{REDD-BSL,SS,i,pool\#}$	Carbon stock or GHG sources in the REDD baseline scenario (t CO ₂ e)
i	1, 2, 3 ... M strata (unitless)

$$Uncertainty_{REDD-BSL,t*} = \sqrt{Uncertainty_{BSL,RATE,t*}^2 + Uncertainty_{REDD-BSL,SS}^2} \quad (6)$$

Where:

$Uncertainty_{REDD-BSL,t*}$	Cumulative uncertainty in REDD baseline scenario up to year t^* (%)
$Uncertainty_{REDD-BSL,RATE,t*}$	Cumulative uncertainty in the baseline rate of deforestation up to year t (%)
$Uncertainty_{REDD-BSL,SS}$	Total uncertainty in the combined carbon stocks and greenhouse gas sources in the REDD baseline scenario (%)
t	1, 2, 3, ... t^* time elapsed since the start of the project activity (years)

$$Uncertainty_{REDD-BSL,SS} = \frac{\sqrt{\sum_{i=1}^M (U_{REDD-BSL,SS,i} \times E_{REDD-BSL,SS,i})^2}}{\sum_{i=1}^M E_{REDD-BSL,SS,i}} \quad (5)$$

Where:

$Uncertainty_{REDD-BSL,SS}$	Total uncertainty in the combined carbon stocks and greenhouse gas sources in the REDD baseline scenario (%)
$U_{REDD-BSL,SS,i}$	Percentage uncertainty in the combined carbon stocks and greenhouse gas sources in stratum i in the REDD baseline scenario (%)
$E_{REDD-BSL,SS,i}$	Sum of combined carbon stocks and GHG sources in the REDD baseline scenario (t CO ₂ -e)
i	1, 2, 3, ... M strata (unitless)

The Non-permanence tool provides the procedures for conducting the non-permanence risk analysis and buffer determination required for REDD+ projects. To determine the number of buffer credits that shall be deposited in the AFOLU pooled buffer account, the overall risk rating shall be converted to a percentage. The outcome of the analysis was 10%, so the expected VCU of the project are as follows:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
08-November-2021 to 31-December-2021	37.568	11.271	3.371	2.293	20.634
01-January-2022 to 31-December-2022	233.165	69.950	20.921	14.229	128.065
01-January-2023 to 31-December-2023	279.693	83.908	25.096	17.069	153.621

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
01-January-2024 to 31-December-2024	326.221	97.866	29.270	19.908	179.176
01-January-2025 to 31-December-2025	372.749	111.825	33.445	22.748	204.731
01-January-2026 to 31-December-2026	419.277	125.783	37.620	25.587	230.286
01-January-2027 to 31-December-2027	465.804	139.741	41.795	28.427	255.841
01-January-2028 to 31-December-2028	512.332	153.700	45.969	31.266	281.397
01-January-2029 to 31-December-2029	558.860	167.658	50.144	34.106	306.952
01-January-2030 to 31-December-2030	605.388	181.616	54.319	36.945	332.507
01-January-2031 to 31-December-2031	651.916	195.575	58.494	39.785	358.062
01-January-2032 to 31-December-2032	691.658	207.497	62.060	42.210	379.891
01-January-2033 to 31-December-2033	697.474	209.242	62.582	42.565	383.085

Vintage period	Estimated baseline emissions (tCO2e)	Estimated project emissions (tCO2e)	Estimated leakage emissions (tCO2e)	Estimated buffer pool allocation (tCO2e)	Estimated total VCU issuance (tCO2e)
01-January-2034 to 31-December-2034	703.290	210.987	63.103	42.920	386.280
01-January-2035 to 31-December-2035	709.106	212.732	63.625	43.275	389.474
01-January-2036 to 31-December-2036	714.922	214.477	64.147	43.630	392.668
01-January-2037 to 31-December-2037	720.738	216.221	64.669	43.985	395.863
01-January-2038 to 31-December-2038	726.554	217.966	65.191	44.340	399.057
01-January-2039 to 31-December-2039	732.370	219.711	65.713	44.695	402.252
01-January-2040 to 31-December-2040	738.186	221.456	66.234	45.050	405.446
01-January-2041 to 31-December-2041	744.002	223.201	66.756	45.404	408.640
01-January-2042 to 31-December-2042	748.848	224.655	67.191	45.700	411.302
01-January-2043 to 31-December-2043	748.848	224.655	67.191	45.700	411.302

Vintage period	Estimated baseline emissions (tCO2e)	Estimated project emissions (tCO2e)	Estimated leakage emissions (tCO2e)	Estimated buffer pool allocation (tCO2e)	Estimated total VCU issuance (tCO2e)
01-January-2044 to 31-December-2044	748.848	224.655	67.191	45.700	411.302
01-January-2045 to 31-December-2045	748.848	224.655	67.191	45.700	411.302
01-January-2046 to 31-December-2046	748.848	224.655	67.191	45.700	411.302
01-January-2047 to 31-December-2047	748.848	224.655	67.191	45.700	411.302
01-January-2048 to 31-December-2048	748.848	224.655	67.191	45.700	411.302
01-January-2049 to 31-December-2049	748.848	224.655	67.191	45.700	411.302
01-January-2050 to 31-December-2050	748.848	224.655	67.191	45.700	411.302
01-January-2051 to 07-November-2051	748.848	224.655	67.191	45.700	411.302
Total	18.380.908	5.514.272	1.649.245	1.121.739	10.095.652

3.3 Monitoring

3.3.1 Data and Parameters Available at Validation (VCS, 3.16)

Data / Parameter	$\Delta C_{BSL,unplanned}$
Data unit	t CO2e
Description	Net greenhouse gas emissions in the baseline from unplanned deforestation
Source of data	Module BL-UP
Value applied	11,217,391
Justification of choice of data or description of measurement methods and procedures applied	See section 3.2 developed under Module BL-UP guide.
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	Regional Forest Cover / Non-Forest Cover Benchmark Map
Data unit	ha
Description	Map that shows the location of forest and non-forest areas in the Reference Region RRD at the beginning of the reference period).
Source of data	Satellite imagery used is adequate in terms of spatial resolution (less than 30 meters) and an appropriate scale (Landsat and Planet Scope)
Value applied	357,893
Justification of choice of data or description of	Satellite imagery used is adequate in terms of spatial resolution (less than 30 meters) and an appropriate scale.

measurement methods and procedures applied	
Purpose of Data	The images were used for all the purposed listed below: • Determination of baseline scenario • Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage baseline scenario • Calculation of leakage emissions
Comments	
Used in Equations	3

Data / Parameter	Project Forest Cover Benchmark Map
Data unit	ha
Description	Map showing the location of forest within the project area at the beginning of each monitoring period. The benchmark map will show the deforested areas at each monitoring event (Map of spatial limits)
Source of data	Satellite imagery used is adequate in terms of spatial resolution (less than 30 meters) and an appropriate scale (Landsat and Planet Scope)
Value applied	391,225
Justification of choice of data or description of measurement methods and procedures applied	Satellite imagery used is adequate in terms of spatial resolution (less than 30 meters) and an appropriate scale.
Purpose of Data	The project area forest benchmark map for year 2018 is used to: • Determine baseline scenario • Calculate baseline emissions • Calculate project emissions
Comments	
Used in Equations	3, 5, 7

Data / Parameter	Leakage Belt Forest Cover Benchmark Map
Data unit	Ha
Description	Map showing the location of forest within the leakage belt at the beginning of each monitoring period. The benchmark map will show the deforested areas at each monitoring event
Source of data	Satellite imagery used is adequate in terms of spatial resolution (less than 30 meters) and an appropriate scale (Landsat and Planet Scope)
Value applied	350,973
Justification of choice of data or description of measurement methods and procedures applied	Satellite imagery used is adequate in terms of spatial resolution (less than 30 meters) and an appropriate scale.
Purpose of Data	The leakage belt forest cover benchmark map is used to • Calculate project emissions • Calculate leakage
Comments	Non-applicable
Used in Equations	3, 6, 8

Data / Parameter	CF _j
Data unit	t C t-1 d.m
Description	Carbon fraction of biomass for tree species j
Source of data	Species- or family-specific values from the literature (e.g. IPCC 2006 INV GLs AFOLU Chapter 4 Table 4.3) shall be used if available, otherwise default value of 0.47 t C t-1 d.m. can be used.
Value applied	0.47 t C t-1 d.m
Justification of choice of data or description of measurement methods and procedures applied	Default value 0.47 t C t-1 d.m. can be used, or species specific values from the literature (e.g. IPCC 2006 INV GLs AFOLU Chapter 4 Table 4.3)

Purpose of Data	The Carbon fraction for dry wood was used to: • Calculate baseline emissions • Calculate project emissions • Calculate leakage
Comments	Where new species are encountered in the course of monitoring, new carbon fraction values must be sourced from the literature or otherwise use the default value.
Used in Equations	1 – Module CP-AB

Data / Parameter	D _j
Data unit	t d.m. m ⁻³
Description	Basic wood density in t d.m. m ⁻³ for species <i>j</i>
Source of data	Different information and studies sources were used
Justification of choice of data or description of measurement methods and procedures applied	According to good practice procedures, project should prioritize the use of local information when available, otherwise national or regional studies may be used.
Purpose of Data	The basic wood density was used for: • Calculate baseline emissions • Calculate project emissions • Calculate leakage emissions
Comments	Non-applicable
Used in Equations	1 – Module CP-AB

Data / Parameter	F _j (X,Y)
Data unit	BA tree ⁻¹
Description	Allometric equation for species <i>j</i> linking measured tree variable(s) to aboveground biomass of living trees, expressed as t d.m. tree ⁻¹
Source of data	Allometric models appropriate for tropical rain forest for aboveground tree biomass where AGB is aboveground biomass (Mg ha ⁻¹ dry weight), D is diameter (cm), H is height (m) and WD is wood density (g

	cm-3). * Saldarriaga's model for palm tree biomasses in all types of forests (2014).	
Value applied	Forest Stratification	Model
	Dense forest	(BA)=a+B1*Ln(D^2*H*ρ) where a=-2,413, b1=0,932, r2=0,948
	Palms	BA=139,48+7,308*H^1,133
Justification of choice of data or description of measurement methods and procedures applied	Applied to other REDD+ projects in the same regional area.	
Purpose of Data	The allometric equation for tree biomass was used to: • Calculate baseline emissions • Calculate project emissions • Calculate leakage	
Comments		
Used in Equations	1 – Module CP-AB	

Data / Parameter	R
Data unit	t root t ⁻¹ shoot
Description	Root to shoot ratio appropriate to forest type
Source of data	IPCC 2006
Value applied	0.37
Justification of choice of data or description of measurement methods and procedures applied	Default value for tropical rain forests

Purpose of Data	The root ratio is used to estimate below ground trees biomass, and was used to: • Calculate baseline emissions • Calculate project emissions • Calculate leakage
Comments	Where new species are encountered in the course of monitoring, new carbon fraction values must be sourced from the literature or otherwise use the default value.
Used in Equations	5,7,13 – Module CP-AB

3.3.2 Data and Parameters Monitored (VCS, 3.16)

Data / parameter	Project Forest Cover Monitoring Map
Data unit	ha
Description	Map showing the location of forest land within the project area at the beginning of each monitoring period. If within the Project Area some forest land is cleared, the benchmark map must show the deforested areas at each monitoring event
Source of data	Satellite images
Description of measurement methods and procedures to be applied	By using satellite images and remote sensing to map forest and non-forest covering the Project Area it would be determined if there are any variations in the forest in the project area.
Frequency of monitoring/recording	Every 2 years (or less) with images. Verification of deforested areas will be continually monitored in field by the project participants.

Value applied	N/A
Monitoring equipment	Computers and SIG software
QA/QC procedures to be applied	
Purpose of data	Calculation of project emissions
Calculation method	N/A
Comments	
Data / parameter	Project Forest Cover Monitoring Map
Data unit	ha
Description	Map showing the location of forest land within the project area at the beginning of each monitoring period. If within the Project Area some forest land is cleared, the benchmark map must show the deforested areas at each monitoring event
Source of data	Leakage Belt Forest Cover Monitoring Map
Description of measurement methods and procedures to be applied	ha
Frequency of monitoring/recording	Map showing the location of forest land within the leakage belt at the beginning of each monitoring period.

Value applied	Satellite imagery used is adequate in terms of spatial resolution (less than 30 meters) and an appropriate scale (Landsat and Planet Scope)
Monitoring equipment	By using satellite images and remote sensing to map forest and non-forest covering the Project Area it would be determined if there are any variations in the forest in the project area.
QA/QC procedures to be applied	Every 2 years (or less) with images. Verification of deforested areas will be continually monitored in field by the project participants.
Purpose of data	N/A
Calculation method	Computers and SIG software
Comments	

Data / parameter	ADistPA,t
Data unit	ha
Description	Area impacted by natural disturbance in the project area at time t; ha
Source of data	Satellite images
Description of measurement methods and procedures to be applied	Analysis of natural disturbance in project area like fires or permanent flooding.
Frequency of monitoring/recording	This will be monitored at least every two years.

Value applied	N/A
Monitoring equipment	Computer and software
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of the non-permanence risks
Calculation method	Interpretation of satellite images.
Comments	
Data / parameter	A _{sp}
Data unit	ha
Description	Area of sample plots in ha
Source of data	Recording and archiving size of sample plots
Description of measurement methods and procedures to be applied	The methodology for plot establishment and the results obtained are presented in folder and is based on the Protocol, that defines that 42 plots over 1,240 m ² is appropriate to deliver an accurate biomass estimation in natural forest.
Frequency of monitoring/recording	This will be monitored at least every ten years for baseline renewal.
Value applied	Metric line - Calculator
Monitoring equipment	N/A

QA/QC procedures to be applied	Calculation of baseline emissions
Purpose of data	Area estimation
Calculation method	
Comments	
Data / parameter	N
Data unit	ha
Description	Number of sample plots in ha
Source of data	Recording and archiving of number of sample plots
Description of measurement methods and procedures to be applied	The methodology for plot establishment and the results obtained are presented in folder. The methodology for plot establishment and the results obtained are presented in folder and is based on the Protocol, that defines that 42 plots over 1,240 m ² is appropriate to deliver an accurate biomass estimation in natural forest with an uncertainty that should be under 15% and is expected to be less than 10% with these number of plots.
Frequency of monitoring/recording	This will be monitored at least every ten years for baseline renewal.
Value applied	
Monitoring equipment	N/A

QA/QC procedures to be applied	Calculation of baseline emissions
Purpose of data	Counting plots
Calculation method	
Comments	
Data / parameter	DBH
Data unit	cm
Description	Diameter at breast height of a tree in cm
Source of data	Field measurements
Description of measurement methods and procedures to be applied	Measured at 1.3 m. above ground, when possible.
Frequency of monitoring/recording	This will be monitored at least every ten years for baseline renewal.
Value applied	Metric line
Monitoring equipment	Protocol for field measurement
QA/QC procedures to be applied	Calculation of baseline emissions
Purpose of data	Field measure at breast high

Calculation method	
Comments	
Data / parameter	H
Data unit	m
Description	Total height of tree
Source of data	Field measurements
Description of measurement methods and procedures to be applied	Total height measure
Frequency of monitoring/recording	This will be monitored at least every ten years for baseline renewal.
Value applied	Metric line
Monitoring equipment	Protocol for field measurement
QA/QC procedures to be applied	Calculation of baseline emissions
Purpose of data	Field measure of total height
Calculation method	
Comments	

3.3.3 Monitoring Plan (VCS, 3.16, 3.20)

The monitoring plan follows the requirements for a REDD+ project according to VCS Standard VM0007, as well as those stated in CCB Standard. Data, parameters and procedures to carry out carbon accounting described in sections 3.3.1 and 3.3.2 are going to be the same during the monitoring period. All details regarding the process involved in the monitoring scheme are described in the annex *Procedimiento QC-QA Jutai.pdf*. Some important elements of the information management system of the project are described above.

Data control and activity recording:

To ensure that the GHG emissions estimations and activity data reflect the characteristics and activities of the REDD+ project in a consistent, precise, complete and transparent manner, a series of activities and procedures have been defined and are implemented during the monitoring period. In this sense, quality assurance and quality control (QA/QC) activities are an integral part of the monitoring process. The quality control activities correspond to the methods used to verify that the collection and acquisition of data and calculations, as well as GHG emissions estimations, uncertainties, information storing and reports development, are carried out correctly and reflect the reality of the project. These activities also involve technical reviews of emission sources, activity data, emission factors, and other parameters used for carbon accounting. The results of the quality control activities allow emissions estimations and uncertainty to be adjusted and corrected, as well as identifying opportunities for improvement.

The main quality control activity consists of validating the quality of the information sources associated with the emission sources. In this sense, all the data of the activities will come from primary information sources (workshops, surveys, field observations, direct records, among others), satellite images, land use interpretations and other national socio-environmental studies, mainly.

To guarantee the quality of the information, the people responsible for providing the information on each technical line gives any evidence to other members of the team, who oversee and verify the characteristics and quality of the information, in order to subsequently consolidate and prepare any type of monitoring report. Once the final versions of documents are defined, the Project Coordinator at the Management Mechanism is in charge of uploading the information in the project's online shared folders. The information on field activities is supported by clear and transparent evidence. The members of the community and the team of professionals that work in the field are the ones who prepare and present any progress made, and afterwards they share it with other members of the team who are in charge of report writing and submitting to third parties.

Regarding the process to review information, verifications consist of randomly validating the data used (according to the sample that the reviewer considers appropriate, which should not be less than 10% of the records) and compare them against the respective supports (sources of

information). In this step, any inconsistency that is detected must be corrected. If necessary, the person responsible of recording the information should be contacted and any concerns or discrepancies that may arise should be clarified.

After that, formulas in the calculation sheets and activity data consolidation are corroborated and it is verified that used information corresponds to the correct numbers and entries that were registered in the original records (primary information). Following, the results are compared against the maximum expected results, which allows to validate magnitudes and whether the numbers are within the expected range. In this way, transcription and parameterization errors, among others, are detected and adjusted. As a final result of these quality control activities, the project assures that the information declared in the monitoring reports is true and adequately supported.

All records and physical and digital files are in charge of the professionals that work as part of the project developer team. All documentation related to the project (forms, spreadsheets, geographic information, templates, reports and other evidence of implementation) is physically stored (if applicable) and digitized to be uploaded in the online digital archive of the project. All information generated during the development of the project must be kept for at least ten years. Each member of the team must provide the evidence, supporting documentation and final versions of the documents in digital format to the Management Mechanism, who is in charge of managing the digital archive of the project that rests on a web platform.

Responsibilities and competencies for monitoring activities:

Project Coordination: The Management Mechanism assumes leadership for the development of monitoring reports. Appoints and communicates to those responsible for each topic to supply and consolidate information on project activities, based on the defined data and parameters to be monitored. It contributes to verify the application of the principles of control and quality of the project. The person in charge of project coordination is responsible for managing the digital archive of the project that is stored on a web platform.

Geographic Information Systems professionals: they are responsible for obtaining the necessary geographic information to generate cartographic analysis and process data and parameters related to land cover and the changes observed during the monitoring periods. They keep and apply principles to reduce uncertainty in emission estimations and maintain quality control and quality assurance procedures. All documentation and files to support geographic information reports are delivered to the project coordination in digital format to be uploaded in the shared folders.

Social and environmental professionals: they assist implementation of project activities in the field and in the office. They oversee compilation of evidence regarding project progress and achievements in the territory, as well as facilitating the communication between the members of the community and the project coordination. They provide the information to generate monitoring reports, as well as documents and files to support data and parameters. All archives and final versions of the documents are delivered to the project coordination in digital format to be upload in the project shared folders.

Communities' members: they are active participants in planning and activities implementation in the field, as well as assisting the registration and compiling of evidence of activities development in the territories.

External consultants: they assist the professional's project team and are included according to the needs identified at each stage. They support the implementation of activities in the field, compile evidence and generate activities' reports, maintaining the quality and control of the information required by the REDD+ Committee. All supports and final versions of the documents are delivered to the project coordination in digital format to be uploaded in project shared folders.

Assessment and identification of project boundaries:

The correct identification of the geographical and operational limits of the project is essential to reflect the economic and operative reality of the initiative. The project is developed in legally constituted territories of indigenous communities and the activities to address and control deforestation drivers and contribute to sustainable development have been defined and agreed with the members of the communities.

The geographical limits remain fixed and have been determined with forest cover that has been stable during at least ten years before the start date of the project. The exact location of these forests is georeferenced and stored in digital files that guarantee the integrity of the information. The changes observed in these forests are the basis to determine project performance and activities impact and correspond to the information reported in the monitoring reports.

REDD+ activities include a series of elements that address different social and environmental aspects that constitute a comprehensive strategy to reduce forest loss. Each activity has its own indicators and procedures that offer an appropriate follow-up and allows impact assessment of project implementation.

Every time a project monitoring report is made, the team of professionals must validate that the project limits correspond to the same ones that are defined in the project design document. In the scenario in which there are changes in the geographical areas, either to increase or decrease the project area, or the activities proposed to stop deforestation are modified and this represents that project emissions change, these new elements must be incorporated as part of the project design and emissions accounting should be adjusted to reflect the new reality of the initiative. To adjust the baseline emissions or project emissions, all procedures and methodology requirements defined by the carbon standard to update the baseline must be followed.

Accuracy assessment and improvement opportunities:

All activity data must come from reliable sources such as good-resolution satellite images, records, logs, invoices, lists, templates submitting, receipts or contractual execution reports, among others. Special attention and care are always taken when dealing with any premises or assumptions that

may affect results or influence data consolidation and reporting of project activities, thus everything must be duly supported or demonstrated. The important thing is that the coordination of the project and the team in charge of the monitoring report have the technical and procedural certainty that the data and parameters used to report the project's emissions and activities have clear traceability, that calculations and results can be replicated and that they reflect the reality of the actions that have been implemented. Any errors that are detected by the members of the project must be corrected and the information must be adjusted accordingly.

During the process to consolidate the information for each monitoring report, the project coordination can identify the opportunity to improve the mechanisms for recording and managing relevant information. The improvements usually involve strategies aimed at strengthening and increasing the accuracy and reliability of the project activity data. Currently, the information sources used are robust and easy to obtain (images, official cartographic information files, attendance lists, records, spreadsheets, photographs, videos, contractual execution reports, among others). However, taking into account that the monitoring reports are going to be prepared approximately every two years, the project coordination must review the relevance and appropriate timing to adjust procedures and mechanisms that may facilitate consolidation and reporting tasks for each monitoring report, thus, maintaining the objective of delivering accurate and precise information in each project report.

3.3.4 Dissemination of Monitoring Plan and Results (VCS, 3.18; CCB, CL4.2)

Communication channels described in section 2.1.3 will be used to disseminate Monitoring plan and activities implementation reports. These channels include sending physical and digital documents and information to communities' members emails, print documents and place them in communities' headquarters' offices, distribute printed documents in community settlements, post documents online in the public domain of the CCB and VCS websites in accordance with each program's procedures and also will be transmitted orally during meetings and workshops with the communities. Also, during community general assemblies in which REDD+ activities planning, and approval takes place, executive reports concerning budget execution and activities' progress are going to be presented to all participants. In this last case, all members that are part of the Management Mechanisms are responsible to present this information.

Other dissemination means may be agreed or consider with the communities and can include channels such as especial printed reports, bulletins, infographic material, website and social media posts, delivered documents via email, among other ways.

Summaries of monitoring results will be socialized with stakeholders and communities' members prior to verification processes during workshops and local meetings.

The monitoring plans and results, and additional information will be available in Portuguese (local common language).

3.4 Optional Criterion: Climate Change Adaptation Benefits

The project does not seek to be validated to the Gold Level for climate change adaptation benefits.

4 COMMUNITY

4.1 Without-Project Community Scenario

4.1.1 Descriptions of Communities at Project Start (CCB, CM1.1)

Description of the three Indigenous Territories are presented in section 2.1.6 and 2.1.8. As mentioned above the 12 communities (Aldeias) conforming the 3 Indigenous Territories (Terra Indígena) correspond to indigenous population of the Kokama, Tikuma and Kambeba people (Povos).

As important cultural elements, home remedies represent a way of maintaining the beliefs and cultural knowledge of the indigenous group. Among the products used for this purpose stand out the copaiba peel, cocoa leaf, dried papaya flower, banana-apple milk, pariri eye, sucuba peel, chicory and capeba roots, holy grass, chilli pepper, carapanaúba, purple cotton sheet and the root or the pasting killer flower.

Likewise, language is an important cultural element for the communities, considering great adaptive capacity of these peoples, for whom miscegenation was one of the instruments of migration and territorial expansion. However, the Tikúna language is considered isolated, having no link with any other known linguistic family, being a tonal language, a characteristic not found among the indigenous peoples who live close.

Indigenous territoriality based on the low density of occupation, in the dispersion even in small settlements, in a low-impact technology based on the knowledge of the dynamics and respect for the given environmental conditions, in a process of adaptation to the limiting factors. The territorial management undertaken by the group is organized by community, holding some control of part of their territory, with areas of exclusive use in the vicinity of the settlements and areas of common use where there is an abundance of certain species that are preferably of the entire group.

Housing areas and family farm lots are characterized by ancestral occupation, which has been transmitted from generation to generation. Houses built follow a linear planning along the banks of the water sources, facilitating the access of each family unit.

Current conditions of the community reflect the historical absence of the State in the region. The community lack of social infrastructure, sanitation systems, education, and health facilities. Some of the problems presented include:

- Poor health system coverage, lack of medical centres and sanitation
- Lack of transportation means and roads and difficulties in transporting goods produced within the territory, as these territories are unconnected to the road network, surrounded by an extensive mostly intact forested area
- Need for recreational and cultural areas and proper community infrastructure
- Lack of social investment and government presence
- Lack of education facilities
- Lack of communication means

Despite of being located within a natural area rich in biodiversity Indigenous Territories (Terras Indígena) have environmental issues related to:

- Vulnerability to climate related events such as flooding, landslides, sedimentation, and forest fires
- Water pollution
- Lack of knowledge of proper environmental management (especially regarding solid and liquid wastes)
- Loss of fauna and flora

These issues are related to the principal economic activities carried out in the territory such as artisanal and mechanized mining, agriculture, and forestry.

Currently the community is willing to recover their tradition and cultural values that have been lost due to economic and social conditions.

4.1.2 Interactions between Communities and Community Groups (VCS, 3.19; CCB, CM1.1)

Interactions between the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, as well as the correspondent indigenous communities (*Aldeias*) rely in commercial and cultural activities principally. Through COPIJU initiatives cultural interactions, education initiatives and important matters regarding indigenous and territory aspects are shared. They are collaborating in the structuring and formulation of the project.

4.1.3 High Conservation Values (CCB, CM1.2)

For the definition of High Conservation Values in the project area, the biological, ecological, social, and cultural attributes that stand out for the goods and services they provide were identified. The project zone is located within habitats of ecological and cultural importance for the country and for local communities and traditional cultures. The defined HCV for the Indigenous Territories corresponds to:

High Conservation Value	Traditional production practices
Qualifying Attribute	Historical occupation over the territory has led to the establishment of traditional production practices, which have allowed the community to develop their own life project and define a unique cultural identity. It involves traditional agricultural and mining production that allows food stability and income generation for community members.
Focal Area	Traditional culture and environment

High Conservation Value	Traditional language
Qualifying Attribute	Protecting traditional language represent a way of maintaining and reinforcing the beliefs and cultural identity of the indigenous group, through oral and written traditions and to preserve the history of the communities. However, this knowledge is threatened by interactions with other cultures, Portuguese speaking peoples and the movement of communities' members seeking for economic opportunities.
Focal Area	Traditional culture and environment

4.1.4 Without-Project Scenario: Community (CCB, CM1.3)

Table 3. Problem flow analysis: community conditions in absence of the project

Project Intervention Areas	Contributing Factors	Contributing Factors	Direct Factors	Community Focal Issues
Support for productive activities: training, extension, crop yield improvement, processing facilities, value chain development.	Insufficient training opportunities and technical extension, capital.	Limited technical capacity to develop alternatives	Lack of sustainable livelihood and economic alternatives	Extreme poverty
		Insufficient access to financing for value added facilities, crop improvements		
		Limited infrastructure and connectivity with potential markets		
Integrated development planning, long-term investment in infrastructure and programs.	Limited access to alternative capital for infrastructure. Low capacity for integrated planning	Lack of funding transfers, government support	Lack of funding for community infrastructure, education and health	Insufficient community infrastructure and health & education programs
		Lack of integrated planning		
Management and administration training, environmental education, funding for salaried project personnel and implementation logistics.	Lack of capacity and resources for administration, project management. and environmental awareness	Natural resources utilized without long-term planning, to help meet short-term economic and capital needs	Unsustainable timber exploitation	Long-term loss of natural capital and associated ecosystem services
		Limited environmental awareness.	Unsustainable mining activities	
		External actors that take advantage of poor community conditions and limited management capacities	External actors through illicit mining activities that exert direct pressure on natural resources	

Poverty, insufficient infrastructure and programs and presence of external actors and decline in ecosystem services have been identified to be important focal issues for the project communities. The factors most directly responsible for these issues are lack of sustainable livelihood and economic alternatives, lack of infrastructure, health and education programs, and the unsustainable exploitation of natural resources to meet short-term economic and capital needs. In the absence of the project there are no plans, and no anticipation of future interventions that would be sufficient to address the various factors that contribute to these problems. Therefore, for example, a continued lack of training opportunities, technical

extension, capital for value-added facilities, and expertise for value chain development will result in limited to no improvements to poverty conditions, livelihood alternatives and unsustainable forest exploitation to meet short-term needs, in the baseline.

Similarly, a lack of resources and capacity for integrated planning, infrastructure and project management in the absence of the project will most likely result in limited or no future improvement to community health and education programs and infrastructure.

Within the implementation of the project, it is expected that well-being conditions significantly improve, especially regarding a sustainable development and economic opportunities for the community, that will also improve the social dimension.

Social empowerment, women enrollment and capacity building will be fundamental for the improvement of the living conditions, covering topics related to a proper environmental management, biodiversity recovery, strengthening of economic activities such as a sustainable agricultural culture.

By having a new income, the community will be able to develop community projects to improve education and health systems. Moreover, it is expected that ventures are supported.

4.2 Net Positive Community Impacts

4.2.1 Expected Community Impacts (CCB, CM2.1)

Community group	Indigenous Territories (Terra Indigena) of Acapuri de Cima, Estrela da Paz and Riozinho
Impact(s)	Improve capacity building and training
Type of benefit/cost/risk	The project will implement training and capacity building activities on topics such as sustainable productive systems implementation and management, forest management, project management, community governance, among others, though, will deliver direct benefits for the community.
Change in well-being	320 of community members improve their capacities by 2051.

Community group	Indigenous Territories (Terra Indigena) of Acapuri de Cima, Estrela da Paz and Riozinho
Impact(s)	Improve economic conditions and employment of community members
Type of benefit/cost/risk	Through sustainable productive activities the project will deliver direct and indirect income opportunities and new income sources. The project will offer economic resources to invest in community projects and productive initiatives that will benefit people and promote sustainable livelihoods. Specific activities will be defined with the community.
Change in well-being	160 members of the community improve their economic conditions by 2051.

Community group	Indigenous Territories (Terra Indigena) of Acapuri de Cima, Estrela da Paz and Riozinho
Impact(s)	Improved access to education services and educational infrastructure
Type of benefit/cost/risk	A direct benefit from project implementation is to provide facilities and scholarships for community members to access formal education (school, university). It is expected that access to education provides a basis to improve social and economic development, environmental awareness and access to job opportunities.
Change in well-being	480 members of the community have improved access or quality in education.

	At least 1 facility is improved and otherone is built by 2051.
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Community group	Indigenous Territories (Terra Indigena) of Acapuri de Cima, Estrela da Paz and Riozinho
Impact(s)	Improved access to and conditions of health services.
Type of benefit/cost/risk	The project will provide facilities and elements that facilitates access and increased quality of health services available to the community members. Improving health services delivers direct benefits to community welfare.
Change in well-being	320 members of the community improve their access or quality conditions of health services by 2051. At least 1 facility built or improved by 2051.

Community group	Indigenous Territories (Terra Indigena) of Acapuri de Cima, Estrela da Paz and Riozinho
Impact(s)	Improved household services and conditions
Type of benefit/cost/risk	The project will improve access to communications, clean water, sanitation and electric energy. It is also expected to improve housing infrastructure conditions. This represents a direct benefit to people well-being and facilitates social development.
Change in well-being	1,280 members of the community improve their housing conditions or services by 2051. Community members households are improved.

	Household services operation or infrastructure is improved.
Community group	Indigenous Territories (Terra Indigena) of Acapuri de Cima, Estrela da Paz and Riozinho
Impact(s)	Territorial Governance strengthened
Type of benefit/cost/risk	The project will directly support the construction of the Indigenous ethno-development Plan of the community, socialize it and strengthen the indigenous communities (Aldeias). Other community plans may be defined and implemented to strength social structure and land management. Community plans guide efforts, investment and actions according to community needs and social expectations, so this activity will benefit all the people. Activities will be defined with the community.
Change in well-being	At least 3 community plan are developed or updated by 2051 At least 3 community plan are under implementation by 2051 At least 480 members of the indigenous are trained by 2051
Community group	Indigenous Territories (Terra Indigena) of Acapuri de Cima, Estrela da Paz and Riozinho
Impact(s)	Cultural identity strengthening
Type of benefit/cost/risk	The project will directly support the strengthening of the cultural identity through the development of actions aiming to

	preserve traditions, costumes and sacred spaces. Activities will be defined with the community.
Change in well-being	Conservation and transfer of traditional knowledge and cultural identity

Community group	Indigenous Territories (Terra Indigena) of Acapuri de Cima, Estrela da Paz and Riozinho
Impact(s)	Recreation activities and sports infrastructure improved
Type of benefit/cost/risk	Project implementation will finance activities to provide or improve recreational or sport facilities, as well as elements and community infrastructure related with these issues. Recreation activities and sports promote social cohesion and community strengthening. Specific activities will be defined with the community.
Change in well-being	At least 3 sport facility built or improved by 2051. Recreation activities are developed.

Community group	Indigenous Territories (Terra Indigena) of Acapuri de Cima, Estrela da Paz and Riozinho
Impact(s)	Transportation condition improved
Type of benefit/cost/risk	Project activities will finance transportation means and infrastructure improvement, recognizing that these are key elements to support productive activities and community interaction with markets and processes outside the communities. Facilitate access and transportation services

	for people and goods increase community well-being and productive competitiveness. Specific activities will be defined with the community.
Change in well-being	Transport vehicles are increased or improved. Connectivity is improved. Transportation services improved.

Community group	Indigenous Territories (Terra Indigena) of Acapuri de Cima, Estrela da Paz and Riozinho
Impact(s)	Project planning and activities definition are made with community members
Type of benefit/cost/risk	Ensuring the participation of community members during project activities definition and planning represents a direct benefit to people because it guarantees the possibility to address their needs and interests. In this way the project gains community appropriation and commitment and facilitates implementation stages.
Change in well-being	Community members participate in project planning and activities definition.

4.2.2 Negative Community Impact Mitigation (VCS, 3.19; CCB, CM2.2)

It general terms, the expectations throughout the project development are positive for all the stakeholders. Nonetheless, potential negative impacts must be assessed in order to identify mitigation measures to diminish and neutralize them, if applicable.

Risk	Consideration	Mitigation measures
Leakage to outer communities	Increase of population migration to offsite spaces for the development of current productive activities; these activities include access to HCV' areas	Leakage management plan*
Income reductions	Community could perceive a reduction in revenue due to increased governance in the project zone. In this case is likely to perform deforestation activities in nearby areas.	Proposal of alternative livelihoods and constant adaptation of governance instruments.
Increase of cost of living		Import of daily commodities using available space on boats and road transport, when available.

4.2.3 Net Positive Community Well-Being (VCS, 3.19; CCB, CM2.3, GL1.4)

The project will generate important benefits for the community, related to the increase in welfare and quality of life conditions. The indigenous communities are considered to be poor, with restricted access to basic services such as energy, health care, education, transport, water and sanitation. The project will invest in social infrastructure, and in income generating productive activities, so that incentives for deforestation are reduced. It is expected that the project will help raise the economic conditions of the community while forest governance capacity is enhanced. The following are the main community related objectives:

- Improve quality of life through better basic service provisions: different investments in terms of direct social benefits, including transportation, communication, infrastructure, education, medical care, basic sanitation.
- Strengthen governance of territory: It is expected that communities strengthen land use planning and mechanisms to guarantee food security and protect forests and natural resources.
- Enhance livelihoods: Community beneficiaries will be trained and assisted to improve its productive activities (i.e.: agriculture, pisciculture, poultry farming), while committing to conservation of forest areas.

- Financial conditions: Cash flows from carbon revenues and from cash crops will help improve the financial conditions of the community over the long term.

Moreover, the project is generating net positive impacts on the welfare of women and ensuring that women participate in decision-making together with Indigenous Territories (*Terra Indígena*) of Acapuri de Cima, Estrela da Paz and Riozinho initiatives aiming to boost participation of women in all the process.

Considering that traditionally women are already involved in agricultural and fishing activities because they have access to land as mentioned in section 2.1.6, they are involved in productive activities and, due to the roles that women have been developing; they participate and have control over resources and are not excluded from decision-making processes.

4.2.4 High Conservation Values Protected (CCB, CM2.4)

As mentioned in section 4.1.3, Community HCV are related with subsistence dynamics and cultural identity. While delivering new economic opportunities for the community, it is expected that the communities' gain stability for the people in the territory so opportunities to strengthen their cultural identity will come with the project activities.

The project intends to improve ecosystem services and goods (forests, soils or land-use fertility, water sources) that help to achieve a sustainable livelihood with unpolluted water and a significant improvement in sanitation and health systems, along with forest protection and all the biodiversity that is present in the territory. Local fauna and flora are direct beneficiaries of natural ecosystem protection, including vulnerable and threatened species.

4.3 Other Stakeholder Impacts

4.3.1 Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.1)

Positive impacts on other stakeholder may include an increased commercial activity with other municipalities and major availability of goods (agricultural).

Potential negative impacts on other stakeholders are related principally to leakage to outer communities.

4.3.2 Mitigation of Negative Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.2)

As mentioned in section 4.2.2 the negative potential impacts on other stakeholders can manifest in leakage or an increase population migration outside the project area to develop productive activities that result in deforestation. Mitigation measures include promoting active participation of all community members in the project activities, deliver direct benefits to most of the local people and increase possibilities to generate incomes through sustainable productive initiatives.

4.3.3 Net Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.3)

The project activities are not anticipated to result in net negative impacts on the well-being of other stakeholders. In case any leakage situation is presented, it is expected to be controlled by the management plan and by a constant consultation within other stakeholders.

4.4 Community Impact Monitoring

4.4.1 Community Monitoring Plan (CCB, CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

As the project's implementer, the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, will coordinate and manage monitoring procedures over the project lifetime. The General Assembly is the prime decision organism, which is the maximum decision authority; however, due to its large number of members, it will unlikely be able to perform and coordinate the necessary procedures to conduct the Monitoring Plan. Thus, the Board of the Indigenous Territories is the most suitable organism to carry on this task, because of its reduced number of members and leading roles assigned to each one, based on the report provided by the Management Mechanism.

The Indigenous Territories board of Directors is composed of prepared and trained members who can make acute choices that benefit the entire community. Engagement of the local community for setting inventory plots and community training in MRV and administrative tasks is necessary.

Impacts of the project activities on the project zone communities will be closely monitored, reported, and evaluated according to the indicators presented below. Monitoring results will help to evaluate the progress of community-based activities, lessons learned and community inputs, and to implement adaptive management plan.

Community groups	Members of Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	People attending workshops for identification and prioritization of the required investments in productive systems, social investment, governance and monitoring
Unit	Number
Sampling methods	The number of community members who attend the training days for identification and prioritization of the required investments in productive systems, social investment, governance and monitoring is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Women attending workshops for identification and prioritization of the required investments in productive systems, social investment, governance and monitoring
Unit	Number
Sampling methods	The number of women who attend the training days for the management of the prioritized production systems is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	People attending workshops to strengthen the capacities in management of the prioritized production systems
Unit	Number

Sampling methods	The number of community members who attend the training workshops for the management of the prioritized production systems is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Women attending workshops to strengthen the capacities in management of the prioritized production systems
Unit	Number
Sampling methods	The number of women who attend the training workshops for the management of the prioritized production systems is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Households receiving technical assistance to improve the management of the prioritized production systems
Unit	Number
Sampling methods	The number of households receiving technical assistance to improve the management of the prioritized production systems is determined.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	People attending workshops related to transportation conditions

Unit	Number
Sampling methods	The number of community members who attend workshops or meetings related to transportation is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Women attending workshops related to transportation conditions
Unit	Number
Sampling methods	The number of women who attend workshops or meetings related to transportation is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Elements or activities that improve the transport conditions of people
Unit	Number
Sampling methods	The number of elements or activities that facilitate the mobilization of people is determined and the value is reported
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	People attending workshops related to the improvement of educational infrastructure, services and access to education
Unit	Number

Sampling methods	The number of community members who attend workshops related to the improvement of educational infrastructure, services and access to education is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Women attending workshops related to the improvement of educational infrastructure, services and access to education
Unit	Number
Sampling methods	The number of women who attend workshops related to the improvement of educational infrastructure, services and access to education is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Built/improved facilities for education purposes
Unit	Number
Sampling methods	The number of built/improved facilities for education purposes is determined and the value is reported
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Community members with access to education
Unit	Number

Sampling methods	The number of community members with access to education is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Women with access to education
Unit	Number
Sampling methods	The number of women with access to education is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Scholarships awarded to community members
Unit	Number
Sampling methods	The number of scholarships awarded to community members is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Scholarships awarded to women
Unit	Number
Sampling methods	The number of scholarships awarded to women is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	People attending workshops related to health services and access
Unit	Number
Sampling methods	The number of community members who attend workshops related to health services and access is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Women attending workshops related to health services and access
Unit	Number
Sampling methods	The number of women who attend workshops related to health services and access is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Health facilities built/improved
Unit	Number
Sampling methods	The number of health facilities that are built or improved is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	People with improved access to health services
Unit	Number
Sampling methods	The number of people that have improved the access to health services.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	People attending workshops on housing, communications, water and sanitation, and energy issues
Unit	Number
Sampling methods	The number of community members who attend workshops on housing, communications, water and sanitation, and energy issues is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Women attending workshops on housing, communications, water and sanitation, and energy issues
Unit	Number
Sampling methods	The number of women who attend workshops on housing, communications, water and sanitation, and energy issues is determined, and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Houses with energy or clean water or sanitation
Unit	Number
Sampling methods	Number of houses that have energy, clean water or sanitation.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Indigenous ethno-development plan developed/updated
Unit	Number
Sampling methods	The number of indigenous ethno-development plan developed or updated.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Indigenous ethno-development plan implemented
Unit	Number
Sampling methods	The number of indigenous ethno-development plan that are being implemented.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Actions and/or elements related to cultural identity developed/improved

Unit	Number
Sampling methods	Quantification of the number of actions and/or elements executed and/or improved to promote cultural identity conservation
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	People attending workshops to improve territorial management, and legal and financial capacities
Unit	Number
Sampling methods	The number of community members who attend the training days to improve territorial management, and legal and financial capacities is determined and the value obtained is reported.
Frequency	Annual

Community groups	Members of the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho, which includes the entire project zone beneficiaries (collective property)
Variable	Women attending workshops to improve management, legal and financial capacities
Unit	Number
Sampling methods	The number of women who attend the training days to improve management, legal and financial capacities is determined and the value obtained is reported.
Frequency	Annual

4.4.2 Monitoring Plan Dissemination (CCB, CM4.3)

Monitoring plan has been designed with the communities' participation and according to REDD+ strategy implementation and expected outcomes. Though, this plan is disseminated through the same communication channels described in section 2.1.3 and mentioned in section 3.3.4. These channels include sending digital files to community members through emails, print documents and place them in community headquarters' offices, distribute printed documents in communities' settlements, post documents online in the public domain of the CCB and VCS websites in accordance with each program's procedures and they will also be transmitted orally during meetings and workshops with the community. Also, during communities' general assemblies in which REDD+ activities are planned and approved; the monitoring plan is also presented. Other dissemination means can be used such as bulletins, infographic material, community websites and social media posts.

The monitoring plan has been socialized with stakeholders and communities' members during workshops and local meetings, prior to verification process.

Monitoring plan, as it is included in the present project description and the results are presented in the monitoring report, these documents will also be available in public domain on the CCB and VCS websites.

4.5 Optional Criterion: Exceptional Community Benefits

Brazil is the largest country in South America and one of the most populous countries in the world. In spite of having the strongest economy in Latin America, Brazil still has extremely high levels of poverty and inequality (Guedes, y otros, 2012). The project zone is in the municipalities of Jutai, Fonte Boa and Jurua, which belongs to the Amazon State. Most of this state corresponds to tropical jungle, where cities are clustered along navigable waterways and are accessible only by boat or plane. Amazonas is the Brazilian state with the biggest indigenous population. In the north of the country where the Amazon state is located, there are high levels of poverty but relatively low levels of income inequality. This is partially explained by a relatively high prevalence of smallholders in the rural areas and the unique quality of Amazonian urbanization where city dwellers maintain a strong link to rural areas. Rural poverty has a synergistic relation with environmental change. (Guedes, y otros, 2012).

Indigenous communities' part of the project own and have management rights over their territory as mentioned in section 2.5. Project activities will generate climate, community and biodiversity benefits that can be attributed to the communities of Acapuri de Cima, Estrela da Paz and Riozinho, as stated in sections 3.1.5, 4.2 and 5.2, considering that indigenous communities are considered as the natural guardians of forests and that project activities seek to strengthen communities culture and capacities for forest management, promoting communities permanence in the territory by improving living conditions that will positively impact climate, community and biodiversity in the project area.

4.5.1 Short-term and Long-term Community Benefits (CCB, GL2.2)

The project will provide short- and long-term well-being benefits for the community members, as it invests in social infrastructure needs (i.e., health, education, sanitation, housing, communication, transport, energy), and in income generating productive activities, according to the theory of change model jointly built with the project beneficiaries.

Net positive impacts can be seen in investments in short term productive activities (eg, cassava, acai, chicken farming) and leveraging in the mid to long term (eg, pisciculture). In addition, benefits for community members to be engaged in forest surveillance and environmental education, as well as for forest community monitoring groups represent additional work opportunities that generate family income.

In order to keep detailed traceability for community welfare, indicators are included not only in Theory of Change' Model but also in monitoring plan. Most of these indicators are focused on net changes in the community well-being throughout development of project activities; it is also intended to perform assessments of impacts in community well-fare and define, if applicable, the type and degree of impacts.

4.5.2 Community Participation Risks (CCB, GL2.3)

The project will provide short- and long-term well-being benefits for the community members, as it invests in social infrastructure needs (i.e., health, education, sanitation, housing, communication, transport, energy), and in income generating productive activities, according to the theory of change model jointly built with the project beneficiaries.

Net positive impacts can be seen in investments in short term productive activities (eg, cassava, acai, chicken farming) and leveraging in the mid to long term (eg, pisciculture). In addition, benefits for community members to be engaged in forest surveillance and environmental education, as well as for forest community monitoring groups represent additional work opportunities that generate family income.

In order to keep detailed traceability for community welfare, indicators are included not only in Theory of Change' Model but also in monitoring plan. Most of these indicators are focused on net changes in the community well-being throughout development of project activities; it is also intended to perform assessments of impacts in community well-fare and define, if applicable, the type and degree of impacts.

4.5.3 Marginalized and/or Vulnerable Community Groups (CCB, GL2.4)

Community Group

Members of the indigenous territories of Estrela da Paz, Acapuri da Cima and Riozinho, which includes the entire project zone beneficiaries (collective property)

Net positive impacts	According to workshops and definition of the REDD activities it is expected to improve the education, housing, transportation, productive activities of the community, and creation of jobs. See folder Annex 7. Workshops
Benefit access	According to the Project Management Mechanism one of the criteria used to define investments is to deliver benefits to as many community members as possible. First set of beneficiaries to participate at the first implementation stages were identified during the participatory workshops and meetings. These groups will help guide further interventions of the project in the future gradually involving additional participants. Social investments benefit larger groups of population.
Negative impacts	No negative impacts are expected to occur as a result of the project implementation.

4.5.4 Net Impacts on Women (CCB, GL2.5)

The project is generating net positive impacts on the welfare of women and ensuring that women participate in decision-making together with the Indigenous Territories (*Terra Indigena*) initiatives aiming to boost participation of women in all the process.

Traditionally, women are already involved in agricultural and fishing activities because they have access to land as mentioned in section 2.1.6. Since women have access to land, they are involved in productive activities and, due to the roles that women have been developing; they participate and have control over resources and are not excluded from decision-making processes.

Project implementation will help empower women to a higher extent, with regards to environmental education, as they are the best and closer channel to the young and vulnerable population. Social control is an important element to keep the REDD+ project in the long term, and women will play an important role through their natural social engagement and community-oriented skills.

4.5.5 Benefit Sharing Mechanisms (CCB, GL2.6)

Benefit sharing mechanism is presented in section 2.5.8 above.

4.5.6 Benefits, Costs, and Risks Communication (VCS, 3.18; CCB, GL2.7)

The social engagement process with local communities has included a participatory discussion of the positive and negative facts regarding the development of a REDD+ project. Discussion included issues such as volatility in prices, the need to comply with commitments (technical, social and economic facts), and on how to manage disagreements within community members and their level of engagement with the project, among others.

Initial workshops carried out gather expectations, expected benefits and risks with the development of the project. Recently, the consultant team has carried out activities with the community to expand and deepen the social engagement exercise with additional materials and techniques that build upon the successful experience of other REDD+ projects. During these socialization meetings, the project team explained the benefits, pros and cons, risks and costs of developing and merchandizing a REDD+ project. The process of socialization and participation has been supported by printed material, prepared according to the literacy level of beneficiaries.

During implementation, meetings will continue to take place to keep the community updated about the use of funds, targets met, and need for adjustment. REDD+ is a long-term process, that need continuous feedback and adjustment to ensure that the objectives are met. This requires a continuous risk assessment and updating of risk management strategies.

4.5.7 Governance and Implementation Structures (CCB, GL2.8)

The project governance and implementation structure is guided by the existing self-governance structures in the Indigenous Territories and communities.

Based on a formal definition by national constitution and later laws, Indigenous Territories (*Terra Indigena*) act as the collective land property and self-government basic structure. Land title has been given by the National Government to the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho; this means that the communities own the land and forests and is entitled to self-government and administration of the territory. Through a democratic process, the General Assembly of the Community appoints a Board for a pre-defined term (i.e. three year), led by a President or Chairman, and represented by a Legal Representative. Decisions are taken by the Governing Board and are endorsed by General Assembly meetings.

In regard to the legal framework, Indigenous Territories follow: (i) national laws issued by the government, and (ii) internal rules of procedures defined and established by each community, according to traditions, customs, norms and duties; accepted by all community members that share an identity; within the framework of the Brazilian Constitution, laws and norms.

The Local Board evaluates Community Assembly performance and discusses topics of general interest, throughout regular meetings along the year. If needed, special meetings are carried out according to internal community regulations. Decision-making process is in charge of General Assembly by majority of attendees or by consensus.

In accordance with governance structures of the Indigenous Territories, most aspects of the project will be approved by Community Assembly. The different normative or governance instruments, at minimum, must include: (i) management plans and economic development projects, and (ii) economic governance of land and resources within the territory.

Regarding to REDD+ projects, the law does not exclude or prohibit the possibility that Indigenous Territories are involved and/or constituted in business developments or income-generating mechanisms, which are aimed at produce and merchandize jointly and efficiently, goods and/or services to accomplish the economic needs of its members. The productive activities of the project will differentiate between the actions of individuals or families and those of the community, which have collective benefits, administered by the Indigenous Community (*Aldeia*) or Territory (*Terra Indígena*). Neither of these options precludes the creation of new associations or businesses to meet the specific interests of groups within a community so long as these activities are implemented on communal resources, such as forests.

4.5.8 Smallholders/Community Members Capacity Development (CCB, GL2.9)

Capacity assessment is made under indicators related to: (i) administrative capacity and (ii) organizational strength. It is important that throughout project development, a detailed evaluation of these indexes takes place, and thus determines the profits and gain derived from capacity building processes.

Moreover, it is important to develop different training processes and schemes, oriented to develop particular skills and abilities necessary for a specific community enterprise effort. The training processes (contents and tools) will be adapted to cultural conditions. Training areas shall focus on technical, financial and managerial activities, adjusted to project activities.

Under project' scope, similar trainings will be based on existing capacity and needs. Other capacity building efforts could include: (i) strategic project planning, (ii) development of results chains, (iii) monitoring, among others. Specific training to support project activities will include improved forest protection, creation of livelihood programs, crop diversification, and formal training; to perform tasks such as agriculture techniques, harvesting of natural harvest acai, product processing, and packaging. The project also will provide training associated with post-harvest management and to the Indigenous Territories Governing Board members in disciplines such as leadership, accounting, and management.

5 BIODIVERSITY

5.1 Without-Project Biodiversity Scenario

5.1.1 Existing Conditions (VCS, 3.19; CCB, B1.1)

The Brazilian Amazon state is well known for the vast biodiversity, full of endemic species and complex ecosystem functions and species dynamics. High rainfall, tropical conditions contribute to the notably high biodiversity (flora and fauna). Information for the amazon state and the Solimoes river is available, however, for the project area in the municipality of Jutai, Juruá and Fonte Boa, this has not been widely documented and detail information on biodiversity at local project-level scales is scarce in most of the cases.

Currently, within the Project zone there is no formal regulation or policy framework for the forest management. However, there are some adopted practices including the Amazon Indigenous Guard (forest rangers), capacity building for training community members to act as environmental agents within the indigenous communities, socialize and reinforce the importance of the sustainable use of natural resources (COPIJU, 2011).

Additionally, the forest governance despite of relying on political influence, has been strengthening in the country. They sought to identify and prioritize municipalities with high deforestation rates for the development of initiatives with the aim of preventing, controlling and combating illegal deforestation (González, 2018).

During the last ten years, the community has observed a significant reduction of wood species, reduction of vegetation and a drawdown of fauna species and the establishment of subsistence crops.

A vast variety of biodiversity is found in the project area. The following tables present some of the flora and fauna species found in these territories according to the databases of the IUCN redlists¹⁴:

Flora Species		
<i>Rhipsalis baccifera</i>	<i>Clusia viscida</i>	<i>Oenocarpus mapora</i>
<i>Cedrela odorata</i>	<i>Swartzia laurifolia</i>	<i>Guatteria megalophylla</i>
<i>Zannichellia palustris</i>	<i>Eumachia podocephala</i>	<i>Licania parviflora</i>
<i>Manihot brachyloba</i>	<i>Blakea rosea</i>	<i>Oryza alta</i>

¹⁴ <https://www.iucnredlist.org/es/search/list>

Flora Species		
<i>Echinochloa polystachya</i>	<i>Coussapoa nítida</i>	<i>Hevea spruceana</i>
<i>Licania hypoleuca</i>	<i>Alchorneopsis floribunda</i>	<i>Cordia sebestena</i>
<i>Unonopsis duckei</i>	<i>Bertiera guianensis</i>	<i>Eschweilera juruensis</i>
<i>Peltogyne campestris</i>	<i>Protium stevensonii</i>	<i>Cordia alliodora</i>
<i>Moronobea pulchra</i>	<i>Froesiodendron amazonicum</i>	<i>Scleria cyperina</i>
<i>Quiina cidiana</i>	<i>Trichilia mazanensis</i>	<i>Bocageopsis canescens</i>
<i>Heisteria amphoricarpa</i>	<i>Trichilia pleeana</i>	<i>Panopsis rubescens</i>
<i>Eugenia agathopoda</i>	<i>Parkia igneiflora</i>	<i>Virola decorticans</i>
<i>Sterigmapetalum obovatum</i>	<i>Parkia panurensis</i>	<i>Scleronema micranthum</i>
<i>Aniba williamsii</i>	<i>Diclinanona calycina</i>	<i>Varrolnia bullata</i>
<i>Nectandra pulverulenta</i>	<i>Macoubea guianensis</i>	<i>Solanum obliquum</i>

Birds

Common name	Scientific name
Lineated woodpecker	<i>Hylatomus lineatus</i>
Jabiru	<i>Jabiru mycteria</i>
Grey-winged Trumpeter	<i>Psophia crepitans</i>
Harpy Eagle	<i>Harpia harpyja</i>
Osprey	<i>Pandion haliaetus</i>
Great White Egret	<i>Ardea alba</i>
Scarlet Macaw	<i>Ara macao</i>
Roseate Spoonbill	<i>Platalea ajaja</i>
Rufous-sided Crake	<i>Laterallus melanophaius</i>
Collared Sand Martin	<i>Riparia riparia</i>
Nacunda Nighthawk	<i>Chordeiles nacunda</i>
Turkey Vulture	<i>Cathartes aura</i>
Common Barn-owl	<i>Tyto alba</i>

Common name	Scientific name
Picui Dove	<i>Columbina picui</i>
Sanderling	<i>Calidris alba</i>
Green-backed Heron	<i>Butorides striata</i>
Crested Oropendola	<i>Psarocolius decumanus</i>
Blue and yellow Macaw	<i>Ara ararauna</i>
Barn Swallow	<i>Hirundo rustica</i>
Giant Cowbird	<i>Molothrus oryzivorus</i>
King Vulture	<i>Sarcoramphus papa</i>

Mammals

The amazon state has a vast variety of mammals, some in the category of vulnerable, including:

Common name	Scientific name
Kinkajou	<i>Potos flavus</i>
Lesser long-tailed bat	<i>Choeroniscus minor</i>
Silky Short-tailed bat	<i>Carollia brevicauda</i>
Dark-fruit eating bat	<i>Artibeus obscurus</i>
Seba's short-tailed bat	<i>Carollia perspicillata</i>
Great Stripe-faced bat	<i>Vampyroides caraccioli</i>
Southern naked-tailed armadillo	<i>Cabassous unicinctus</i>
Silky anteater	<i>Cyclopes didactylus</i>
Southern tamandua	<i>Tamandua tetradactyla</i>
Jaguarundi	<i>Herpailurus yagouaroundi</i>
Ocelot	<i>Leopardus pardalis</i>
Giant Otter	<i>Pteronura brasiliensis</i>
Tayra	<i>Eira barbara</i>
South American Coati	<i>Nasua nasua</i>

Common name	Scientific name
Black-capped Capuchin	<i>Sapajus apella</i>
Collared Titi Monkey (endemic)	<i>Cheracebus torquatus</i>
Black-headed Night Monkey	<i>Aotus nigriceps</i>
Nine-banded Armadillo	<i>Dasypus novemcinctus</i>
Emilia's Short-tailed Opossum	<i>Monodelphis emiliae</i>
Green Acouchi	<i>Myoprocta pratti</i>
Giant Anteater (vulnerable)	<i>Myrmecophaga tridactyla</i>
Northern Tiger Cat (vulnerable)	<i>Leopardus tigrinus</i>
White-lipped Peccary (vulnerable)	<i>Tayassu pecari</i>
Giant Armadillo (vulnerable)	<i>Priodontes maximus</i>
Bald-headed Uacari (vulnerable)	<i>Cacajau calvus</i>
Lowland Tapir (vulnerable)	<i>Tapirus terrestris</i>
Nancy Ma's Night Monkey (vulnerable)	<i>Aotus nancymae</i>
Goeldi's Monkey (vulnerable)	<i>Callimico goeldii</i>
Tucuxi (endangered)	<i>Sotalia fluviatilis</i>

Amphibians:

Common Name	Scientific name
South American Common Toad	<i>Rhinella margaritifera</i>
Rusty Treefrog	<i>Boana boans</i>
Small-headed Treefrog	<i>Dendropsophus microcephalus</i>
Bolivian Bleating Frog	<i>Hamptophryne boliviana</i>
Tiger-striped Leaf Frog	<i>Callimedusa tomopterna</i>
Common Lesser Toad	<i>Rhinella granulosa</i>
Surinam Horned Frog	<i>Ceratophrys cornuta</i>
White-lined Leaf Frog	<i>Phyllomedusa vaillantii</i>
Bolivian White-lipped Frog	<i>Leptodactylus bolivianus</i>
Three-striped Arrow-poison Frog	<i>Ameerega trivittata</i>

Common Name	Scientific name
Red Snouted Treefrog	<i>Scinax ruber</i>
South American Bullfrog	<i>Leptodactylus pentadactylus</i>
Cane Toad	<i>Rhinella marina</i>
Spot-legged Poison Frog	<i>Ameerega picta</i>
Basin Treefrog	<i>Boana lanciformis</i>
Rufous Frog	<i>Leptodactylus fuscus</i>

Reptiles

Common name	Scientific name
Spectacled Caiman	<i>Caiman crocodilus</i>
Black Caiman	<i>Melanosochus niger</i>
Common Green Iguana	<i>Iguana iguana</i>
Massurana	<i>Clelia clelia</i>
Big-scaled Blind Snake	<i>Trilepida macrolepis</i>

Fish

Common name	Scientific name
Sardinia	<i>Microschemobrycon melanotus</i>
Pirarucu	<i>Arapaima gigas</i>
Pirillo	<i>Rhinodoras boehlkei</i>
Rio Negro Pygmy Anchovy	<i>Amazonsprattus scintilla</i>
Carriker's Anchovy	<i>Anchoviella carrikeri</i>
Jurua Anchovy	<i>Jurengraulis juruensis</i>
Tambaqui	<i>Colossoma macropomum</i>
Sorubim	<i>Sorubim</i>
Pacú	<i>Myleus pacu</i>

Flora and fauna Species	The project is not intended to negatively impact threatened, or endangered species. On the contrary, through forest management the project seeks to protect and improve current biodiversity within the project area
Habitats	No protected or endangered habitats are to be impacted within the project area nor due to project activities. The project sought for the forest governance and conservation

5.1.2 High Conservation Values (CCB, B1.2)

The following High Conservation Values were identified in the project area in accordance with the CCB Standard and represent strategic elements that deliver important services to community and its well-being as well as indicators of project achievement and ecosystem health indicators.

Flora

High conservation value	Tropical Rain Forest
Qualifying attribute	Tropical Rain Forest are essential to tackle climate change impacts and help the stabilization of global and local temperature. They are one of the most species-rich environments considering that they house several species of mammals, reptiles, insects, amphibians, and birds, in addition they have the largest living biomass. This type of forest also provides environmental services such as regulation of hydrological cycles, carbon storage, biological diversity, and reduction of soil erosion, as well as social services such as food security of communities settled in the influence area of the ecosystems. Protecting the Tropical Rain Forest in the project area contributes to protect endangered and endemic species and the amazon biome considered to be one of the most important biomes in the world.

Focal area	Tropical Rain Forest in the project area
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Fauna

High conservation value	Jaguar (<i>Panthera onca</i>)
Qualifying attribute	This important specie serves as a quality indicator of the ecosystem health and is an emblem for tropical forest conservation initiatives. As the biggest cat present in South America and the continuous habitat destruction that threats its population around its distribution range, its protection is a key element for ensuring feasible populations and protect many other species that live in the same type of ecosystem. The jaguar is an umbrella species and any efforts regarding its protection will benefit many others and requires vast areas of forest where endangered species are present.
Focal area	Project zone, dense highland forest

5.1.3 Without-project Scenario: Biodiversity (CCB, B1.3)

The assessment of the land use without-project scenario has also been carried out, considering how biodiversity conditions will be affected in the project zone. For this, biodiversity trends of the Amazon Region, changes in natural forest cover, and causal factors were analyzed.

Within the project area, biodiversity has been highly affected due to the economic activities carried out in the territory and the region that surrounds it. Direct biodiversity threats associated with forest loss, and their contributing factors are identified in the next table. Mainland forest, key conservation target for the project, are impacted by a number of direct threats from land-use conversion for crops, pastures and settlements, fuelwood and domestic timber extraction and mining activities. These pressures are exerted by local and external agents. Threatened species and forest dependent species in general, are at risk of further decline without significant intervention. Factors identified to be contributing to these declines are not being addressed in the absence of the project and there are no signs currently of them being addressed in any significant way in the future. For these reasons, it is projected that in the absence of the project, the amount and connectivity as well and the structure, function and available habitat provided by project area forests will continue to decline.

Table 4. Problem flow analysis: biodiversity conditions in absence of the project.

Project Intervention Areas	Contributing Factors	Contributing Factors	Direct Threat	Conservation Target, Focal Issues
Governance: Strengthening forest governance. Provide tools that contribute to the improvement of the management by the group of forest rangers, the monitoring of biodiversity and the control of deforestation. Development of the indigenous ethno-development plan. Follow-up and monitoring of biodiversity in the indigenous territories (<i>Terras indigenas</i>).	Limited local resources, legal framework, governance and capacity for land-use planning	Communities' laws do not limit forest exploitation	Conversion to crop land	Mainland forests (diminished forest area, fragmentation, degradation of forest composition, structure, function and habitat) Biodiversity HCVs
		Ineffective land-use planning	Conversion to settlements	
	Limited land-use implementation capacity	Limited permits issued for forest utilization, mining activities and natural resources management	Conversion for mining activities	
		Presence of external actors exerting pressure on natural resources	Logging for construction of infrastructure	
Support for productive activities, alternative livelihood	Limited alternatives to land clearing (for crops, pastures or settlements) and timber extraction (fuelwood)	Unmet economic, sustenance needs for families. High demand and price for minerals (gold) and so for mining activities	Fuelwood for domestic use	
Capacity building and training		Limited local capacity for environmental leadership project implementation		

5.2 Net Positive Biodiversity Impacts

5.2.1 Expected Biodiversity Changes (VCS, 3.19; CCB, B2.1)

Changes in biodiversity in both the project scenario and the projected without-project scenario have been analyzed, taking into account current deforestation trends in the project zone as well as recognition of biodiversity changes in the project stated by the communities. Project activities have been designed to prevent deforestation and forest degradation, and to achieve forest conservation in the long term. Forest conservation will lead to positive biodiversity changes, as the habitats and ecosystems will not only be protected, but also will be allowed to recover and regenerate biodiversity corridor dynamics that had been lost or affected with deforestation.

Biodiversity element	Flora and Fauna
Estimated change	Positive
Justification of change	The project objectives include the conservation and recovery of forests, which will enable species to be preserved, and recovered under a forest structure. REDD by definition is oriented to conserve and protect biodiversity associated with the forests. Monitoring in general is linked to the forest status monitoring. This applies both to fauna and flora which habitat and ecosystem dynamics will result from the conservation and restoration efforts.

5.2.2 Mitigation Measures (VCS, 3.19; CCB, B2.3)

The project will not have negative impacts on biodiversity. Sustainable agriculture activities will take place in already degraded lands prepared for this land use purpose and the project will not allow the expansion of the agriculture frontier. On the contrary there will be areas, currently degraded that will recover under passive forest restauration practice, and that will add to the forest areas in the long term.

By preventing deforestation and forest degradation to take place, the project will have positive impacts in the recovery and conservation of endemic and threatened species and will contribute to the conservation of the amazon biome.

5.2.3 Net Positive Biodiversity Impacts (CCB, B2.2, GL1.4)

A comparison of the project scenario conditions over time to projected biodiversity conditions that would have existed without the project makes possible to demonstrate the net-positive biodiversity impact.

Reduced deforestation and forest degradation will result in the improved maintenance of forest biodiversity and in many cases the recuperation (through natural regeneration) of forest biodiversity attributes as compared to conditions in the without-project scenario. Some of the identify improvements are:

- Increase in the amount (measured in hectares) of intact or semi-intact forest vegetation
- Continuation of functional biodiversity attributes
- Recuperation of landscape connectivity which contributes to processes including seed dispersion, reproduction, gene flow and associated capacity to adapt to climate change.
- Reduced change of land use for unplanned and illegal use activities
- Greater awareness of biodiversity and of the importance of land-use practices
- Local production and employment alternatives.
- Community members capable of project management

Together these changes correlate to a net biodiversity benefit for flora in the project zone and provide habitat conditions which are conducive to the well-being of a greater number of faunal species than would be the case in the without-project scenario.

5.2.4 High Conservation Values Protected (CCB, B2.4)

As mentioned, the project does not intend to negatively affect any HCVs related to biodiversity. The project intends to improve the related ecosystem services and goods (forests, soils or land-use fertility, water sources) that assures that communities could have a sustainable livelihood with unpolluted water resources. Moreover, through forest conservation and restoration of previously intervened areas (other than the project area), vegetation and fauna species are expected to recover and eventually enhanced. Economic activities to be developed are planned in a sustainable way to contribute to the communities to continue to inhabit their territories.

5.2.5 Species Used (VCS, 3.19; CCB, B2.5, B2.6)

Within the Indigenous Territories (*Terra Indigena*), restoration of previously intervened areas (other than the project area) may be promoted. In case of this, species to be used are going to be native and will be defined by the community.

At the moment, species defined to be used in productive systems are:

Species introduced	Classification	Justification for use	Adverse effects and mitigation
Acai (<i>Euterpe oleracea</i>)	Native	is a species of palm tree cultivated for its fruit (berries), hearts of palm (vegetable), leaves, and trunk wood. The species is native of the Amazonia. Açaí palms are tall, slender trees growing to more than 25 m tall, with pinnate leaves up to 3 m long. The fruit is small, round, and black purple in colour	There are no adverse effects nor mitigation measures to be addressed as it corresponds to a native species
Cassava (<i>Manihot esculenta</i>)	Native	is a woody shrub of the spurge family, native to South America. Although a perennial plant, cassava is extensively cultivated as an annual crop in tropical and subtropical regions for its edible starchy root tuber, a major source of carbohydrates. Cassava is predominantly consumed in boiled form, but substantial quantities are used as an edible coarse flour. Cassava is the third-largest source of food carbohydrates in the tropics.	There are no adverse effects nor mitigation measures to be addressed as it corresponds to a native species

Establishment of these species will be developed in areas that had been previously intervened, as stated in section 2.1.16. All these species are commonly used in the project zone and the region and are widely considered to be non-invasive. Nevertheless, sugarcane is an introduced species and is described in section 5.2.7.

Eventually, other species may be considered to establish new productive areas, but no invasive or GMO species would be used.

5.2.6 Invasive Species (VCS, 3.19; CCB, B2.5)

As mentioned above, expected species to be used in the project are widely considered to be non-invasive and native.

5.2.7 GMO Exclusion (CCB, B2.7)

Since organic fertilization and pest control methods are the ones which community members are currently accustomed to use, no Genetically Modified Organisms (GMO's) will be implemented by the project.

5.2.8 Inputs Justification (VCS, 3.19; CCB, B2.8)

No fertilizers, chemical pesticides or biological control agents are expected to be used. In case the project encounters a need for the use of any chemical methods at a future date then there will be a prior assessment of possible adverse effects and standard operating procedures produced to ensure their safety to the environment and community members.

5.2.9 Waste Products (VCS, 3.19; CCB, B2.9)

Specific operating procedures for identifying, classifying and managing all waste products resulting from project are the following:

Activity	Waste Management
Sustainable Agriculture	Use empty containers according with current regulations on hazardous waste. Reduce application of fertilizers and pesticides. If needed use IPM practices. Avoid burning of waste biomass. If done, use it as fertilizer.

5.3 Offsite Biodiversity Impacts

5.3.1 Negative Offsite Biodiversity Impacts (CCB, B3.1) and Mitigation Measures (CCB, B3.2)

The project activities are designed to deliver positive biodiversity effects and they are not anticipated to lead to negative impacts, either on site or off.

5.3.2 Net Offsite Biodiversity Benefits (VCS, 3.19; CCB, B3.3)

The project activities are designed to deliver positive biodiversity impacts inside and outside the project zone. Agricultural production systems inside de project zone will incorporate good management practices, thus, reducing soil erosion, water pollution and eutrophication processes that affect

downstream conditions and biodiversity. Progressive substitution of mining activities for other productive activities that doesn't imply natural resources degradation, will reduce water pollutants and improve quality downstream. In this sense, rivers and streams that cross through the project area are expected to receive less contamination load due to project interventions, representing a net benefit to biodiversity.

Project forest conservation will also contribute to regional landscape and ecosystem protection initiatives. During project implementation, land use planning within the Indigenous Territories of Acapuri de Cima, Estrela da Paz and Riozinho will consider recommendations from protected or reserve natural areas (like The Cujubim Sustainable Development Reserve, the Rio Jutáí Extractive Reserve and Mamirauá Sustainable Development Reserve) for proper management practices. The objective is to ensure that no land use change inside the project area may deteriorate or affect biodiversity conservation practices in the protected area, and that conservation areas should be defined near this boundary to serve as a buffer zone for the protected area. Land use planning will also consider other surrounding areas that have standing forests to stablish or define conservation areas inside the territory that will contribute to maintain or even increased the extension of continuous forest in the long term, which will benefit ecological connectivity and biodiversity protection.

Other project activities that involve infrastructure improving or building, improving housing infrastructure or services, will be developed with the correct and legal environmental considerations to reduce waste, water consumption, avoid hazardous materials and protect natural resources. These considerations will reduce environmental impacts that otherwise would go beyond project boundaries and affect biodiversity.

Under these considerations, no potential unmitigated negative impacts on biodiversity outside the project area. REDD+ activities will contribute to biodiversity conservation inside and outside the project area. The project will generate favorable conditions for water, forest and biodiversity protection. Biological connectivity, ecosystem integrity and hydrological conditions will be maintained and improved during project lifetime. If any harm or negative impact over biodiversity is acknowledged during project implementation, mitigation, compensation or termination actions will be taken.

5.4 Biodiversity Impact Monitoring

5.4.1 Biodiversity Monitoring Plan (CCB, B4.1, B4.2, GL1.4, GL3.4)

At the ecosystem level, forest biomass monitoring will demonstrate the efficacy of activities designed to reduce degradation and deforestation, resulting in forest structure and composition maintenance. This will be done using the same remote sensing monitoring techniques described in the Climate Monitoring Plan (sections 3.3.2 and 3.3.3 of the PD).

Data / Parameter	Area of Tropical Rain Forest
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Data unit	ha
Description	Quantification of the area of forest land within the project area at the beginning and the end of each monitoring period.
Source of data	Satellite images
Description of measurement methods and procedures to be applied	By using satellite images and remote sensing to map forest cover in the Project Area.
Frequency of monitoring/recording	Annually
Value applied	N/A
Monitoring equipment	Computers and SIG software
QA/QC procedures to be applied	
Purpose of Data	Monitoring of HCVs. It would be determined if there are any variations in the forest area.
Calculation method	N/A
Comments	

Data / Parameter	Avoided Tropical Rain Forest loss
Data unit	ha
Description	Quantification of the area of forest loss avoided compared to the estimated in the baseline scenario.
Source of data	Satellite images
Description of measurement methods and procedures to be applied	By using satellite images and remote sensing to map forest cover in the Project Area.
Frequency of monitoring/recording	Annually
Value applied	N/A
Monitoring equipment	Computers and SIG software
QA/QC procedures to be applied	
Purpose of Data	Monitoring of HCVs
Calculation method	N/A

Comments	
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Fauna

The monitoring of the jaguar will be done through direct and indirect methods as indicated below. If sighting of individuals or traces are reported by community members, semi-structured interviews will be conducted to gather the information and contribute to the species monitoring. The interviews may collect information related to date of sighting, location, number of individuals or type of traces observed, estimated size and behavior, if available. Camera trap will be placed, and the records will be analyzed to quantify the number of sightings of the specie. Also, indirect evidence of the individuals is going to be used to contribute the monitoring activity.

Data / Parameter	Number of sightings reported
Data unit	Number
Description	Quantification of the sightings reported by the community members within the limits of the Indigenous Territories Quantification of the sightings recorded within the project area using camera trap.
Source of data	Direct counting
Description of measurement methods and procedures to be applied	The sightings reported by the communities will be informed and interviews will be carried out to generate a record of evidence of these. Sightings on records of camera trap will be counted and reported.
Frequency of monitoring/recording	Annually
Value applied	N/A
Monitoring equipment	Camera trap, binoculars, GPS, field book, record forms.
QA/QC procedures to be applied	Project proponents will store the interviews performed. If available, photographic and/or audiovisual record will be stored as well.
Purpose of Data	Monitoring of HCVs.
Calculation method	N/A
Comments	

Data / Parameter	Number of traces observed
Data unit	Number

Description	Quantification of traces identified during monitoring trails offer an indirect way to identify the presence of the species in the territory.
Source of data	Direct identifications of traces
Description of measurement methods and procedures to be applied	Quantification of traces such as footprints, droppings, marks, urine and victims that are observed during trails or other activities may be registered in a form to enable reporting and follow up this type of sighting.
Frequency of monitoring/recording	Annually
Value applied	N/A
Monitoring equipment	Camera, GPS, gypsum, binoculars, field book and track record form.
QA/QC procedures to be applied	Project proponents will store the record forms filled out during the monitoring trails.
Purpose of Data	Monitoring of HCVs.
Calculation method	N/A
Comments	

It is anticipated that project activities will lead to positive enhancement of HCV areas.

5.4.2 Biodiversity Monitoring Plan Dissemination (CCB, B4.3)

Dissemination of Monitoring Plan and Implementation Reports consider the guidelines described in sections 2.3.4 and 3.3.4.

5.5 Optional Criterion: Exceptional Biodiversity Benefits

The project is not seeking Gold Level for exceptional biodiversity benefits.

APPENDIX 1: STAKEHOLDER DESCRIPTION TABLE

Stakeholder	Rights, interest, and overall relevance to the project
Indigenous communities (Aldeias) of the three indigenous territories (Terra Indigena)	Rights to land and to VCR Beneficiaries Implementing parties
Mining developers	Agents dedicated to mining activities in the Indigenous Territories. One of the goals of the project is to promote the transition from extractive activities to sustainable activities and demonstrate in the short term the possibility to make a good living out of nature-based production mechanisms.
Municipality of Jutai	Does not have direct rights to carbon benefits but are articulated with the project and local interventions.
Municipality of Fonte Boa	Does not have direct rights to carbon benefits but are articulated with the project and local interventions.
Municipality of Jurua	Does not have direct rights to carbon benefits but are articulated with the project and local interventions.
Jutai Indigenous Council (COPIJU)	Does not have direct rights to carbon benefits but are articulated with the project and local interventions.
Institute of Environmental Protection of the Amazonas State	Does not have direct rights to carbon benefits but are articulated with the project and local interventions.
Secretary of the Environment of Jutai	Does not have direct rights to carbon benefits but are articulated with the project and local interventions.
Brazilian Institute of Environment and Renewable Natural Resources IBAMA	Does not have direct rights to carbon benefits but are articulated with the project and local interventions.

APPENDIX 2: PROJECT ACTIVITIES AND THEORY OF CHANGE TABLE

Use this appendix, if applicable, to identify project activities and fulfill the requirements of Section 2.1.17 above. This is an example of just one method of representing the theory of change. Results chains/flow diagrams are another effective way to represent the theory of change. Modify the table, if necessary, to suit the project activities, or delete if not used.

Activity description	Expected climate, community, and/or biodiversity			Relevance to project's objectives
	Outputs (short term)	Outcomes (medium term)	Impacts (long term)	
Sustainable Productive activities (cash crops such as acai, pineapple, watermelon)	Crops in place (ha) Technical assistance provided (number of trainees)	Harvest Produce (tons/year) Revenues (\$/year)	Forest conservation as incentives for mining and expansion of agricultural frontier are reduced	Forest conservation is the main project objective. Emissions reduction by avoided deforestation (climate objective). Community benefits delivered. Biodiversity protection improved.
Natural forest restoration	Secondary forest zoning and assisted natural regrowth (ha)	Secondary forest standing (ha)	Forest cover increase	Carbon sequestration adding to carbon credit production (climate objective) Biodiversity protection improved.

Activity description	Expected climate, community, and/or biodiversity			Relevance to project's objectives
	Outputs (short term)	Outcomes (medium term)	Impacts (long term)	
Social investments (water and sanitation, health, education, energy, transport)	Social infrastructure being planned and built	Social infrastructure built and functioning Social control on conservation commitments	Services related to social infrastructure provided and contributing to communities welfare Incentives for deforestation reduced	Community benefits delivered. Biodiversity protection improved. Forest conservation is favored due to community awareness (climate objectives)
Forest governance (indigenous ethno-development plan)	Workshops and consultants to help produce indigenous ethno-development plan	Plan drafted, adopted and socialized	Framework for indigenous ethno-development in place to guide investments according to cultural values	Forest conservation is included in territory planning. Community benefits delivered. Biodiversity protection improved
Forest governance (surveillance)	Forest rangers hired and equipment procured	Rangers monitoring forests and reporting	Deforestation agents are deterred from changing land use	Forest is protected. Community benefits delivered. Biodiversity protection improved
Forest governance (education)	Trainers are trained and support materials produced	Environmental education is provided to community members and other stakeholders	Communities members aware of deforestation problems, and about benefits of forest protection	Forest is protected. Community benefits delivered. Biodiversity protection improved

Activity description	Expected climate, community, and/or biodiversity			Relevance to project's objectives
	Outputs (short term)	Outcomes (medium term)	Impacts (long term)	
Monitoring (forest plots and monitoring)	Monitoring plan in place, and human resources hired and equipment acquired	Forest inventory plots undertaken, and forest cover assessed	Monitoring of forest cover reported	Forest is protected. Community benefits delivered. Biodiversity protection improved

APPENDIX 3: PROJECT RISKS TABLE

Use this appendix to identify and summarize the project risks and fulfill the requirements of Sections 2.3.6, 2.3.16, 2.5, and **¡Error! No se encuentra el origen de la referencia..1– ¡Error! No se encuentra el origen de la referencia.,** above. Where no risk is identified, write 'No risk identified' and provide justification in the third column. Add rows as needed.

Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) needed and designed to mitigate the risk
Potential increase in the severity of storms, flooding, fires and sedimentation	• Loss of forest covers • Ecosystem loss • Community vulnerability increase • Emission of GHG in the project scenario	For risks like extreme events such as fires, storms or flooding, forest conservation contributes to mitigate negative impacts. It is also important to point out that these risks correspond to natural phenomena and it is difficult to establish mitigation protocols prior to the manifestation of the events in the area. It is expected that reducing forest loss and degradation in the project area, these risks are mitigated.
Former deforestation activities (agriculture, pastures, mining and settlements)	• Loss of forest covers • Ecosystem loss • High dependence on income resources • Emission of GHG in the project scenario	To mitigate these risks, the project is going to generate income opportunities, implement sustainable productive activities, create jobs, deliver social investment (education, infrastructure, sanitation, health, electricity, food security), strengthen and build capacities of the indigenous communities, forest rangers and patrols, along with surveillance over the territory. These activities tackle the main cause of deforestation that relates with economic and educational reasons. Acknowledging the historical land use practices, it is expected to induce transition from deforestation activities to sustainable activities in a progressive way, as a result of delivering new and real opportunities for people to make a living and improve their livelihoods through conservation activities. This integrated approach represents the REDD+ strategy, and it will address individual members and progressively extend to the whole community, as project develops. Monitoring REDD+ activities and adjust the intervention based on observed results, is part of the project implementation and

Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) needed and designed to mitigate the risk
		improvement scheme that contributes to reduce the risk of former activities to increase or extend through the territory.
Weak governance capacity of the indigenous communities	- Increasing deforestation within project boundaries - Low territorial control - Persistence of livelihoods related to deforestation - Weak land use planning	To mitigate this risk, the project has two components related with strengthen governance and monitoring. Access to formal education and technical formation in administrative topics, business management and development, as well as defining and implementation of the indigenous ethno-developing plan, land use plan and biodiversity monitoring plan are part of the elements that increase governance capacity of the local councils and that will be implemented through specific project activities.
Changes in the socioeconomic dynamics	- Forest loss - Community lack of participation in the project activities - GHG emissions within the project area	The project includes governance improvement and sustainable production. The project is going to generate income opportunities, implement sustainable productive activities, create jobs, deliver social investment (education, infrastructure, sanitation, electricity, food security), as well as build capacities to improve territory governance and facilitate access to education programs. These activities address possible socioeconomic dynamics that result in forest loss, so it is expected to mitigate this risk through the explained integrated approach. In addition, community engaging and development of participatory processes help to ensure that community members involve in an active way and the expected outcomes of the project in the short, middle and long-term are fulfill.
Land use changes in the project area due to deforestation and forest degradation	- Forest loss - Ecosystem fragmentation - Emission of GHG within the project area	Monitoring activities will be carried out to identify areas where deforestation and forest degradation take place. Drivers and agents have been identified and a complete and integrated REDD+ strategy has been defined to reduce the trends of deforestation and degradation. REDD+ activities are oriented to improve land use of already degraded areas and contribute to stop forest loss. Establishing production systems that support livelihoods and offer income opportunities for the people, reduces the need to clear forest areas and mitigate land use changes trends.

Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) needed and designed to mitigate the risk
		Specific REDD+ activities that generate and implement land use plans of the territory will allow to increase control and promote an efficient land use, so deforestation and forest degradation are expected to reduce as much as possible.
Increase in the development of mining activities in the territory of the Indigenous communities conforming the Estrella Acapuzihno REDD+ Project	• Increasing deforestation within project boundaries • Low territorial control • Persistence of livelihoods related to deforestation • Community lack of participation in the project activities • GHG emissions within the project area	The execution of activities that offer income generation, job creation and economic stability out of sustainable productive activities are part of the strategy of the project to promote a gradual transition from extractive mining to sustainable livelihoods and mitigate this risk. The community needs to make a living out of sustainable practices in order to reduce mechanized mining. Delivering community and individual benefits and economic opportunities through the project activities implementation is the best incentive and strategy to reduce the possibility for mining expansion. As long as the community experience that conservation of their territory represents a good and decent livelihood, social control over expanding mining activities will also increase and the risk of mining expansion will be mitigated.

APPENDIX 4: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification
Benefit distribution	Commercial and Development Agreement between the project developers and the community	The project region has multiple actors, so this information could be use in a way that may affect the project participants.
Financial health	Project's financial model.	The project region has multiple actors, so this information could be use in a way that may affect the project participants.

APPENDIX 5: REFERENCES

- Alves, T. (2023). *Brazil strengthens environmental preservation efforts, signs new Indigenous land demarcations*. Obtenido de Brazil Reports: <https://brazilreports.com/brazil-strengthens-environmental-preservation-efforts-signs-new-indigenous-land-demarcations/5287/>
- Brazilian NR. (2023). *Brazilian Regulatory Standards*. Obtenido de <https://www.braziliannr.com/brazilian-regulatory-standards/>
- Caravela Datos y Estadísticas. (2023). *Jutaí - AM*. Obtenido de <https://www.caravela.info/regional/juta%C3%AD-am>
- Conservation International; Government of Amazonas; British Embassy. (s.f.). *Sustainable Development Reserve Cujubim*. Obtenido de https://www.conservation.org/docs/default-source/brasil/rdscujubimpdf.pdf?sfvrsn=d8f4dd8d_2
- COPIJU. (2011). *Plano de vida do movimento indígena de Jutaí/ AM e Região*. Obtenido de Visao Mundial, OPAN, USAID .
- Global Forest Watch . (2023). *Jutaí*. Obtenido de <https://www.globalforestwatch.org/dashboards/country/BRA/4/33/?category=land-cover&dashboardPrompts=eyJzaG93UHJvbXB0cyl6dHJ1ZSwicHJvbXB0c1ZpZXdlZCI6WyJkb3dubG9hZERhc2hib2FyZFN0YXRzIl0sInNldHRpbmdzljp7InNob3dQcm9tcHRzljp0cnVILCJwcm9tcHRzVmllld2VkljbmlmRvd25>
- Global Forest Watch. (2023). *Fonte Boa*. Obtenido de <https://www.globalforestwatch.org/dashboards/country/BRA/4/24/?category=summary&dashboardPrompts=eyJzaG93UHJvbXB0cyl6dHJ1ZSwicHJvbXB0c1ZpZXdlZCI6WyJkb3dubG9hZERhc2hib2FyZFN0YXRzIl0sInNldHRpbmdzljp7InNob3dQcm9tcHRzljp0cnVILCJwcm9tcHRzVmllld2VkljbmlmRvd25sb2>
- González, J. (2018). Analyzing the South American policy of Climate Change and forests through REDD+: The cases of Argentina, Brazil and Bolivia. *Latin American Journal of International Affairs* , 01-17.
- Guedes, G., Brondízio, E., Barbieri, A., Resende, A., Penna-Firme, R., & D'Antona, Á. (2012). *Poverty and Inequality in the Rural Brazilian Amazon: A Multidimensional Approach*. Obtenido de <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3426830/>
- Human Rights Watch. (2023). *Brazil: Reject Harmful Bill on Indigenous Rights*. Obtenido de <https://www.hrw.org/news/2023/05/29/brazil-reject-harmful-bill-indigenous-rights#:~:text=Brazil's%20Constitution%20recognizes%20Indigenous%20people's,territories%20and%20to%20protect%20them>.
- IBGE. (2023). *Fonte Boa*. Obtenido de <https://cidades.ibge.gov.br/brasil/am/fonte-boa/panorama>
- IBGE. (2023). *Historia y fotos*. Obtenido de <https://cidades.ibge.gov.br/brasil/am/jutai/historico>
- IBGE. (2023). *Jutaí*. Obtenido de <https://cidades.ibge.gov.br/brasil/am/jutai/panorama>
- IUCN. (s.f.). *Strengthening Voices for Better Choices in Brazil*. Obtenido de https://www.iucn.org/sites/default/files/import/downloads/brazil_1.pdf
- Ministry of the Environment . (2016). *ENREDD+*. Obtenido de National Strategy for Reducing Emissions from Deforestation and Forest Degradation, and the role of Conservation of Forest Carbon Stocks, Sustainable Management of Forests and Enhancement of Forest Carbon Stocks: <http://redd.mma.gov.br/en/>
- Ministry of the Environment. (2014). *Brazil's submission of a Forest Reference Emission Level (FREL) for reducing emissions from deforestation in the Amazonia biome for REDD+ results based paymentes under UNFCCC*.
- NREF Brazil . (2022). *NREF Brazil*.
- Sandy, M. (s.f.). *HE AMAZON RAIN FOREST IS NEARLY GONE*. Obtenido de TIME: <https://time.com/amazon-rainforest-disappearing/>